

## **STAMPEDE**

### **Systemic Therapy in Advancing or Metastatic Prostate Cancer: Evaluation of Drug Efficacy**

**A multi-arm multi-stage randomised controlled  
trial**

**Version: 17.0**

**Date: 19-Oct-2017**

|                           |                            |
|---------------------------|----------------------------|
| <b>MRC CTU AT UCL ID:</b> | <b>PR08</b>                |
| <b>ISRCTN #:</b>          | <b>ISRCTN78818544</b>      |
| <b>NCT #:</b>             | <b>NCT00268476</b>         |
| <b>EUDRACT #:</b>         | <b>2004-000193-31</b>      |
| <b>CTA #:</b>             | <b>00316/0026/001-0001</b> |
| <b>MREC #:</b>            | <b>04/MRE07/35</b>         |

**Authorised by:**

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Name:</b>      | <b>PROFESSOR NICHOLAS D JAMES</b> |
| <b>Role:</b>      | <b>CHIEF INVESTIGATOR</b>         |
| <b>Signature:</b> |                                   |

|              |                           |
|--------------|---------------------------|
| <b>Name:</b> | <b>MATTHEW SYDES</b>      |
| <b>Role:</b> | <b>TRIAL STATISTICIAN</b> |

## GENERAL INFORMATION

This document was constructed using the MRC CTU at UCL Protocol Template Version 4.0. It describes the STAMPEDE trial, coordinated by the Medical Research Council (MRC) Clinical Trials Unit (CTU) at University College London (UCL), and provides information about procedures for entering patients into it. The protocol should not be used as an aide-memoire or guide for the treatment of other patients. Every care has been taken in drafting this protocol, but corrections or amendments may be necessary. These will be circulated to the registered investigators in the trial, but sites entering patients for the first time are advised to contact the Cancer And Other Non-Infectious Diseases Group, MRC CTU at UCL, London, UK, to confirm they have the most up-to-date version.

## COMPLIANCE

The trial will be conducted in compliance with the approved protocol, the Declaration of Helsinki 1996, the principles of Good Clinical Practice (GCP), Commission Directive 2005/28/EC with the implementation in national legislation in the UK by Statutory Instrument 2004/1031 and subsequent amendments, the UK Data Protection Act (DPA number: Z6364106), and the National Health Service (NHS) Research Governance Framework for Health and Social Care (RGF). International sites will comply with the principles of GCP as laid down by the ICH topic E6 (Note for Guidance on GCP), Commission Directive 2005/28/EC (the European Directive 2001/20/EC [where applicable]) and applicable national regulations.

## SPONSOR

Medical Research Council, 2nd Floor, David Phillips Building, Polaris House, North Star Avenue, Swindon, SN2 1FL, UK.

On 01-Aug-2013, the MRC CTU became part of University College London (UCL). The MRC maintains sponsorship for the trial however UCL is the legal entity responsible for the running of the trial. This responsibility is delegated to the coordinating trial unit, the MRC CTU at UCL.

## FUNDING

Cancer Research UK's Clinical Research Committee (formerly the Clinical Trials Advisory Awards Committee), Medical Research Council, and educational grants from Novartis, Sanofi-Aventis, Pfizer, Janssen Pharma NV, Astellas, Clovis Oncology.

## AUTHORISATIONS AND APPROVALS

The following persons are authorised to sign the final protocol and protocol amendments for the sponsor: Professor Nicholas James (Chief Investigator) and Matthew Sydes (Trial Statistician).

## TRIAL REGISTRATION

This trial has been registered with the ClinicalTrials.gov Clinical Trials Register, where it is identified as NCT00268476.

### REGISTRATION AND RANDOMISATION

Call MRC CTU at UCL, Monday to Friday 0900-1700  
Excluding public holidays or dates when notice has been given by the Unit.  
Tel: +44 (0) 20 7670 4777

### SAE REPORTING

**Fax** to 020 7670 4818 within 24 hours of becoming aware of the event  
Or send via **encrypted** email to [mrcctu.stampede@ucl.ac.uk](mailto:mrcctu.stampede@ucl.ac.uk)

## TRIAL ADMINISTRATION

### COORDINATING SITE

MRC Clinical Trials Unit at UCL  
90 High Holborn  
2nd Floor  
London WC1V 6LJ  
UK

Switchboard: 020 7670 4700  
Fax: 020 7670 4818  
Email: [mrcctu.stampede@ucl.ac.uk](mailto:mrcctu.stampede@ucl.ac.uk)  
Website: <http://www.stampedetrial.org/>

### MRC CTU AT UCL STAFF

|                              |                      |      |                              |
|------------------------------|----------------------|------|------------------------------|
| Trial Manager:               | Mazna Anjum          | Tel: | 0207 670 4772                |
| Trial Manager:               | Michelle Buckner     | Tel: | 0207 670 4840                |
| Trial Manager:               | Joanna Calvert       | Tel: | 0207 670 4916                |
| Trial Manager:               | Claire Murphy        | Tel: | 0207 670 4620                |
| Trial Manager:               | Christopher Wanstall | Tel: | 0207 670 4882                |
| Trial Manager:               | Arlen Wilcox         | Tel: | 0207 670 4822                |
| Data Manager:                | Carly Au             | Tel: | 0207 670 4794                |
| Data Manager:                | Lina Bergstrom       | Tel: | 0207 670 4693                |
| Data Manager:                | Danielle Johnson     | Tel: | 0207 670 4604                |
| Data Manager:                | Saba Khan            | Tel: | 020 670 4865                 |
| Data Manager:                | Tasmin Philips       | Tel: | 0207 670 4947                |
| Data Manager:                | Peter Vaughan        | Tel: | 0207 670 4655                |
| Clinical Project Manager:    | Claire Amos          | Tel: | 0207 670 4771                |
| Clinical Project Manager:    | Nafisah Atako        | Tel: | 0207 670 4896                |
| Statistician:                | Chris Brawley        | Tel: | 0207 670 4677                |
| Statistician:                | Fiona Ingleby        | Tel: | 0207 670 4696                |
| Senior Trial Statistician:   | Matthew Sydes        | Tel: | 0207 670 4798                |
| Transdermal Oestradiol Lead: | Duncan Gilbert       | Tel: | Please contact STAMPEDE team |
| Clinical Research Fellow     | Clare Gilson         | Tel: | Please contact STAMPEDE team |
| Director, MRC CTU at UCL:    | Max Parmar           | Tel: | Please contact STAMPEDE team |

### CHIEF INVESTIGATOR

Prof Nicholas James

Clinical Oncology, The Medical School  
University of Birmingham  
Birmingham, B15 2TJ  
UK

Tel: 0121 371 3615 (Secretary)  
Email: [n.d.james@bham.ac.uk](mailto:n.d.james@bham.ac.uk)

## CO-INVESTIGATORS AND TMG MEMBERS

|                                                |                        |                                                 |
|------------------------------------------------|------------------------|-------------------------------------------------|
| Prof Paul Abel<br>Imperial College             | London, UK             | Co-CCI "Transdermal oestradiol comparison"      |
| Dr Gerhardt Attard<br>Oncologist               | London, UK             | CCI "Enzalutamide+Abiraterone comparison"       |
| Dr Simon Chowdhury<br>Oncologist               | London, UK             |                                                 |
| Prof Noel Clarke<br>Urologist                  | Manchester, UK         | TMG Vice-Chair<br>Co-CCI "Metformin comparison" |
| Mr William Cross<br>Urologist                  | Leeds, UK              |                                                 |
| Prof David Dearnaley<br>Oncologist             | London, UK             |                                                 |
| Prof Silke Gillesen<br>Oncologist              | St Gallen, Switzerland | Co-CCI "Metformin comparison"                   |
| Prof Rob Jones<br>Oncologist                   | Glasgow, UK            |                                                 |
| Dr Zafar Malik<br>Oncologist                   | Liverpool, UK          |                                                 |
| Prof Malcolm Mason<br>Oncologist               | Cardiff, UK            | TMG Vice-Chair                                  |
| Prof Ruth Langley<br>Oncologist                | London & Brighton, UK  | Co-CCI "Transdermal oestradiol comparison"      |
| Dr Chris Parker<br>Oncologist                  | London, UK             | CCI "M1 RT comparison"                          |
| Mr Alastair Ritchie<br>Surgeon                 | Edinburgh, UK          |                                                 |
| Dr Martin Russell<br>Oncologist                | Glasgow, UK            |                                                 |
| Prof George Thalmann<br>Urologist              | Bern, Switzerland      |                                                 |
| Robin Millman<br>Patient representative        | UK                     |                                                 |
| David Matheson,<br>Patient representative      | UK                     |                                                 |
| Prof Mark Sculpher<br>Health Economics Advisor | York, UK               |                                                 |
| Dr Amanda Adler<br>Diabetologist               | Cambridge, UK          | Diabetologist "Metformin comparison"            |
| <b>SAAK Coordinating Centre</b>                |                        |                                                 |
| Corinne Schär                                  | Bern, Switzerland      |                                                 |

**Trial Advisory Groups**

|                           |                             |
|---------------------------|-----------------------------|
| Biological Research Group | Prof Malcolm Mason (Chair)  |
|                           | Prof David Waugh (Co-Chair) |

|                               |                          |
|-------------------------------|--------------------------|
| Metabolic Translational Group | Prof Noel Clarke (Chair) |
|-------------------------------|--------------------------|

|                       |                          |
|-----------------------|--------------------------|
| Clinical Safety Group | Prof Noel Clarke (Chair) |
|-----------------------|--------------------------|

## SUMMARY OF TRIAL

| SUMMARY INFORMATION TYPE             | SUMMARY DETAILS                                                                                                                               |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Acronym                              | STAMPEDE                                                                                                                                      |
| Long Title of Trial                  | Systemic Therapy in Advancing or Metastatic Prostate Cancer: Evaluation of Drug Efficacy: A multi-arm multi-stage randomised controlled trial |
| Version                              | 17.0                                                                                                                                          |
| Date                                 | 19-Oct-2017                                                                                                                                   |
| MRC CTU at UCL ID                    | PR08                                                                                                                                          |
| NCT #                                | NCT00268476                                                                                                                                   |
| EudraCT #                            | 2004-000193-31                                                                                                                                |
| Study Design                         | Multi-arm multi-stage platform randomised controlled trial                                                                                    |
| Type of Participants to be Studied   | Men starting long-term hormone therapy for metastatic or high-risk non-metastatic prostate cancer                                             |
| Setting                              | Tertiary care                                                                                                                                 |
| Interventions to be Compared         | Various - see comparison-specific tables                                                                                                      |
| Study Hypothesis                     | Various - see comparison-specific tables                                                                                                      |
| Definitive Primary Outcome Measure   | Overall survival (unless stated)                                                                                                              |
| Intermediate Primary Outcome Measure | Failure-free survival (unless stated)                                                                                                         |
| Secondary Outcome Measure(s)         | Toxicity<br>Symptomatic skeletal events<br>Quality-of-life<br>Cost-effectiveness                                                              |
| Randomisation                        | Minimisation using a random element across a number of stratification factors                                                                 |
| Number of Participants               | See comparison-specific tables                                                                                                                |
| Duration                             | See comparison-specific tables                                                                                                                |
| Sponsor                              | Medical Research Council                                                                                                                      |
| Funders                              | Cancer Research UK<br>Medical Research Council<br>Astellas<br>Clovis Oncology<br>Janssen<br>Novartis<br>Pfizer<br>Sanofi-Aventis              |

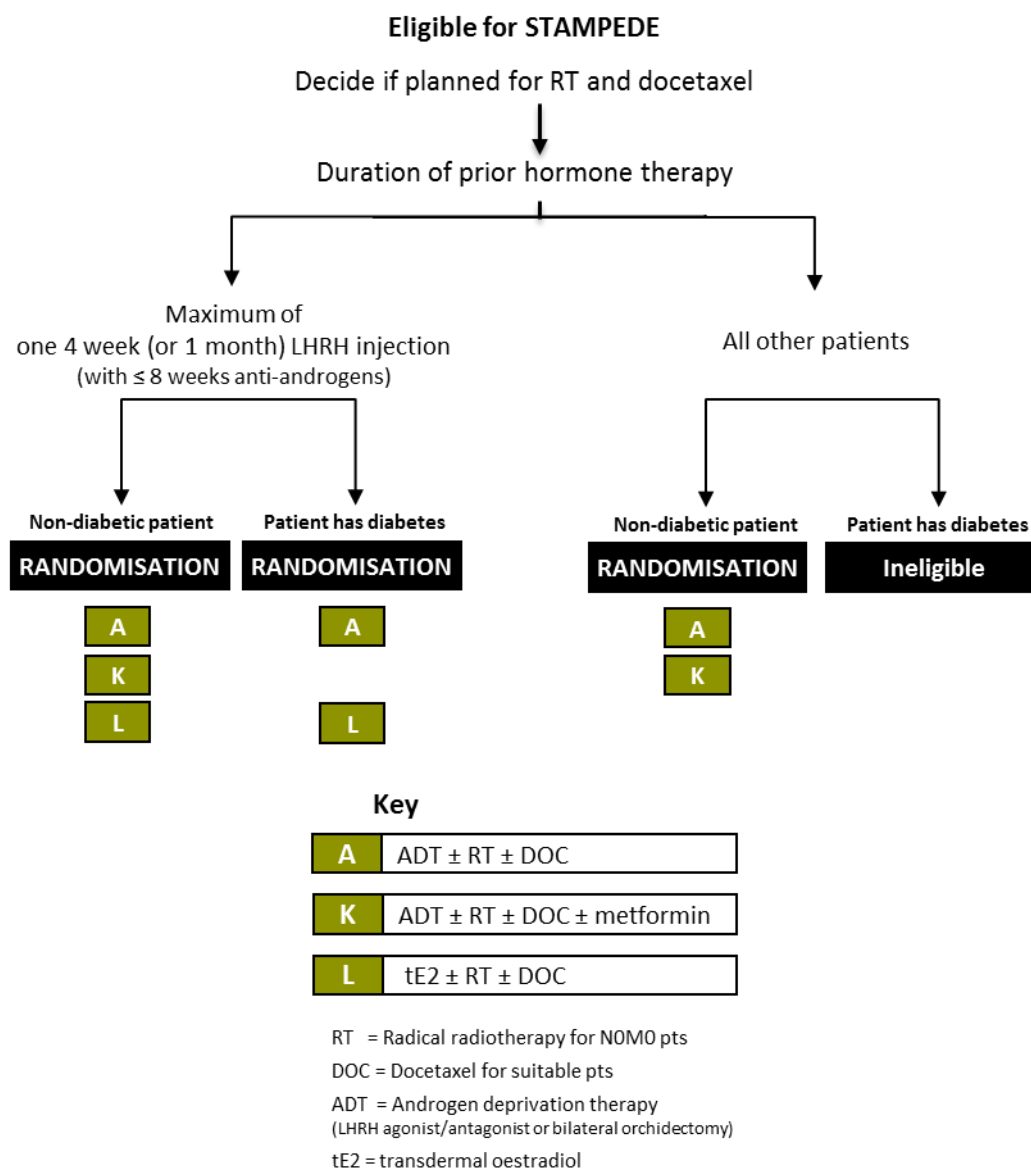
| SUMMARY INFORMATION TYPE             | SUMMARY DETAILS                                                                                                                                                          |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>“Original comparisons”</b>        |                                                                                                                                                                          |
| Type of Participants to be Studied   | Men starting long-term hormone therapy for metastatic or high-risk non-metastatic prostate cancer                                                                        |
| Control Arm                          | Arm A: Standard-of-care (SOC)<br>Androgen-deprivation therapy (ADT) ± prostate RT ± docetaxel                                                                            |
| Interventions to be Compared         | Arm B: SOC + zoledronic acid<br>Arm C: SOC + docetaxel<br>Arm D: SOC + celecoxib<br>Arm E: SOC + zoledronic acid + docetaxel<br>Arm F: SOC + zoledronic acid + celecoxib |
| Allocation ratio                     | 2 control arm : 1 research arm [2A:1B:1C:1D:1E:1F]                                                                                                                       |
| Study Hypothesis                     | Research interventions will improve survival over SOC                                                                                                                    |
| Definitive Primary Outcome Measure   | Overall survival                                                                                                                                                         |
| Intermediate Primary Outcome Measure | Failure-free survival                                                                                                                                                    |
| Number of Participants               | Sufficient for 400 control arm definitive primary outcome measure events (in practice, >3000 patients)                                                                   |
| Duration                             | 10 years                                                                                                                                                                 |
| Status                               | Primary results published (1, 2)                                                                                                                                         |
| <b>“Abiraterone comparison”</b>      |                                                                                                                                                                          |
| Type of Participants to be Studied   | Men starting long-term hormone therapy for metastatic or high-risk non-metastatic prostate cancer                                                                        |
| Control arm                          | Arm A: Standard-of-care (SOC)                                                                                                                                            |
| Intervention to be Compared          | Arm G: SOC + abiraterone                                                                                                                                                 |
| Allocation ratio                     | 1 control arm : 1 research arm [1A:1G]                                                                                                                                   |
| Study Hypothesis                     | Addition of abiraterone to SOC will improve survival over SOC alone                                                                                                      |
| Definitive Primary Outcome Measure   | Overall survival                                                                                                                                                         |
| Intermediate Primary Outcome Measure | Failure-free survival                                                                                                                                                    |
| Number of Participants               | Around 1,800 patients for 267 control arm definitive primary outcome measure events                                                                                      |
| Duration                             | 6 to 8 years                                                                                                                                                             |
| Status                               | Primary results published (3)                                                                                                                                            |
| <b>“M1   RT comparison”</b>          |                                                                                                                                                                          |
| Type of Participants to be Studied   | Men starting long-term hormone therapy for newly-diagnosed metastatic prostate cancer with no contraindication to prostate radiotherapy                                  |
| Control arm                          | Arm A: Standard-of-care (SOC)                                                                                                                                            |
| Intervention to be Compared          | Arm H: SOC + radiotherapy to the prostate (RT)                                                                                                                           |



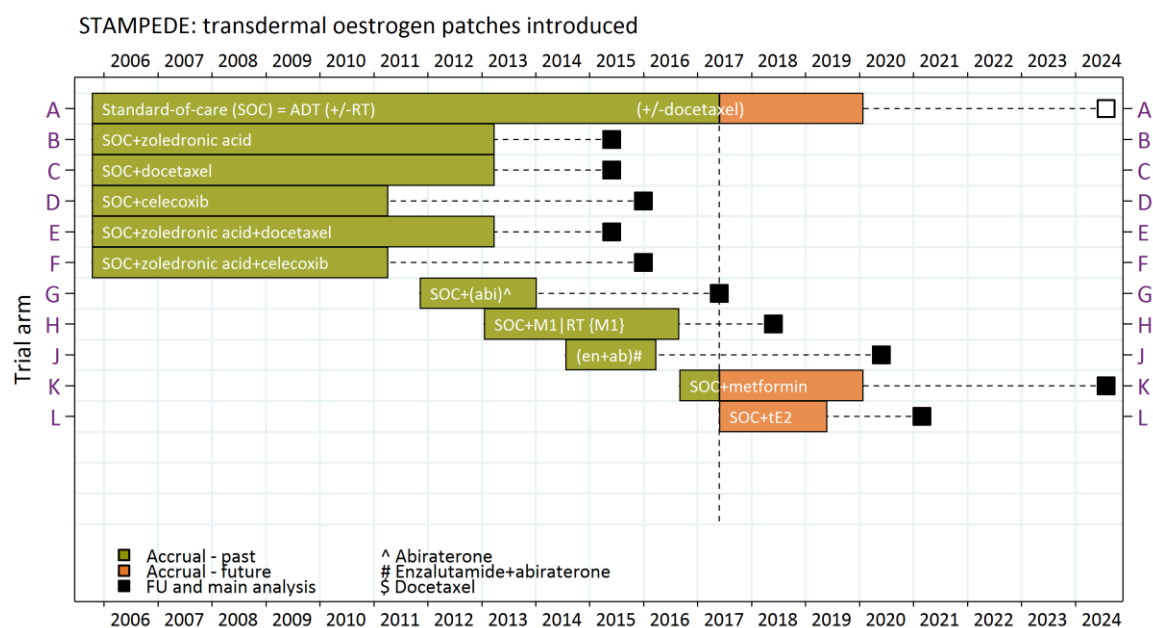
| SUMMARY INFORMATION TYPE                       | SUMMARY DETAILS                                                                                                                                        |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Allocation ratio                               | 1 control arm : 1 research arm [1A:1H]                                                                                                                 |
| Study Hypothesis                               | Addition of RT to SOC will improve survival over SOC alone                                                                                             |
| Definitive Primary Outcome Measure             | Overall survival                                                                                                                                       |
| Intermediate Primary Outcome Measure           | Failure-free survival                                                                                                                                  |
| Number of Participants                         | Around 1,800 patients for 267 control arm definitive primary outcome measure events                                                                    |
| Duration                                       | 6 to 8 years                                                                                                                                           |
| Status                                         | In follow-up                                                                                                                                           |
| <b>“Enzalutamide + abiraterone comparison”</b> |                                                                                                                                                        |
| Type of Participants                           | Men starting long-term hormone therapy for metastatic or high-risk non-metastatic prostate cancer                                                      |
| Control Arm                                    | Arm A: Standard-of-care (SOC)                                                                                                                          |
| Interventions to be Compared                   | Arm J: SOC + enzalutamide + abiraterone                                                                                                                |
| Allocation ratio                               | 1 control arm : 1 research arm [1A:1J]                                                                                                                 |
| Study Hypothesis                               | Addition of enzalutamide, in combination with abiraterone, to SOC will improve survival over SOC alone                                                 |
| Definitive Primary Outcome Measure             | Overall survival                                                                                                                                       |
| Intermediate Primary Outcome Measure           | Failure-free survival                                                                                                                                  |
| Number of Participants                         | Around 1,800 patients for 267 control arm definitive primary outcome measure events                                                                    |
| Duration                                       | 6 to 8 years                                                                                                                                           |
| Status                                         | In follow-up                                                                                                                                           |
| <b>“Metformin comparison”</b>                  |                                                                                                                                                        |
| Type of Participants to be Studied             | Non-diabetic men, with no contraindication to metformin, starting long-term hormone therapy for metastatic or high-risk non-metastatic prostate cancer |
| Control arm                                    | Arm A: Standard-of-care (SOC)                                                                                                                          |
| Intervention to be Compared                    | Arm K: SOC + metformin                                                                                                                                 |
| Allocation ratio                               | 1 control arm : 1 research arm [1A:1K]                                                                                                                 |
| Study Hypothesis                               | Addition of metformin to SOC will improve survival over SOC alone                                                                                      |
| Definitive Primary Outcome Measure             | Overall survival                                                                                                                                       |
| Intermediate Primary Outcome Measure           | Overall survival                                                                                                                                       |
| Number of Participants                         | Around 1,800 patients, including around 1,100 M1 (metastatic) patients, for 374 control arm definitive primary outcome measure events among            |

| SUMMARY INFORMATION TYPE                   | SUMMARY DETAILS                                                                                                                                                                                                                         |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | M1 patients                                                                                                                                                                                                                             |
| Duration                                   | 10 years                                                                                                                                                                                                                                |
| Status                                     | Recruiting                                                                                                                                                                                                                              |
| <b>“Transdermal oestradiol comparison”</b> |                                                                                                                                                                                                                                         |
| Type of Participants to be Studied         | Men starting long-term hormone therapy for metastatic or high-risk non-metastatic prostate cancer, having had no more than one 4-week (or one-month) LHRH (Luteinizing hormone releasing hormone) injection & 8 weeks of anti-androgens |
| Control arm                                | Arm A: Standard-of-care (SOC)                                                                                                                                                                                                           |
| Intervention to be Compared                | Arm L: Transdermal oestradiol ± RT ± docetaxel                                                                                                                                                                                          |
| Allocation ratio                           | 1 control arm : 1 research arm [1A:1L]                                                                                                                                                                                                  |
| Study Hypothesis                           | Transdermal oestradiol will be non-inferior to standard hormone therapy, while having fewer side-effects and improved quality-of-life                                                                                                   |
| Definitive Primary Outcome Measures        | Co-primary endpoints of progression-free survival and overall survival                                                                                                                                                                  |
| Intermediate Primary Outcome Measure       | Progression-free survival                                                                                                                                                                                                               |
| Number of Participants                     | Around 500 to include within a meta-analysis with the PATCH trial, which will include around 2,000 patients overall                                                                                                                     |
| Duration                                   | 4 to 6 years                                                                                                                                                                                                                            |
| Status                                     | Recruiting                                                                                                                                                                                                                              |

Figure 1: Recruiting arms of the STAMPEDE trial from Protocol version 17.0



**Figure 2: Arms of the STAMPEDE trial open to recruitment over time**



**Table 1a: Schedule of Assessments**

| ALL PATIENTS                                                                                      | PRE-REGISTRATION <sup>a</sup> | PRIOR TO RANDOMISATION |
|---------------------------------------------------------------------------------------------------|-------------------------------|------------------------|
| Confirmation of metastatic status to determine eligibility for registration & biomarker screening | X                             |                        |
| Full staging                                                                                      |                               | X <sup>b</sup>         |
| Biochemistry and Haematology <sup>c</sup>                                                         | X                             | X                      |
| Pre-ADT PSA (can be obtained up to 6 months before randomisation)                                 |                               | X                      |
| ECG                                                                                               |                               | X                      |
| BP                                                                                                |                               | X                      |
| FFPE tumour block sent for biomarker-screening <sup>a</sup>                                       |                               | X <sup>a</sup>         |

<sup>a</sup> Only patients participating in biomarker-screening, see [Section 4.3.1](#) for details and the [Biomarker-Screening Manual](#) available via the website.

<sup>b</sup> Cross-sectional imaging of pelvis and abdomen (e.g. CT or MRI), Bone Scan (or equivalent e.g. whole body MRI, choline-PET-CT, PSMA-CT-PET), Chest X-ray (**only** if chest was not included in cross sectional imaging)

<sup>c</sup> Full blood count, urea and electrolytes, liver function tests, creatinine or estimated glomerular filtration rate

**Table 1 b: Patients randomised before 5th Sep 2016**

| ARM A/G/J                                                              | BASELINE | ASSESSMENT WEEK         |    |    |    |    |                |    |                |                |    |     | ALL FURTHER FU VISITS <sup>a</sup> | PROGRESSION | END OF TRT | PRIOR TO 2 <sup>ND</sup> LINE TRT |
|------------------------------------------------------------------------|----------|-------------------------|----|----|----|----|----------------|----|----------------|----------------|----|-----|------------------------------------|-------------|------------|-----------------------------------|
|                                                                        |          | 4-6                     | 12 | 18 | 24 | 36 | 48             | 60 | 72             | 84             | 96 | 104 |                                    |             |            |                                   |
| Baseline bloods <sup>1</sup>                                           | X        |                         |    |    |    |    |                |    |                |                |    |     |                                    |             |            |                                   |
| Blood collection cell-free DNA Streck <sup>TM</sup> tubes <sup>2</sup> |          |                         |    |    |    |    | X <sup>b</sup> |    | X <sup>b</sup> | X <sup>b</sup> |    |     |                                    | X           | X          | X                                 |
| Saliva sample <sup>3</sup>                                             |          | Any time point          |    |    |    |    |                |    |                |                |    |     |                                    |             |            |                                   |
| FFPE block <sup>3</sup>                                                |          | At the point of request |    |    |    |    |                |    |                |                |    |     |                                    |             |            |                                   |
| PSA                                                                    | X        | X                       | X  | X  | X  | X  | X              | X  | X              | X              | X  | X   | X                                  |             |            |                                   |
| Waist circumference                                                    | X        | X                       | X  | X  | X  | X  | X              | X  | X              | X              | X  | X   | X                                  |             |            |                                   |
| Weight                                                                 | X        | X                       | X  | X  | X  | X  | X              | X  | X              | X              | X  | X   | X                                  |             |            |                                   |
| Height                                                                 | X        |                         |    |    |    |    |                |    |                |                |    |     |                                    |             |            |                                   |
| QL + HE <sup>3,4</sup>                                                 | X        | X                       | X  | X  | X  | X  | X              | X  | X              | X              | X  | X   | X                                  |             |            |                                   |

| ARM-SPECIFIC TESTING |                  | BASELINE | ASSESSMENT WEEK |                |    |    |    |    |                |                |                |                |                | ALL FURTHER FU VISITS <sup>a</sup> |
|----------------------|------------------|----------|-----------------|----------------|----|----|----|----|----------------|----------------|----------------|----------------|----------------|------------------------------------|
|                      |                  |          | 4-6             | 12             | 18 | 24 | 36 | 48 | 60             | 72             | 84             | 96             | 104            |                                    |
| G&J                  | Blood pressure   |          | X <sup>c</sup>  | X <sup>c</sup> | X  | X  | X  | X  | X <sup>c</sup> | X <sup>c</sup> | X <sup>c</sup> | X <sup>c</sup> | X <sup>c</sup> | X <sup>c</sup>                     |
| G&J                  | LFTs & potassium |          | X <sup>d</sup>  | X <sup>d</sup> | X  | X  | X  | X  | X <sup>d</sup> | X <sup>d</sup> | X <sup>d</sup> | X <sup>d</sup> | X <sup>d</sup> | X <sup>d</sup>                     |

<sup>a</sup> Follow-up visits after year 2 need to be carried out every 6 months for the first 5 years. At year 6 and onwards visits should be every 12 months whilst follow-up continues.

<sup>b</sup> Sample only required for patient with metastatic disease at trial entry (M1)

<sup>c</sup> For patients receiving research abiraterone blood pressure should be checked every 2 weeks until 12 weeks on treatment, then monthly until 12 months on treatment. Then every 2-months providing measurements have been stable and normal. It is acceptable to review documented patient-monitored BP values or those obtained via the patients GP.

<sup>d</sup> For patients receiving research abiraterone, liver function tests and serum potassium monitoring is required 2-weekly in the first 12-weeks, then monthly until 12 months on treatment. For patients who have not experienced toxicity following 12 months of treatment this may be reduced to every 2 months, to continue whilst receiving research abiraterone. Increased monitoring is required in patients experiencing toxicity; see [Tables 7,8,9](#) for details.

<sup>1</sup> Phosphate, Magnesium, Albumin, calcium and testosterone pre-ADT if available

<sup>2</sup> Only patients participating in sequential blood sampling sub-study, see the [Sample Collection & Handling Manual](#) available via the STAMPEDE website for details

<sup>3</sup> Only for patients that have consented to participate in the relevant sub-study

<sup>4</sup> 1st 700 pts and from Protocol version 8.0 onwards all patients who have consented to participate in the Quality of life (QL) sub-study. QL and Health-economic data collection is to continue until 5 years post randomisation or progression, whichever is sooner.

**Table 1c: Patients randomised after 5th Sep 2016**

| ARM A/K/L                                                                 | BASELINE       | ASSESSMENT WEEK         |    |    |                |    |                |    |                |                |    |                | ALL<br>FURTHER<br>FU VISITS <sup>a</sup> | PROGRESSION | END<br>OF<br>TRT | PRIOR TO<br>2 <sup>ND</sup> LINE<br>TRT |
|---------------------------------------------------------------------------|----------------|-------------------------|----|----|----------------|----|----------------|----|----------------|----------------|----|----------------|------------------------------------------|-------------|------------------|-----------------------------------------|
|                                                                           |                | 4-6                     | 12 | 18 | 24             | 36 | 48             | 60 | 72             | 84             | 96 | 104            |                                          |             |                  |                                         |
| Baseline bloods <sup>1</sup>                                              | X              |                         |    |    |                |    |                |    |                |                |    |                |                                          |             |                  |                                         |
| Blood collection cell-free DNA<br>Streck <sup>TM</sup> tubes <sup>2</sup> | X <sup>b</sup> |                         |    |    |                |    | X <sup>b</sup> |    | X <sup>b</sup> | X <sup>b</sup> |    |                |                                          | X           | X                | X                                       |
| Saliva sample <sup>3</sup>                                                |                | Any time point          |    |    |                |    |                |    |                |                |    |                |                                          |             |                  |                                         |
| FFPE block <sup>3</sup>                                                   |                | At the point of request |    |    |                |    |                |    |                |                |    |                |                                          |             |                  |                                         |
| PSA                                                                       | X              | X                       | X  | X  | X              | X  | X              | X  | X              | X              | X  | X              | X                                        |             |                  |                                         |
| Waist circumference                                                       | X              | X                       | X  | X  | X              | X  | X              | X  | X              | X              | X  | X              | X                                        |             |                  |                                         |
| Weight                                                                    | X              | X                       | X  | X  | X              | X  | X              | X  | X              | X              | X  | X              | X                                        |             |                  |                                         |
| Height                                                                    | X              |                         |    |    |                |    |                |    |                |                |    |                |                                          |             |                  |                                         |
| QL & HE <sup>3,4</sup>                                                    | X              | X                       | X  | X  | X              | X  | X              | X  | X              | X              | X  | X              | X                                        |             |                  |                                         |
| HbA1c                                                                     | X              |                         |    |    | X <sup>c</sup> |    | X <sup>c</sup> |    |                |                |    | X <sup>c</sup> | X <sup>c</sup>                           |             |                  |                                         |
| Fasting Glucose                                                           | X              |                         |    |    | X <sup>c</sup> |    | X <sup>c</sup> |    |                |                |    | X <sup>c</sup> |                                          |             |                  |                                         |
| Fasting Triglycerides                                                     | X              |                         |    |    | X <sup>c</sup> |    | X <sup>c</sup> |    |                |                |    | X <sup>c</sup> |                                          |             |                  |                                         |
| Lipid profile                                                             | X              |                         |    |    | X <sup>c</sup> |    | X <sup>c</sup> |    |                |                |    | X <sup>c</sup> | X <sup>c</sup>                           |             |                  |                                         |

| ARM-SPECIFIC TESTING |                                           | BASELINE | ASSESSMENT WEEK |    |    |    |    |    |    |    |    |    |     | ALL<br>FURTHER<br>FU VISITS <sup>a</sup> |
|----------------------|-------------------------------------------|----------|-----------------|----|----|----|----|----|----|----|----|----|-----|------------------------------------------|
|                      |                                           |          | 4-6             | 12 | 18 | 24 | 36 | 48 | 60 | 72 | 84 | 96 | 104 |                                          |
| K                    | Renal function <sup>5</sup>               | X        |                 |    |    | X  |    | X  |    | X  | X  | X  | X   | X                                        |
| L                    | Testosterone &<br>Oestradiol <sup>6</sup> |          | X               | X  |    | X  |    | X  |    | X  |    | X  | X   | X                                        |

<sup>a</sup> Follow up visits after year 2 need to be carried out every 6 months for the first 5 years. At year 6 and onwards visits should be every 12 months.

<sup>b</sup> Sample only required for patient with metastatic disease at trial entry (M1)

<sup>c</sup> If missed, samples can be obtained +/-12 weeks of the scheduled FU visit, maintaining 10-12 weeks in between the tests due at week 24 and 48 weeks.

<sup>1</sup> Phosphate, Magnesium, Albumin, calcium and testosterone pre-ADT if available

<sup>2</sup> Only patients participating in sequential blood sampling sub-study, see the [Sample Collection & Handling Manual](#) available via the STAMPEDE website for details

<sup>3</sup> Only for patients that have consented to participate in the relevant sub-study

<sup>4</sup> 1st 700 pts and from Protocol v8.0 onwards all patients who have consented to participate in the Quality of life (QL) sub-study. Data collection is to continue until 5 years or progression, whichever is sooner.

<sup>5</sup> Monitoring of renal function (Creatinine or estimated GFR) required at least every 6 months (week 24, week 104 etc.) while receiving metformin.

<sup>6</sup> Hormone tests are required whilst the patient is still receiving research transdermal oestradiol. Note that additional tests may be necessary as detailed in [Section 6.2.8.B](#) and [Section 6.2.8.C](#).

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## ABBREVIATIONS & GLOSSARY

| ABBREVIATION | EXPANSION                                      |
|--------------|------------------------------------------------|
| AA           | Anti-androgen                                  |
| AAH          | Amalgamated Anthracite Holdings                |
| ACE          | Angiotensin-Converting Enzyme                  |
| ACTH         | Adrenocorticotrophic hormone                   |
| ADT          | Androgen deprivation therapy                   |
| AR           | Androgen receptor                              |
| AS           | Activity Stage                                 |
| AUC          | Area under the plasma concentration–time curve |
| BID          | Twice a day (bis in die)                       |
| BP           | Blood pressure                                 |
| BRG          | Biological Research Group                      |
| BSA          | Body surface area                              |
| CCI          | Comparison Chief Investigator                  |
| CF           | Consent Form                                   |
| CI           | Chief Investigator                             |
| CI           | Confidence interval                            |
| Co-CCI       | Comparison Co-Chief Investigator               |
| Cox-2        | Cyclooxygenase 2                               |
| CRF          | Case Report Form                               |
| CRN          | Clinical Research Network                      |
| CRUK         | Cancer Research UK                             |
| CRPC         | Castration Resistant Prostate Cancer           |
| CT           | Computerised tomography                        |
| CTA          | Clinical Trials Authorisation                  |
| CTAAC        | Clinical Trials Advisory and Awards Committee  |
| CTC          | Common Toxicity Criteria                       |
| CTU          | Clinical Trials Unit                           |
| CTV          | Clinical Tumour Volume                         |
| CXR          | Chest X-ray                                    |
| DAB          | Dual Androgen Blockade                         |
| DHT          | Dihydrotestosterone                            |
| DNA          | Deoxyribonucleic Acid                          |

| ABBREVIATION | EXPANSION                                                 |
|--------------|-----------------------------------------------------------|
| DPA          | Data Protection Act                                       |
| ES           | Efficacy Stage                                            |
| IB           | Investigator Brochure                                     |
| ICH          | International Conference on Harmonization                 |
| ECG          | Electro cardiogram                                        |
| FBC          | Full Blood Count                                          |
| FFS          | Failure-Free Survival                                     |
| FFPE         | Formalin Fixed Paraffin Embedded                          |
| GCP          | Good Clinical Practice                                    |
| GFR          | Glomerular Filtration Rate                                |
| GP           | General Practitioner                                      |
| HbA1c        | Glycated haemoglobin                                      |
| Hb           | Haemoglobin                                               |
| HE           | Health Economics                                          |
| HES          | Hospital Episode Statistics                               |
| Hr           | Hour                                                      |
| HR           | Hazard Ratio                                              |
| HSCIC        | Health & Social Care Information Centre                   |
| HT           | Hormone Therapy                                           |
| IDMC         | Independent Data Monitoring Committee                     |
| IM           | Intramuscular                                             |
| IMRT         | Intensity Modulated Radiation Therapy                     |
| IR           | Immediate-Release                                         |
| ISRCTN       | International Standard Randomised Controlled Trial Number |
| IU           | International Units                                       |
| IV           | Intravenous                                               |
| LFTs         | Liver Function Tests                                      |
| LHRH         | Luteinising Hormone Releasing Hormone                     |
| LREC         | Local Research Ethics Committee                           |
| m            | Month                                                     |
| mcg          | Microgram                                                 |
| MHRA         | Medicine and Healthcare Products Regulatory Agency        |
| min          | Minutes                                                   |
| MRC          | Medical Research Council                                  |

| ABBREVIATION | EXPANSION                                                                                 |
|--------------|-------------------------------------------------------------------------------------------|
| MREC         | Multi-Centre Research Ethics Committee                                                    |
| MRI          | Magnetic resonance imaging                                                                |
| M0           | Non-metastatic                                                                            |
| M1           | Metastatic                                                                                |
| NCI          | National Cancer Institute (USA)                                                           |
| NHS          | National Health Service                                                                   |
| N0           | Node-negative                                                                             |
| N+           | Node-positive                                                                             |
| NSAID        | Non-Steroidal Anti-inflammatory Drugs                                                     |
| OD           | Once per day (omne in die)                                                                |
| ONS          | Office for National Statistics                                                            |
| OS           | Overall Survival                                                                          |
| PFS          | Progression-free survival                                                                 |
| PI           | Principal Investigator                                                                    |
| PIS          | Patient Information Sheet                                                                 |
| po           | Orally (per orum)                                                                         |
| PSA          | Prostate Specific Antigen                                                                 |
| pts          | Patients                                                                                  |
| PTV          | Planned Tumour Volume                                                                     |
| QALY         | Quality-adjusted Life Years                                                               |
| qds          | Four times each day (quater die sumendus)                                                 |
| QL           | Quality-of-life                                                                           |
| R&D          | Research and Development                                                                  |
| RECIST       | Response Evaluation Criteria In Solid Tumours                                             |
| SAE          | Serious Adverse Event                                                                     |
| SAR          | Serious Adverse Reaction                                                                  |
| sc           | Under skin (sub-cutaneous)                                                                |
| SmPC         | Summary of Product Characteristics                                                        |
| SOC          | Standard-of-Care                                                                          |
| SR           | Sustained-Release                                                                         |
| SSA          | Site Specific Assessment                                                                  |
| STAMPEDE     | Systemic Therapy in Advancing and Metastatic Prostate Cancer: Evaluation of Drug Efficacy |
| SUSAR        | Suspected Unexpected Serious Adverse Reactions                                            |

| ABBREVIATION | EXPANSION                            |
|--------------|--------------------------------------|
| SWOG         | South West Oncology Group            |
| tE2          | Transdermal Oestradiol               |
| TMG          | Trial Management Group               |
| TMT          | Trial Management Team                |
| TURP         | Trans-Urethral Resection of Prostate |
| TSC          | Trial Steering Committee             |
| UCL          | University College London            |
| ULN          | Upper Limit of Normal                |
| U+E          | Urea and Electrolytes                |
| WHO          | World Health Organisation            |

| TERM                                      | DEFINITION                                                                                                                                                                                                                                                |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ADT                                       | Androgen deprivation therapy given in the form of LHRH agonists/antagonists (abbreviated to LHRH) or alternatively, transdermal oestradiol                                                                                                                |
| Hormone Therapy                           | Refers to all forms of hormone therapy given in the first line setting and includes LHRH, anti-androgens, transdermal oestradiol, GnRH Agonists and antagonists. This term does not include novel AR-targeted agents such as abiraterone or enzalutamide. |
| PSA nadir                                 | For trial purposes, this refers to the lowest PSA value detected between randomisation and week 24 on trial. This is used to derive the PSA progression value.                                                                                            |
| Protocol Research treatment               | Additional treatments patients allocated to research arms (B-K) receive as part of the STAMPEDE protocol e.g. metformin for patients allocated to arm K, or alternative in the case of transdermal oestradiol (arm L)                                     |
| Protocol standard-of-care (SOC) treatment | Standard forms of background treatment permitted as part of the STAMPEDE protocol which include licenced ADT (e.g. LHRH analogues) given in the setting of hormone-naïve prostate cancer and first-line use of docetaxel                                  |
| Non-protocol treatments                   | All prostate cancer treatments given following disease progression in the management of CRPC                                                                                                                                                              |
| Prednisolone                              | In Swiss sites this maybe referred to as prednisone.                                                                                                                                                                                                      |



## 1 LAY SUMMARY

Prostate cancers depend upon the male hormone testosterone for their growth. Lowering testosterone levels (either by removing all or part of both testes, or by giving anti- male hormone treatment) slows the growth of prostate cancers. This type of treatment is called hormone treatment or androgen deprivation therapy (ADT) and is often used when prostate cancers have spread outside the prostate gland. Although hormone treatment is usually successful at stopping the cancer growing for a period of time, the cancer will begin to grow again in most men. In addition, standard hormone treatment with injections (LHRH Analogues) can cause a range of side-effects which may become serious and affect quality-of-life, particularly since some men could remain on treatment for a decade or longer.

The overall aim of this trial, which is called STAMPEDE, is to assess novel approaches for the treatment of men with prostate cancer who are starting long-term ADT for the first time. Since opening to accrual in Oct-2005, the trial has tested many ways of treating prostate cancer and some results are now already known. More than 10,000 men will join the trial with answers becoming available throughout the trial. The trial will also look at the effects each treatment has on quality-of-life, and which treatment provides the greater value for money for the health service.

New patients joining the trial from Protocol version 16.0 onwards may be eligible to join one of two treatment comparisons, metformin (treatment group K; the “metformin comparison”) and transdermal oestradiol (treatment group L; the “transdermal oestradiol comparison”). A computer program will be used to allocate which treatment each participant receives, using a chance process.

**Table 2: Summary of treatment groups currently open to recruitment (Protocol version 17.0)**

| TREATMENT BEING TESTED | TREATMENT GROUP | SUMMARY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | FROM PROTOCOL VERSION |
|------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Metformin              | Arm K           | This anti-diabetic medication is proposed to have both anti-cancer effects and may help prevent the adverse metabolic effects of long-term ADT. STAMPEDE will investigate whether adding metformin to the current standard-of-care for non-diabetic men can improve all-cause survival.                                                                                                                                                                                                                                                                                                                                                                  | 15.0                  |
| Transdermal oestradiol | Arm L           | This is a form of hormone treatment which can suppress testosterone as effectively as standard ADT and has been shown to avoid some of the side-effects. For example, treatment with transdermal oestradiol does not appear to cause the bone to thin, a common problem with standard forms of ADT which might lead to the bones becoming fragile (osteoporosis) and more likely to break. It may also help to avoid some of the side effects and therefore improve overall quality of life compared with standard forms of ADT. STAMPEDE will investigate whether transdermal oestradiol can treat the cancer as well as current standard forms of ADT. | 16.0                  |

Further results are expected in the next few years from other treatments tested in STAMPEDE, which have completed recruitment. These include treatments currently used in different settings, including abiraterone and enzalutamide, both currently used when hormone treatment is no longer

effective and the cancer has started to grow again, termed castrate resistant prostate cancer (CRPC). Prostate radiotherapy, which is a treatment used in localised prostate cancer, has also been tested as an additional treatment for men with cancer that has spread to other parts of the body (metastatic prostate cancer). The results relating to these questions are expected in the next few years.

**Table 3: Summary of treatment groups closed to recruitment; results awaited**

| TREATMENT BEING TESTED                | TREATMENT GROUP | SUMMARY                                                                                                                                                                                                                                                                                                                                                                                                                           | FROM PROTOCOL VERSION |
|---------------------------------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Prostate radiotherapy                 | Arm H           | This is treatment with high-energy x-rays targeted to the prostate gland. This treatment is now mandatory within STAMPEDE for patients with cancer that is confined to the prostate gland as large trials have shown it improves life expectancy. We are not certain whether we should give radiotherapy to the prostate if the cancer has already spread and so we are investigating this in STAMPEDE.                           | 9.0                   |
| Enzalutamide (given with abiraterone) | Arm J           | Enzalutamide is another novel hormone treatment, similar to abiraterone, which is also used in advanced prostate cancer, when standard hormone therapy has stopped working. Enzalutamide works by blocking androgen receptors and this may complement abiraterone. STAMPEDE is testing whether this treatment combination is a more effective way of controlling prostate cancer growth for longer and improving life expectancy. | 12.0                  |

In the past STAMPEDE also tested whether adding docetaxel chemotherapy, zoledronic acid, celecoxib, alone or in combination, was beneficial in controlling prostate cancer growth and improving life expectancy. Recruitment has now been completed to all of these original treatment groups and the results have been presented. For further information relevant to these treatment groups, refer to the STAMPEDE website where you can see earlier versions of the protocol and find summaries of the results and links to the scientific publications.

**Table 4: Summary of treatment (groups) closed to recruitment; results reported**

| TREATMENT TESTED | TREATMENT GROUP | SUMMARY OF RATIONALE AND RESULTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | PROTOCOL VERSION ADDED |
|------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Zoledronic acid  | Arm B           | <p>Prostate cancer cells can spread to bones and weaken them. Zoledronic acid is a drug that reduces bone destruction and hardens bones cells.</p> <p>The results of STAMPEDE show that the addition of zoledronic acid does not prolong life expectancy. These results were compared with data from other similar trials that have tested this treatment, these data also support the findings of STAMPEDE.</p>                                                                                                                                                                                                                                                                                                                                      | 1.0                    |
| Docetaxel        | Arm C           | <p>Docetaxel is a type of chemotherapy which stops cells replicating. It has been used to treat advanced prostate cancer for some time, and is used in the treatment of lung, breast and ovarian cancer.</p> <p>The results of STAMPEDE show that the addition of docetaxel to hormone treatment does improve survival, most markedly in men with metastatic disease, and delays time to progression for men with locally-advanced and metastatic disease.</p> <p>The results of STAMPEDE were combined with other similar trials testing docetaxel and the results of the meta-analysis support this effect.</p> <p>Docetaxel may now be given as part of standard treatment to all suitable men entering STAMPEDE (from Protocol version 14.0).</p> | 1.0                    |
| Celecoxib        | Arm D           | <p>Celecoxib is an aspirin-like drug that is used to treat arthritis. It slows down the growth of cancer cells in the laboratory. STAMPEDE tested whether the addition of celecoxib could delay the growth of prostate cancer cells. Recruitment stopped early as a planned intermediate analysis failed to demonstrate sufficient effect of this drug. The final results were presented at GU ASCO 2016, a major international congress, and show that alone, celecoxib does not improve life expectancy.</p>                                                                                                                                                                                                                                        | 1.0                    |
| Abiraterone      | Arm G           | <p>This is a novel hormone treatment which works by inhibiting steroid hormone synthesis so blocks prostate cancer cells from generating their own male hormones. This is thought to be a major way in which prostate cancer cells resume growth following anti-hormonal therapies. The results of STAMPEDE have shown that the addition of abiraterone with prednisone improves survival and time to progression or relapse when used earlier, for men with locally-advanced or metastatic disease.</p>                                                                                                                                                                                                                                              | 8.0                    |

Note that the combination of docetaxel and zoledronic acid was assessed in Arm E and, whilst beneficial overall, did not provide additional benefit over docetaxel. The combination of celecoxib and zoledronic acid was assessed in Arm F. No benefit was seen overall, however an improvement in life-expectancy was observed in the group of patients who had metastatic disease at trial entry who received both celecoxib and zoledronic acid (4).

## 2 BACKGROUND

### 2.1 INTRODUCTION AND SETTING

Prostate cancer is a major health problem world-wide and accounts for nearly one fifth of all newly-diagnosed male cancers. In the UK, approximately 46,700 men were diagnosed with prostate cancer in 2014 and over 11,000 men died from the disease(5).

#### 2.1.1 Long-term Androgen Deprivation Therapy

The initial (first-line) treatment for locally-advanced or metastatic prostate cancer is based on androgen deprivation therapy (ADT) achieved either surgically with bilateral orchidectomy, or medically with LHRH agonists or antagonists (6). Long-term use of oral anti-androgens is permitted only when given with LHRH agonists, to achieve dual androgen blockade (previously termed maximum androgen blockade - MAB).

ADT produces responses in up to 95% of patients but it is not curative and disease recurs in virtually all patients treated with ADT as sole therapy, with a median time to progression of 18-24 months (6). Data from the control arm in STAMPEDE has shown that for men with newly-diagnosed metastatic disease, treated with ADT alone, the time to progression is just 11 months. Such progressive disease is referred to as castrate resistant prostate cancer (CRPC).

Another important issue with ADT is the numerous associated side-effects, particularly with prolonged use. Since patients continue on LHRH after disease progression (with additional agents added), many men remain on treatment for a decade or longer, particularly as life expectancy for men with prostate cancer should continue to improve as the number of effective treatments increases. The adverse effects of ADT using LHRH analogues include osteoporosis (leading to an increased risk of fracture), adverse metabolic effects, cognitive decline, sexual dysfunction, hot flushes, physical deterioration and fatigue.

#### 2.1.2 Role Of Radiotherapy For Men With M0 Disease

Two randomised trials, SPCG7 (7) and NCIC PR.3 / MRC PR07 (8-10) have tested the question of whether ADT alone combined with radiotherapy is the best treatment for patients with high-risk localised prostate cancer (NOM0). Both trials demonstrated an improvement in overall and disease specific survival from the addition of radiotherapy to ADT. The size of this overall survival benefit is substantial (hazard ratio 0.68 in SPCG7 and 0.77 in PR07). As these two mature, large, well-conducted randomised trials have demonstrated benefit, we now mandate that radiotherapy be standard for patients with NOM0 disease (i.e. no nodal or metastatic spread). Patients with node-negative M0 prostate cancer will only be allowed to enter the trial if standard radiotherapy is planned. Any patients with NOM0 disease for whom radiotherapy is contra-indicated should be discussed with the STAMPEDE team prior to inclusion. For patients with node-positive, M0 disease there are no randomised data on whether radiotherapy is indicated or not. However the NCIC PR.3 / MRC PR07 trial included patients with unknown nodal status who received whole pelvic radiotherapy (11) and demonstrated a large overall benefit. Additionally, non-randomised data from the STAMPEDE control arm suggests that the benefit observed in patients with NOM0 disease can be extended to those with pelvic nodal involvement. Therefore the STAMPEDE TMG recommends that pelvic nodal radiotherapy be considered for patients with node-positive, M0 disease at the discretion of the treating clinician (12).

### 2.1.3 Role Of Docetaxel For Men With M0 Or M1 Disease

The primary analysis of the "original comparisons" has shown docetaxel to significantly prolong survival (HR 0.78; 95% CI 0.66-0.93)(1). This is in support of the results of the CHAARTED trial which showed docetaxel improved survival in men with metastatic disease(13, 14). There was no evidence of heterogeneity in STAMPEDE in the treatment effect across any patient subgroups and median survival was improved by 10 months, from 71 to 81 months. In a well powered and pre-planned subgroup analysis of men with metastatic disease at randomisation the treatment effect was most apparent with a median survival benefit of 15 months. As a result the STAMPEDE TMG recommends that docetaxel should be strongly considered in all men with metastatic disease at presentation who are commencing ADT for the first time and are fit enough to receive chemotherapy.

Survival data for men without metastases at diagnosis is less mature but a statistically significant improvement in failure-free survival is seen, therefore, docetaxel may also be considered for men with high-risk non-metastatic disease who are commencing ADT for the first time and are fit enough to receive chemotherapy. Therefore, docetaxel is now permitted as part of the standard-of-care for all men entering STAMPEDE at the discretion of the treating clinician and patient.

## 2.2 RATIONALE

There are increasing numbers of treatments which are used post-relapse of first-line ADT in patients with CRPC, but there has been limited evidence as to which is associated with the best response, how they may be combined or sequenced or whether any of them might have a role as first-line treatment. An alternative approach is to investigate the addition of new drugs as part of first-line therapy in patients starting ADT. At this point, patients should be fitter and better able to tolerate treatment than when they have CRPC, and there is the possibility of having a larger and longer-lasting effect. This is the rationale for the evaluation of docetaxel, abiraterone, abiraterone and enzalutamide.

The increasing and widespread use of ADT in prostate cancer management has led to growing awareness of the adverse effects of LHRH. An alternative approach for improving long-term outcomes in patients is therefore to mitigate some of these side-effects. Many of these side-effects can affect quality-of-life as well as result in significant morbidities and potentially life-threatening consequences, particularly with prolonged treatment and in patients with existing co-morbidities.

For these reasons, metformin is being evaluated within the trial as a re-purposed treatment for prostate cancer, because of its potential anti-cancer effects (based on pre-clinical and epidemiological evidence) and the expectation that it may counteract the metabolic effects of long-term ADT. Similarly, transdermal oestradiol, another novel re-purposed treatment approach, is being evaluated as an alternative form of ADT which may be as effective or more effective than LHRH in treating prostate cancer but with fewer side-effects.

## 2.3 DESIGN

STAMPEDE (also known as MRC PR08) is an innovative, multi-arm multi-stage, multi-centre, randomised controlled trial. It initially assessed the effects of a bisphosphonate (zoledronic acid), a cytotoxic chemotherapeutic agent (docetaxel) and a cyclooxygenase (Cox-2) inhibitor (celecoxib), as single agents or combinations, in patients commencing long-term ADT for locally advancing or metastatic prostate cancer. For these questions, each comparison was divided into five stages such

that, for each investigational arm, safety and activity data were generated in the first four stages; an investigational arm could only proceed to the fifth and final stage of recruitment, where it would be assessed for effect on overall survival, if shown to be sufficiently safe and active at all prior activity stages. Patient data from all arms and all stages are, however, included in the final analyses of the primary outcome measure, even if the investigational arm did not proceed to the final stage. Of note, a second, pre-planned interim analysis failed to demonstrate sufficient activity for celecoxib and this agent was removed from trial recruitment in Apr-2011; patients remaining on celecoxib treatment reverted to standard care. Results for all of these “original comparisons” have now been reported(2, 15).

Since the start of the trial, a number of new research arms have been added to STAMPEDE over time to evaluate: abiraterone, a steroid synthesis inhibitor; prostate radiotherapy for patients with newly-diagnosed metastatic disease; enzalutamide, an inhibitor of androgen receptor signalling, given with abiraterone; and metformin, an anti-diabetic medication. In Protocol version 16.0, a new research arm is added for transdermal oestradiol, to be given as an alternative form of ADT.

## 2.4 PREVIOUSLY-REPORTED RESEARCH TREATMENTS

Data have been reported on zoledronic acid, docetaxel, celecoxib and the combination of zoledronic acid with docetaxel or with celecoxib. As such the rationale for these treatments, along with their design and details of treatment administration are no longer covered within the Protocol.

## 2.5 RATIONALE FOR RESEARCH TREATMENTS UNDER EVALUATION

### 2.5.1 Steroid Synthesis Inhibitors

**Note:** recruitment to both the abiraterone containing comparisons has now been completed as the required target accrual was reached.

Recent evidence suggests that an important mechanism for escape from tumour control by androgen ablation is the intracellular conversion of steroid precursors to androgenic steroids by prostate cancer cells. A key enzyme in this process is CYP17, which therefore represents a logical target for therapy in CRPC(16). Abiraterone acetate (3 $\beta$ -acetoxy-17-(3-pyridyl)androsta-5,16-diene, code CB7630; JNJ-212082) is rapidly converted in vivo to abiraterone (JNJ-589485; formerly code named CB7598). It is a selective, irreversible inhibitor of 17 $\alpha$ -hydroxylase/C17,20-lyase (cytochrome P450c17 [CYP17]), an enzyme that is critical in the production of androgens in the testes, adrenal glands and prostate tumour tissue. Inhibition of CYP17 inhibits the conversion of pregnenolone or progesterone into dehydroepiandrosterone (DHEA) or androstenedione, respectively, each of which is a precursor of testosterone. The pharmacodynamic effect is a more effective androgen depletion than can be induced by surgical castration, or medically by gonadotropin releasing (GnRH) hormone analogues used as first-line hormone therapy in prostate cancer.

Approximately 2,280 prostate cancer patients participated in the two Phase 3 RCTs (COU-AA-301 and COU-AA-302), with approximately 1,335 patients receiving abiraterone acetate at 1000mg daily dose continuously, in these studies. These studies have demonstrated abiraterone to prolong survival when given post-docetaxel (HR 0.65) and pre-docetaxel (HR 0.82). As a result it is now approved use in the USA and Europe in CRPC (17, 18).

Side-effects with abiraterone acetate are modest with the main adverse effects being elevated transaminases (usually mild), hypokalaemia and hypertension due to secondary hyperaldosteronism and fluid retention (preventable by low doses of glucocorticoids). In order to prevent secondary hyperaldosteronism, it is recommended that prednisolone (or prednisone) 10mg daily be administered in the CRPC setting. Within more recent studies in earlier stage patients, lower doses (typically 5mg of prednisone/prednisolone) are being used due to concerns about side effects of long-term exposure to glucocorticoid. Within the STAMPEDE trial, we suggest prednisolone/prednisone dose of 5mg OD, which may be increased to 5mg BID at the investigator's discretion if there are any concerns about monitoring or risks for the patient with 5mg OD.

We hypothesise that abiraterone may be more active still, when given up-front in combination with first-line ADT, by preventing or delaying the development of castrate refractory disease.

## 2.5.2 Radiotherapy To The Prostate For Patients With Newly-Diagnosed Metastatic Disease

**Note:** recruitment completed to the radiotherapy arm in Sep-2016 as the revised recruitment target sample was reached. Treatment has been completed in all patients and the results will be reported when the data has matured. See Protocol version 15.0 or older for details on the rationale.

(7, 11, 19, 20)

## 2.5.3 Combination Of Steroid Synthesis Inhibitors And Androgen Receptor Signalling Inhibitor

The most common form of disease progression for men on single-agent abiraterone or enzalutamide is a rise in PSA. This would suggest that the mechanism driving resistance is increased PSA transcription resulting from reactivation of the androgen receptor (AR), or another steroid signalling pathway(21).

The primary pharmacodynamic effect of enzalutamide is inhibition of androgen binding to the AR, AR nuclear translocation in the presence of androgen and AR:chromatin association. In multiple prostate cancer cell lines that specifically model CRPC (LNCaP/AR, VCaP, W741C LNCaP), the consequences of enzalutamide treatment include inhibition of AR-induced gene transcription, reduced cell proliferation, increased cell death by apoptosis and tumour regression.

In a mouse xenograft model of CRPC using prostate cancer cells that overexpress the AR (LNCaP/AR), enzalutamide inhibits tumour growth and reduces tumour size. A major human metabolite of enzalutamide, N-desmethyl enzalutamide, demonstrates key primary pharmacodynamics of similar potency to the parent molecule, while the carboxylic acid derivative metabolite has no known pharmacodynamic effect.

The question under investigation is: can progression be delayed (and survival extended) by using a combination of abiraterone and enzalutamide given up-front in combination with first-line ADT?

### 2.5.3.A Supplementing Abiraterone And Prednisolone With Enzalutamide

Several studies have shown that the AR can become promiscuously activated by very low levels of androgens or other steroid metabolites and drugs that bind the AR (22-25). It is known that very low levels of androgens can persist in patients treated with abiraterone acetate (26). Drugs that bind the AR, may include co-administered glucocorticoids. Furthermore, AR mutations of the sort previously described in CRPC, can be activated by cortisol and other glucocorticoids at levels much lower than those reported in patients treated with abiraterone and prednisolone at a dose of 5mg bid (25, 27). Moreover, abiraterone binds the AR and, although weak antagonism of wild-type and most



previously described AR mutations are observed (27), a similar mechanism to that described with classical anti-androgens, such as bicalutamide, could lead to change-of-function AR mutations associated with AR activation following abiraterone binding. Therefore, concomitant treatment with an androgen receptor signalling inhibitor could prevent “promiscuous” AR activation in patients treated with abiraterone. Enzalutamide is an androgen receptor signalling inhibitor and is approved for use on its own in the treatment of advanced CRPC (28), and there is evidence of activity for hormone-naïve prostate cancer (29).

#### **2.5.3.B Supplementing Enzalutamide With Abiraterone And Prednisolone**

Enzalutamide in combination with ADT is both effective and well tolerated in CRPC(28). However, recent studies have suggested that intra-tumoral testosterone levels increase in patients treated with enzalutamide (30). The implications of this finding are that the increase in intra-tumoral testosterone could be associated with up-regulation of enzymes involved in steroid biosynthesis(31). Although enzalutamide has a high affinity for the AR, this is several-fold lower than both the natural ligands testosterone and DHT (32), which means that enzalutamide would be out-competed at the AR ligand-binding domain if and when androgen levels rise. In vitro, a ten-fold rise in intra-cellular androgen was sufficient to prevent inhibition of AR by 30uM of enzalutamide(27); these levels are representative of the plasma levels of enzalutamide active metabolites, which can be achieved with enzalutamide 160mg po daily (33).

A strategy for preventing the rise in intra-cellular androgens in patients treated with enzalutamide would be inhibition of CYP17A1. Abiraterone is currently the only CYP17A1 inhibitor with proven efficacy. It therefore seems logical to use the combination of enzalutamide and abiraterone to both block a rise of intra-cellular androgens and prevent promiscuous activation of the AR.

#### **2.5.3.C Summary Of Rationale For This Combination**

To date, investigation has focussed on patients with CRPC but there is a strong rationale for the combination of enzalutamide and abiraterone in the hormone treatment-naïve setting in which STAMPEDE is focused.

STAMPEDE is already evaluating abiraterone plus conventional ADT but we will not assess the combination of conventional ADT plus enzalutamide; other trials by industry and other cooperative groups will address that question. The inclusion of an arm with ADT and enzalutamide in STAMPEDE was therefore considered to be a duplication of effort and was not supported by the Trial Management Group.

The combination of enzalutamide and abiraterone is a novel approach and offers considerable promise in delaying progression – it therefore represents an attractive addition to the comparisons under investigation in STAMPEDE, and one that is unlikely to be replicated in other planned trials of this size.

#### **2.5.4 Metformin**

All men joining STAMPEDE are planned for long-term ADT, a treatment associated with an increased risk of insulin resistance, hyperglycaemia, dyslipidaemia and obesity. Over 50% of men receiving long-term ADT will develop Metabolic Syndrome (34) resulting in increased cardiovascular morbidity and mortality. Obesity and high bind insulin C-peptide levels, indicating insulin resistance are independent predictors of increased prostate cancer-specific mortality and the presence of metabolic syndrome and diabetes in men treated with ADT is associated with shorter survival.



Metformin, which in non-diabetic individuals has been shown to lower the incidence of diabetes, counteracts some of these side-effects of ADT, including insulin insensitivity, hyperinsulinaemia and diabetes. It also reduces the levels of cholesterol, LDLs and triglycerides by inhibiting the fatty acid synthesis via activation of Adenosine Monophosphate Activated Kinase (AMPK) and decreases the platelet aggregation factor 1, platelet aggregation, vascular adhesion molecules, CRP and leptin (35-38). Through mitigation of the cardiovascular and metabolic consequences of ADT, metformin is proposed to reduce treatment-associated morbidity and improve all-cause mortality.

In addition, recent data has emerged consolidating the knowledge that cancer progression is linked integrally with metabolic modulators and that modification of this process by metformin has an important effect on cancer progression and survival. Pre-clinical data has shown that metformin is an important stimulator of AMPK which acts as the cellular “masterswitch” for energy regulation. AMPK acts to inhibit the effects of elevated insulin levels which promote metastasis, tumour growth and treatment resistance. Insulin increases mRNA and protein expression of steroidogenic enzymes leading to the up-regulation of intracellular testosterone levels, secreted androgens, thereby activating the AR (39). Metformin also influences the PI3K-AKT pathway and has an anti-proliferative effect via inhibitor of mTOR as well as targeting cancer stem cells. In vitro, metformin has been shown to inhibit androgen-induced IGF-IR up-regulation through disruption of androgen signalling (40).

Evidence in support of this includes a systematic review and meta-analysis of 13,008 men with type 2 diabetes mellitus (T2DM) and concurrent cancer which has shown improved survival in men treated with metformin compared with other anti-diabetic agents. In a systematic review of observational data from over 1 million men, there was a significant association seen between metformin and decreased risk of death from any cancer. Another systematic review found that the use of metformin in diabetic patients was associated with a significantly lower risk of cancer incidence and cancer mortality (41). In a large retrospective cohort study of 3837 diabetic men with prostate cancer, metformin was associated with a decreased risk of prostate cancer specific mortality (HR=0.76 [0.64-0.89]) and death (HR=0.76 [0.70-0.82]). In a prospective non-randomised phase II study in non-diabetic CRPC patients, 36% of patients receiving metformin were progression-free at 3 months and >50% had a prolongation of their PSA doubling time (42).

In summary, metformin is proposed to mitigate many of the adverse side effects of long-term ADT as well as having multiple potential anti-cancer effects and therefore STAMPEDE will evaluate re-purposing this treatment as a novel therapeutic approach in the management of high risk locally-advanced or metastatic prostate cancer.

## 2.5.5 Transdermal Oestradiol

### 2.5.5.A Background & Rationale

ADT with LHRH analogue injections suppresses testosterone to castrate levels, but also depletes oestradiol, since around 80% of oestradiol in men is derived by aromatisation from testosterone. Thus men who are treated with LHRH will have toxicities caused by low levels of both testosterone and oestrogen. The LHRH-associated toxicities which are due to low testosterone include loss of libido, erectile dysfunction and decrease in muscle mass. Other toxicities associated with LHRH such as osteoporosis, increased fracture risk, hot flushes, memory loss, dyslipidemia and increased body fat deposition are thought to be due to oestradiol deficiency. In particular, the adverse effect of LHRH on bone health has been well documented. Oestradiol deficiency prolongs the life-span of bone-resorptive osteoclasts, with the resulting imbalance between osteoclasts and bone-forming osteoblasts increasing the rate of bone thinning. This may lead to osteoporosis and increased risk of fracture, with the rate of fracture increasing with duration of LHRH.

Transdermal oestradiol is a potential alternative to LHRH that may avoid some treatment-related side-effects, therefore improving quality-of-life, which would be advantageous if shown to be equally effective at prolonging survival. Exogenous administration of oestradiol suppresses androgen production through a negative feedback loop involving the hypothalamic-pituitary axis, whilst avoiding the fall in oestradiol associated with castrate levels of testosterone(43). This, in turn, mitigates the toxicities of LHRH associated with oestradiol deficiency. Oral oestrogen was previously used for ADT before the development of LHRH, but discontinued as first-line treatment due to increased thromboembolic toxicity, attributable to first-pass hepatic metabolism (44).

Parenteral administration (e.g. intravenous, intramuscular or transdermal oestradiol) avoids first-pass hepatic metabolism, mitigating the cardiovascular risk, as supported by results so far from the ongoing PATCH (Prostate Adenocarcinoma TransCutaneous Hormones [MRC PR09; ISRCTN70406718]) trial and previous studies evaluating parenteral oestradiol in the form of intramuscular polyestradiol phosphate (43, 45).

To date, there are a number of encouraging results from the PATCH trial demonstrating the safety and early activity of transdermal oestradiol compared to LHRH agonists in men with advanced hormone-naïve prostate cancer (see [Appendix I](#) for further details). In particular, similar rates of cardiovascular events have been observed in the transdermal oestradiol and LHRH arms, as well as equivalent rates of testosterone suppression (based on around 900 patients enrolled up to Oct-2015) (43). Transdermal oestradiol has been shown to avoid the loss in bone mineral density associated with LHRH, and results in improved metabolic profiles and quality-of-life compared to LHRH(46). Furthermore, a pre-planned, confidential, interim analysis of the PATCH trial in Jun-2013 based on progression-free survival (PFS) led to the trial being extended to phase III; that analysis included n=638 participants with 206 PFS events, reviewed against a pre-specified non-inferiority margin hazard ratio of 1.25 and 1-sided alpha 0.25. The phase III evaluation of clinical efficacy for transdermal oestradiol will be based on progression-free and overall survival as co-primary outcome measures.

Demonstrating that transdermal oestradiol is an equally effective approach to ADT would provide a globally important alternative (to LHRH), with the potential to reduce treatment-associated morbidity and improve quality-of-life. In addition, there is a possibility that transdermal oestradiol may improve overall survival compared to standard hormone therapy. First, transdermal oestradiol may reduce treatment-associated morbidity and could potentially benefit overall survival. Second, up to 30% of men with castrate-resistant prostate cancer respond to oral oestrogen as post-relapse therapy, suggesting oestradiol may potentially have additional direct anti-tumour effects(47).

#### **2.5.5.B Meta-Analysis With PATCH Trial**

To further assess the clinical efficacy of transdermal oestradiol, the relevant data from the “transdermal oestradiol comparison” within STAMPEDE will be combined with data from patients recruited into PATCH i.e. the “transdermal oestradiol comparison” within STAMPEDE is not sufficiently powered to form a stand-alone analysis. The evaluation of transdermal oestradiol will be based on a non-inferiority approach (in contrast to the other comparisons within STAMPEDE which are superiority questions), to test the hypothesis that transdermal oestradiol is at least as effective as standard hormone therapy, but with fewer side-effects.

Recruitment of patients to the “transdermal oestradiol comparison” through STAMPEDE enables the transdermal oestradiol research question to be answered more quickly than via PATCH alone. It also reduces the number of patients allocated standard treatment alone in both trials, thereby increasing the proportion of patients receiving a novel treatment approach and improving trial efficiency.

As of Feb-2017, nearly 1,200 patients had been recruited directly to the PATCH trial (also coordinated by MRC CTU at UCL) for the phase III evaluation of clinical efficacy of transdermal oestradiol. The overall recruitment target for the transdermal oestradiol evaluation is approximately 2,000 patients (including initially around 500 to be recruited through STAMPEDE).

## 3 SELECTION OF INSTITUTIONS AND INVESTIGATORS

Centres who wish to participate in the STAMPEDE trial should be registered with the MRC CTU at UCL for this purpose. Before any patients are registered or randomised, the CTU must receive a completed and signed Investigator Statement. The STAMPEDE Investigator Statement is signed by the Principal Investigator for that institution (download from <http://www.stampededtrial.org/>). The return of the Investigator Statement will be taken as confirmation of agreement to adhere to the trial protocol. In addition, a fully-signed model agreement is also required before recruitment can begin.

In compliance with the principles of GCP, all institutions participating in the trial will complete a delegation log and forward this to the CTU. Each person working on the STAMPEDE trial must sign off a section of this log indicating their responsibilities. CTU must be notified of any changes to trial personnel and/or their responsibilities. An up-to-date copy of this log must be stored in the Investigator Site file at the institution and also at CTU.

The Clinical Trial Authorisation (CTA) for the STAMPEDE trial requires that the Medicines and Healthcare Products Regulatory Agency (MHRA) be supplied with the names and addresses of all participating investigators/institutions. Trial staff at the CTU will perform this task; hence, it is vital to receive full contact details for all investigators prior to their entering patients.

Finally, before a patient is entered into the trial, and any trial-related procedures are conducted, written informed consent must be obtained. Approved patient information sheets and informed consent forms are supplied as templates.

Only a limited number of centres participated in the initial Pilot Phase of the original trial; this was to ensure that safety and feasibility data were collected expediently. Subsequent stages of the trial are open to any centre that wishes to participate and has fulfilled the requirements described above. In addition for some comparisons, there will be additional criteria required prior to accreditation, see [Sections 3.1](#) and [3.2](#).

Following substantial amendments and future comparisons opening, sites will be notified of relevant documents and training required and if and when they are able to participate. Further accreditation packs may be circulated as a result to update trial documentation.

### 3.1 COMPARISON-SPECIFIC SITE ACCREDITATION

#### 3.1.1 “Transdermal Oestradiol Comparison”

Only UK centres participating in STAMPEDE will be accredited for the “transdermal oestradiol comparison”, since treatment with transdermal oestradiol is administered using Progynova TS 100mcg/24 hours transcutaneous oestradiol patches (see [Section 6.2.8](#)) which are currently unavailable in Switzerland.

### 3.2 FUTURE PLANNED BIOMARKER-SELECTED COMPARISONS

Initially, only UK centres participating in STAMPEDE will be accredited for the biomarker-selected comparisons, which are in development and will be incorporated in the next version of the protocol.

Sites wishing to gain accreditation for this future comparison should participate in the biomarker-screening pilot, described from Protocol version 16.0 onwards.

Full details of the accreditation procedure for the future biomarker-selected comparisons will be available in the next version of the protocol. Centres participating in the biomarker-screening pilot will need to complete a feasibility assessment and will be provided with additional training prior to activation of the pilot. An additional agreement signed by the Head of Histopathology Service with contact details of a designated secretarial or technical person at each site will be required from each site participating in biomarker testing.

Please refer to the [Biomarker-screening manual](#) for further details.

### 3.3 REQUIRED TRIAL DOCUMENTATION

**Table 5** presents a summary of the required trial documentation for participating centres. Templates are provided on the STAMPEDE website [www.stampedetrial.org](http://www.stampedetrial.org).

**Table 5: Trial documentation required for participating centres**

| TRIAL DOCUMENTATION                                                                     | TIMING                                             |
|-----------------------------------------------------------------------------------------|----------------------------------------------------|
| R&D approval (or local equivalent; including IRMER approval)                            | Before centre participation                        |
| Signed Investigator Statement                                                           | Before centre participation                        |
| Signature list & delegation of responsibilities                                         | Before centre participation                        |
| Trial personnel contact details                                                         | Before centre participation                        |
| PIS, GP & CF on local paper                                                             | Before centre participation                        |
| Signed Clinical Trial Agreement between Trust and Sponsor (or Variation, if applicable) | Before centre participation                        |
| Site initiation training                                                                | Before centre participation                        |
| Signed Pharmacy Pack acknowledgment                                                     | Before centre participation                        |
| Signed Pathology Agreement (for sites participating in biomarker-screening)             | Before centre participation in biomarker screening |

## 4 SELECTION OF PATIENTS

### 4.1 GENERAL INCLUSION CRITERIA

Participants must fulfil both of the criteria in [Section 4.1.1](#) or at least one criterion in [Section 4.1.2](#) or at least one criterion in [Section 4.1.3](#). Additionally, all patients must fulfil the criteria in [Section 4.1.4](#).

#### 4.1.1 High-Risk Newly-Diagnosed Non-Metastatic Node-Negative Disease

Both:

- At least two of: T category T3/4, PSA $\geq$ 40ng/ml or Gleason sum score 8-10
- Intention to treat with radical radiotherapy (unless there is a contra-indication; exemption can be sought in advance of consent, after discussion with CTU)

OR

#### 4.1.2 Newly-Diagnosed Metastatic Or Node-Positive Disease

At least one of:

- Stage T<sub>any</sub> N+ M0
- Stage T<sub>any</sub> N<sub>any</sub> M+

OR

#### 4.1.3 Previously Radically Treated, Now Relapsing (Prior Radical Surgery And/Or Radiotherapy)

At least one of:

- PSA  $\geq$ 4ng/ml and rising with doubling time less than 6 months
- PSA  $\geq$ 20ng/ml
- N+
- M+

AND

#### 4.1.4 For All Patients

- I. Histologically confirmed prostate adenocarcinoma
- II. Intention to treat with long-term androgen deprivation therapy
- III. Treating clinician and patient should have decided if docetaxel is to be part of the standard-of-care prior to randomisation
- IV. Fit for all protocol treatment<sup>1</sup> and follow-up, WHO performance status 0-2<sup>2</sup>
- V. Have completed the appropriate investigations prior to randomisation
- VI. Adequate haematological function: neutrophil count  $>1.5 \times 10^9/l$  and platelets  $>100 \times 10^9/l$
- VII. Adequate renal function, defined as GFR  $>30ml/min/1.73m^2$
- VIII. Serum potassium  $\geq 3.5mmol/L$
- IX. Written informed consent
- X. Willing and expected to comply with follow-up schedule
- XI. Using effective contraceptive method if applicable

<sup>1</sup> Medical contraindications to the trial medications are given in [Section 6](#)

<sup>2</sup> For WHO performance status definitions see [Appendix A](#)

## 4.2 GENERAL EXCLUSION CRITERIA

Patients must not fulfil any of the criteria, below. In addition, see [Sections 4.4.1](#) and [4.4.2](#) for comparison-specific criteria.

- I. Prior systemic therapy for locally-advanced or metastatic prostate cancer except as listed in [Section 4.1.3](#)
- II. Metastatic brain disease or leptomeningeal disease
- III. Abnormal liver functions consisting of any of the following:
  - Serum bilirubin  $\geq 1.5 \times$  ULN (except for patients with Gilbert's disease, for whom the upper limit of serum bilirubin is  $51.3 \mu\text{mol/l}$  or  $3 \text{mg/dl}$ )
  - Aspartate aminotransferase (AST) or alanine aminotransferase (ALT)  $\geq 2.5 \times$  ULN
- IV. Any other previous or current malignant disease which, in the judgement of the responsible clinician, is likely to interfere with STAMPEDE treatment or assessment
- V. Any surgery (e.g. TURP) performed within the past 4 weeks
- VI. Patients with significant cardiovascular disease, including:
  - Severe/unstable angina
  - Myocardial infarction less than 6 months prior to randomisation
  - Arterial thrombotic events less than 6 months prior to randomisation
  - Clinically significant cardiac failure requiring treatment (NYHA II-IV)<sup>3</sup>
  - Cerebrovascular disease (e.g. stroke or transient ischaemic episode) less than 6 months prior to randomisation
  - Patients with uncontrolled hypertension defined as systolic BP greater or equal than  $160 \text{mmHg}$  or diastolic BP greater or equal than  $95 \text{mmHg}$ <sup>4</sup>
  - Or any other significant cardiovascular disease that in the investigator's opinion means the patient is unfit for any of the study treatments.
- VII. Prior chemotherapy for prostate cancer (excluding patients receiving docetaxel as part of the new SOC)
- VIII. Prior exposure to long-term hormone therapy before randomisation (unless as described in [Section 4.5.4](#))
- IX. Prior exposure to systemic treatment for prostate cancer (excluding hormone therapy) e.g. abiraterone and enzalutamide.

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<sup>3</sup> NYHA classifications can be found in [Appendix A](#)

<sup>4</sup> Based on representative values, as judged by the investigator

## 4.3 BIOMARKER-SCREENING PILOT

### 4.3.1 Selection Criteria For Patient Registration

In preparation for the introduction of biomarker-selected comparisons, a Biomarker-Screening Pilot is being undertaken, described in Protocol version 16.0 and onwards. This will be undertaken in a limited number of sites. Participation in the pilot will help to facilitate site accreditation for recruitment to the future biomarker-selected comparisons see [Section 3.2](#).

All patients who fulfil the criteria below should be **registered without delay** in order to proceed to biomarker-screening.

- Registered at a site participating in biomarker-screening
  - Confirmed metastatic disease (M1)
  - Recent FFPE tumour sample available for prompt transfer (sample must have been obtained within 8 months prior to date of registration)
  - Written informed consent provided for the STAMPEDE trial and for biomarker-screening
  - If hormone therapy has started, FFPE tumour blocks should be **sent within**:
    - **8 weeks** of the patient starting LHRH
    - **10 weeks** of starting anti-androgens
- This is to allow for a turnaround time for biomarker analysis results of 4 weeks.

Once eligibility is confirmed complete the registration CRF and contact the CTU to proceed to trial registration, see [Section 5.1](#).

At the point of registration three samples are required to complete the biomarker-screening.

- Expedited retrieval of FFPE tumour block.
- Baseline blood sample collected using cell free DNA Streck™ tubes
- Saliva sample

Please refer to the [Biomarker-screening manual](#) for further details.

Registration will occur **before** the patient is randomised. In the pilot phase, results of biomarker-screening are **not** required prior to randomisation as recruitment has not yet been activated to the “rucaparib comparison”. Therefore patient can be randomised straight after being registered. For further details on results and feedback to patients, please see [Section 4.6.5](#).

Patients who participate in the pilot will continue to be allocated to any of the current open arms for which they are eligible (arm A, L and K). When recruitment is activated to biomarker-selected comparisons, the results of biomarker-screening will determine eligibility and will be required prior to randomisation. The screening pilot will inform the randomisation process for future biomarker-selected comparisons.

## 4.4 COMPARISON-SPECIFIC SELECTION CRITERIA

### 4.4.1 For Randomisation To Include The “Metformin Comparison”

Patients with known diabetes mellitus are not eligible for randomisation to the “metformin comparison”.

All non-diabetic patients require an HbA1c to be performed prior to randomisation (ideal timeline: within 8 weeks prior to randomisation), to confirm their non-diabetic status.



In addition, an assessment of renal function is required to determine glomerular filtration rate (GFR). The method used to determine glomerular filtration rate may vary according local practice. Equations that either estimate glomerular filtration rate (eGFR) or creatinine clearance (CrCl) may be used and the same threshold value applies.

In summary, additional inclusion criteria specifically for the "metformin comparison" are:

- HbA1c <48mmol/mol (equivalent to <6.5%)
- Adequate renal function, defined as  $GFR \geq 45 \text{ ml/min/1.73m}^2$  \*
- No history of lactic acidosis or pre-disposing conditions
- Not current or previous treatment with metformin
- No contra-indications to metformin

\*Except Switzerland, please refer to SAKK appendix for local guidance

Note that if the patient is known to be diabetic or the patient is found to have diabetes mellitus (i.e. HbA1c is 6.5% or higher) following screening, the patient is only eligible for randomisation if they meet all of the selection criteria for the "transdermal oestradiol comparison" (randomisation between Arms A and L only; see [Section 4.4.2](#)).

All patients with abnormal baseline HbA1c (i.e. 6.0% or higher) should be informed and referred to their GP for further management.

Where possible, the screening bloods, including HbA1c, should be performed prior to commencing SOC docetaxel. This is to reduce the likelihood of corticosteroid-related hyperglycaemia impacting on eligibility for the "metformin comparison".

#### **4.4.2 For Randomisation To Include The "Transdermal Oestradiol Comparison"**

Patients who have any of the following are not eligible for the "transdermal oestradiol comparison":

- >8 weeks of anti-androgen use
- >1 dose of monthly or 4 weekly LHRH agonist/antagonist
- Prior LHRH agonist injection with a stated duration of effect greater than 1 month
- >12 weeks since first dose of any hormone therapy
- Bilateral orchidectomy
- Cyproterone acetate started prior to randomisation
- Known porphyria
- Any history of deep vein thrombosis or pulmonary embolism confirmed radiologically
- Known thrombophilic disorder (e.g. Protein C, Protein S, antithrombin deficiency)

Note that patients unsuitable for the "transdermal oestradiol comparison" will only be eligible for randomisation if they meet all of the selection criteria for the "metformin comparison" and therefore may be allocated to control (arm A) or metformin (arm K) only (see [Section 4.4.1](#)).

Patients presenting with relapsed disease who fulfil criteria in [Section 4.1.3](#) are also eligible for the "transdermal oestradiol comparison" providing the neo-adjuvant or adjuvant hormone therapy previously received adheres to criteria outlined in [Section 4.5.4](#).

## 4.5 SCREENING PROCEDURES

**Table 6: Summary of initial required screening and baseline investigations**

| TIMEPOINT              | PATIENTS PARTICIPATING IN BIOMARKER-SCREENING PILOT                                                                                                                                                                                                                                        |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PRE-REGISTRATION       | <ul style="list-style-type: none"> <li>✓ Confirmation of metastatic disease</li> <li>✓ FBC, U&amp;Es, LFTs, Creatinine or estimated GFR</li> <li>✓ Retrieval of recent FFPE tumour block (within 8 months of registration)</li> </ul>                                                      |
| AT REGISTRATION        | <ul style="list-style-type: none"> <li>✓ Transfer of FFPE tumour block</li> <li>✓ Blood collection (cell free DNA Streck™ tubes)</li> <li>✓ Saliva sample</li> </ul>                                                                                                                       |
| TIMEPOINT              | ALL PATIENTS                                                                                                                                                                                                                                                                               |
| PRIOR TO RANDOMISATION | <ul style="list-style-type: none"> <li>✓ Bloods: FBC, U&amp;Es, LFTs, Creatinine, , PSA (including pre-treatment PSA within 6 months of randomisation)</li> <li>✓ Cardiac: ECG, BP</li> <li>✓ Imaging: Bone scan, CT or MRI pelvis and abdomen, CXR if chest not included in CT</li> </ul> |

**Key:** FBC= Full blood count, LFT= liver function test, U&E= Urea and electrolytes, GFR=glomerular filtration rate, BP= blood pressure

### 4.5.1 Investigation Prior to Registration (for patients participating in biomarker-screening pilot)

Sufficient screening investigations must have been completed to ensure that patient fulfil all of the selection criteria for the Biomarker-Screening Pilot, see [Section 4.3.1](#) prior to registration. In addition, confirmation of adequate organ function should be obtained through baseline bloods which at a minimum include FBC, U&Es and LFTs. Metastatic disease should be confirmed radiologically but all the required screening scans may not have been completed. All remaining screening investigations e.g. additional blood and imaging, may be completed following registration prior to randomisation (See [Table 1](#) and [Table 6](#)).

Please refer to the [Biomarker-screening manual](#) for further details relating to sample requirements.

### 4.5.2 Investigations Prior To Randomisation

All patients should have the following examinations performed to confirm eligibility prior to randomisation.

The following standard imaging is required and the latest available scans that reflect the patient's current disease status should be used:

- CT or MRI of pelvis and abdomen
- Bone Scan (or equivalent e.g. whole body MRI, choline-PET-CT, PSMA-CT-PET)
- Chest X-ray (only if chest was not included in CT or MRI which would be preferable)

Any additional imaging such as CT-PET scanning can be performed according to local practice but, for the purposes of the trial, the recorded stage should be the CT stage only; additional information on the CT-PET stage will also be collected.

The following bloods and additional measurements are required prior to randomisation:

- ECG

- Pre-hormone treatment PSA ( this must be obtained within 8 months of randomisation)
- HbA1c
- Full blood count
- Urea and Electrolytes
- Liver function tests
- Serum creatinine
- Systolic and diastolic blood pressure

Patients who initially fail to meet the trial eligibility criteria can be re-screened at a later date. Of note, for patients receiving standard-of-care docetaxel at the point of screening, it is acceptable to use a full blood count measurement prior to chemotherapy to confirm eligibility.

Prior to randomisation:

- Check details of any prior treatments for prostate cancer
- Check any contraindications to radiotherapy or research treatment
- Check concomitant medications

#### 4.5.3 Additional Baseline Investigations

The following blood tests are required at baseline (within 4 weeks before or after randomisation):

- Testosterone (pre-ADT, if available)
- Serum corrected calcium
- Phosphate
- Magnesium
- Albumin
- Fasting glucose (mandatory)
- Fasting triglycerides (mandatory)
- Lipid profile (fasting or non-fasting; total cholesterol, LDL and HDL) (mandatory)

The following additional procedures are required and **mandatory** at baseline:

- Waist circumference measurement
- Weight and height

A trial screening log will be *available* to all centres; copies are *not required* at this time (unless a specific issue is identified at a given site by the STAMPEDE Trial Team).

#### 4.5.4 Androgen Deprivation Therapy Prior To Randomisation

From Protocol version 16.0, patients can potentially be randomised to the “transdermal oestradiol comparison” and it would be preferable for these patients to have had as little exposure to ADT as possible.

Within the separate PATCH trial, patients are randomised within 8 weeks after starting anti-androgens and cannot have received an LHRH injection. This approach is also favoured in STAMPEDE, but patients who have received a single 4-week (or 1-month) LHRH injection remain eligible for randomisation to the “transdermal oestradiol comparison”, as shown in [Figure 3](#).

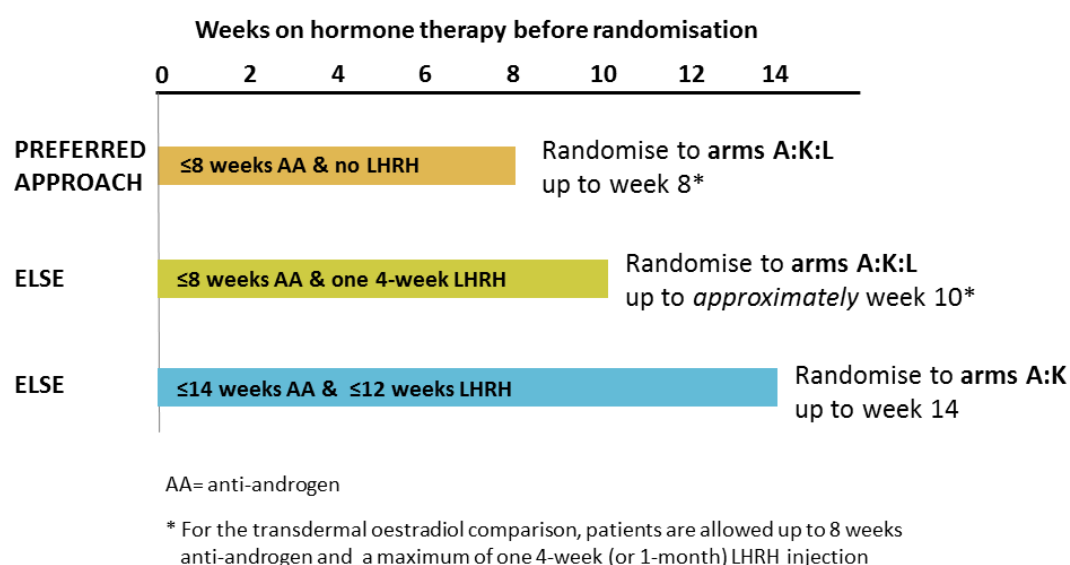
For all other comparisons, if ADT has already started prior to randomisation, the first LHRH injection must have been given within 12 weeks prior to randomisation. Additionally, if anti-androgens are being used, these must have started within 14 weeks prior to randomisation. A PSA level must have been taken prior to starting long-term hormone treatment. Note that baseline testosterone

measurements will not be required in patients who have already commenced hormone manipulation prior to randomisation.

**Figure 3** illustrates the maximum duration of ADT allowed pre-randomisation.

Note that anti-androgen monotherapy is not permitted as a form of long-term hormone therapy. It is accepted that sites participating in the Biomarker-Screening Pilot are unlikely to meet the “preferred timeline for randomisation”.

**Figure 3: Maximum duration of hormone therapy allowed prior to randomisation and the latest point of randomisation**



**Permitted prior hormone therapy for now-relapsing disease:** Any patients now presenting with relapsed disease, previously treated with adjuvant or neo-adjuvant hormone therapy alongside their radical surgery or radiotherapy, must have completed that period of hormone therapy **at least 12 months** before joining STAMPEDE and it must have been **no longer than 12 months in duration**.

#### 4.5.5 Standard-Of-Care (SOC) Radiotherapy

The treating clinician and patient must have decided, prior to randomisation, whether radiotherapy is to be given as part of standard-of-care (SOC).

#### 4.5.6 Standard-Of-Care (SOC) Docetaxel

The treating clinician and patient must have decided, prior to randomisation, whether docetaxel is to be given as part of standard-of-care (SOC). SOC docetaxel treatment should start within 14 weeks after starting hormone therapy (ideal timeline: within 8-10 weeks). Patients can have already started docetaxel treatment when randomised providing this is within 14 weeks after starting hormone therapy.

For those planned for SOC docetaxel and subsequently randomised to receive transdermal oestradiol (Arm L), it is recommended that docetaxel treatment commences *after* patients have

been established on transdermal oestradiol for around 4 weeks, when most patients are likely to have completed the induction period (see [Section 6.2.8](#)).

#### **4.5.7 Starting Trial Treatment**

##### **4.5.7.A Metformin**

For all patients allocated to metformin, treatment should start as soon as possible after randomisation. Investigators should aim that this is at least within 4 weeks post-randomisation and within 12 weeks of starting hormone therapy (see [Section 6.2.7](#)). Metformin can be given in combination with SOC docetaxel.

##### **4.5.7.B Transdermal Oestradiol**

For all patients allocated to transdermal oestradiol, treatment should start as soon as possible after randomisation (and ideally within 1 week after randomisation). It is not necessary to wait for completion of the 4-week (or 1-month) duration of the LHRH injection, if this was previously given prior to randomisation. For those prescribed bicalutamide or flutamide prior to randomisation, this treatment should be discontinued before treatment with transdermal oestradiol can commence (no washout period is needed).

#### **4.5.8 Concomitant Medications**

From Protocol version 15.0 onwards the trial requires the reporting to CTU of information regarding planned or actual long-term (>6 months) use of the following concomitant medications of classes of interest.

- Statins
- Metformin
- Aspirin
- Bisphosphonates or denosumab
- Opiate pain killers
- ACE inhibitors or angiotension II antagonists

This information is of interest both in terms of baseline use and ongoing use through the trial; as such it should be recorded on the Randomisation CRF and will be collected at each follow-up assessment (see [Table 1](#)).

Caution should be exercised when starting any concomitant medications that may result in a worsening of renal function e.g. initiating anti-hypertensive therapies such as ACE inhibitors, diuretics such as frusemide, or starting a non-steroidal anti-inflammatory drug (NSAID). Please refer to [Table 18](#) for more information on drugs which may require additional monitoring of renal function.

All concomitant medications should be continued throughout the trial unless the responsible clinician decides otherwise. If patients continue to require medication for the management of docetaxel-related toxicities, please discuss this with the trial team. See [Section 6.3](#) for more information on concomitant medications and their use with abiraterone, enzalutamide and transdermal oestradiol.

## 4.6 ADDITIONAL DETAILS FOR PATIENTS JOINING SUB-STUDIES

All patients joining STAMPEDE are asked for additional, optional consent to provide samples for three ongoing sub-studies. Since Mar-2016, all consent for additional research is recorded on a separate form to consent for the trial.

For details regarding sample collection, please refer to the [Sample collection and handling manual](#).

For details relating to samples obtained for biomarker-screening please refer to the [Biomarker-screening manual](#).

### 4.6.1 Germline DNA Analysis (Saliva Samples)

In collaboration with Professor Ros Eeles, Institute of Cancer Research, London, DNA is being extracted from saliva samples provided by consenting participants enrolled in STAMPEDE. The aims of this sub-study are to examine the germline (inherited) genetic changes present in men with high-risk localised or metastatic prostate cancer. The aim is to determine the prevalence of germline genetic aberrations present pre-diagnosis and to correlate prostate cancer risk SNP genetic profiles, identified in Genome-wide Association Studies (GWAS) and other sequence variants from next generation sequencing (NGS), with duration of response to ADT and the experimental treatments tested in STAMPEDE.

All patients joining the trial are asked to consent to provide a saliva sample from which germline DNA can be extracted. This has been the case since Protocol version 15.0 and replaces the blood spot collection method used in previous versions of the protocol. Preliminary data has shown saliva to be a feasible method to collect sufficient DNA to conduct analyses of germline (inherited) genetic changes.

All patients who consent to part A (donation of saliva) on the Additional Research Consent Form version 1.0 onwards are eligible for this sub-study. Saliva samples should be provided after randomisation and all consenting patients from all arms can participate. See the [Sample collection and handling manual](#) for further details.

Patients who previously joined the trial prior to Protocol version 13.0 and who consented to provide a blood spot (Consent Form version 4.0 part K) can also be retrospectively approached to provide a saliva sample. These should be collected from patients randomised to the trial from **Nov-2011 onwards** (when recruitment to the abiraterone comparison was activated) who have received the REC approved letter explaining the need for additional saliva sample collection as the DNA extraction using the blood spot method did not work as well as anticipated.

### 4.6.2 Circulating Tumour-DNA Analysis (Sequential Blood Samples)

This sub-study is being conducted in collaboration with Dr Gerhardt Attard, Institute of Cancer Research, London. The aims of this analysis include to identify molecular subgroups with differential treatment effects and, through sequential sampling, identify molecular changes associated with disease progression to explore resistance mechanisms and early detection of treatment failure.

From Protocol version 14.0 (activated from Jan-2014 onwards), sequential blood samples were collected from patients within the “enzalutamide and abiraterone” comparison, i.e. allocated to arm A or J between 29-Jul-2014 and 31-Mar-2016. From Protocol version 16 onwards, **all patients** joining the trial will be asked to donate sequential blood samples from which genetic material shed by the tumour cells can be extracted, enabling tumour DNA analysis.

The sampling schedule is different for M0 and M1 patients and is detailed in the [Sample collection and handling manual](#). Sequential samples are required in order to detect genetic changes within tumours over time. The most important sampling timepoint is at progression, as it is hoped this can inform the potential mechanisms of treatment resistance. From Protocol version 16.0 the sampling schedule has been updated and now includes a baseline sample, obtained as soon as possible after consent is provided. The aim of this additional sampling point is to explore the genetic changes initially present i.e. at the point at which treatment is first started.

From Protocol version 16.0 onwards (activated Sept-2016), all patients allocated to all arms should be asked if they are willing to provide additional consent to participate in this sub-study which is recorded on part B of the Additional Research Consent Form.

#### 4.6.3 Tissue Sample Analysis (FFPE Blocks)

As the clinical outcome data matures for several of the treatments comparisons evaluated within STAMPEDE, correlative analysis of the archival formalin-fixed paraffin-embedded (FFPE) tumour tissue will be undertaken, aiming to identify predictive and prognostic biomarkers. Targeted next-generation sequencing (tNGS) of FFPE tumour samples from selected, consenting STAMPEDE patients will be performed in order to explore the prevalence of genomic aberrations and examine the predictive and prognostic effect of molecular sub-groups. FFPE blocks are currently being collected at selected STAMPEDE sites to support different projects.

All patients joining the trial have been asked to consent for the use of remaining tissue samples e.g. those obtained at prostate biopsy or following surgery, for use in additional research. These samples are usually stored as FFPE tissue blocks at the hospital where the procedure was performed. Randomising sites will be asked to assist in the retrieval of tissue samples stored in pathology stores or referring hospitals when these are required for additional translational sub-studies.

All patients who consent to part C on the additional research consent form are eligible for ongoing sub-studies involving FFPE tumour block analysis. For patients who previously joined the trial prior to 2016, consent for use of remaining samples was provided on the main consent form. Research teams at randomising sites will be required to provide an anonymised copy of the consent form when requesting samples from pathology departments and facilitate the transfer of samples to the trial designated laboratories. Further details on the sample processing and transfer can be found in the [Sample collection and handling manual](#).

#### 4.6.4 Biomarker-Screening Pilot

From Protocol version 16.0 onwards (activated Sept-2016), a Biomarker-Screening Pilot will be activated in selected sites. Patients participating in biomarker-screening will be required to register prior to randomisation, see [Section 4.3](#) for details on the eligibility criteria for registration and see [Section 5](#) for details on registration and randomisation.

At the point of registration three samples are required to complete the biomarker-screening.

- Expedited retrieval of FFPE tumour block
- Baseline blood sample collected using cell free DNA Streck™ tubes
- Saliva sample

Please refer to the [Biomarker-screening manual](#) for further details

The aim of this pilot is to assess the feasibility of rapid pre-randomisation biomarker-screening which will be required for planned future biomarker-selected comparisons. The first biomarker-selected comparison has successfully received independent peer-review through CRUK and is in the late stages of development.

When recruitment is activated to biomarker-selected comparisons, the results of biomarker-screening will be required prior to randomisation and will determine eligibility. The screening pilot will activate before randomisation to biomarker-selected comparisons and inform this process. Therefore patients who participate in the pilot will continue to be allocated to any of the current open arms for which they are eligible (arm A, L and K).

#### **4.6.5 Informed Consent For Genetic Screening**

For patients joining the trial from Protocol version 16.0 onwards (activated Sept-2016), the consent process has been updated. The trial participants are asked to provide explicit informed consent if they wish to receive feedback of any results that arise from research analyses of genetic material extracted from any of the biological samples collected as part of the trial e.g. saliva, FFPE tumour blocks or circulating tumour DNA extracted from blood.

Only results which are of established clinical relevance and for which testing would be available under standard NHS genetic testing guidelines will be fed back e.g. pathogenic BRCA1/2 mutations. Any genetic analysis undertaken as part of additional research associated with STAMPEDE does not replace clinically indicated investigations as only a proportion of STAMPEDE patient will undergo prospective testing and therefore it cannot be guaranteed that results will be fed back in a timely fashion.

STAMPEDE investigators are recommended to refer all patients in whom a clinically relevant genetic result is detected during research analyses to a clinical geneticist. This is to facilitate access to genetic counselling and the required confirmatory testing, this is also necessary in order to offer appropriate advice to biological relatives in the event of detecting a germline (inherited) genetic abnormality. The list of clinically relevant gene mutations to be fed back will be based on current clinical guidelines. The STAMPEDE Biological Research Group will review this periodically to ensure it remains current and oversee this process.

In the information provided to STAMPEDE patients who joined the trial prior to Protocol version 16.0, it was stated that any subsequent genetic results would not be linked to them or their families and therefore results will not be provided in this instance. Going forward, the changes incorporated in Protocol version 16.0 will enable patients who may undergo analyses on genetic material extracted from FFPE tissue, saliva or circulating tumour DNA (extracted from blood) to opt to receive clinically relevant information.



## 5 REGISTRATION AND RANDOMISATION

### 5.1 TRIAL ENROLMENT: DEFINITIONS AND PROCESS

#### 5.1.1 Registration

Currently, registration is only required for patients participating in the Biomarker-Screening Pilot. This is initially being activated in a proportion of STAMPEDE sites.

For sites participating in the Biomarker-Screening Pilot, all patients who are eligible to participate in the Biomarker-Screening Pilot must be registered prior to randomisation. Once registered to participate in the pilot, patients can currently be randomised without waiting for the results of the biomarker-screening providing all the required screening information is known.

See [Section 4.3.1](#) for details on selection criteria, please confirm all criteria are met and complete the Registration CRF prior to contacting the STAMPEDE trial team at CTU.

All participants in the Biomarker-Screening Pilot will be allocated a registration number which relates specifically to the biomarker-screening process. The registration number will be used to identify the patient until the point of randomisation when this will be linked and replaced by the trial number.

Once recruitment is activated to biomarker-selected comparisons, biomarker-screening will be implemented. Registration will be then required for all patients participating in biomarker-screening and patients will only be eligible for randomisation once the results of biomarker-screening are known.

#### 5.1.2 Randomisation

All other patients **not** participating in the Biomarker-Screening Pilot may proceed immediately to randomisation. Eligibility will be confirmed during the randomisation process and patients will be allocated to any of the open research arms for which they are suitable (see [Section 4.4](#) ).

To enter a patient into STAMPEDE (either to register or randomise), the relevant forms should be completed carefully, and CTU contacted by phone:

#### **REGISTRATION & RANDOMISATIONS**

Call MRC CTU at UCL, Monday to Friday 0900-1700  
Excluding public holidays or dates when notice has been given by CTU.  
Tel: +44 (0) 20 7670 4777

A trial number and treatment will be allocated and given over the phone or by return fax. In addition, a letter confirming these details will be sent. The trial number will be the primary way in which the patient will be identified and should be used in all correspondence. Centres should send a letter to the patient's GP to inform them of their trial participation and treatment allocation. The GP letter is supplied as a template and can be downloaded from the trial website [www.stampedetrial.org](http://www.stampedetrial.org).

## 5.2 CO-ENROLMENT GUIDELINES

Ideally, patients should not be participating in any other clinical trial of prostate cancer treatment when they enter STAMPEDE and should not enter any other trials until a failure-free survival (FFS) event has been experienced and reported. After this point, the patient may be entered into further, second-line treatment studies. The primary outcome measure of STAMPEDE is overall survival and follow up reports must continue after co-enrolment. Participation in post-progression studies must be reported to CTU on the Co-enrolment CRF; details of any interventional treatments received for progression in such studies should be reported on the Additional Treatment Log.

Data sharing agreements with “down-stream” trials are encouraged to improve data quality in both trials and to reduce costs to both organisations.

## 6 TREATMENT OF PATIENTS

### 6.1 STANDARD-OF-CARE (SOC)

The standard-of-care for this patient group is **androgen deprivation therapy (ADT)** as per local practice (see [Section 6.1.1](#)). For some patient groups, this should now be supplemented with standard radiotherapy (see [Section 6.1.2](#)). From Protocol version 14.0 onwards the standard-of-care includes permitted use of docetaxel for all suitable patients (see [Section 6.1.3](#)).

In summary, SOC treatment is defined as being **one** of the following combinations:

- ADT alone
- ADT + Radiotherapy
- ADT + Docetaxel
- ADT + Radiotherapy + Docetaxel

#### 6.1.1 Hormone Therapy

Patients will be randomised either to the control arm (Arm A) or to one of the actively recruiting research arms for which the patient is eligible.

With the exception of those allocated to transdermal oestradiol (Arm L), all patients will receive ADT as per local practice to achieve castrate levels of testosterone. Please see [Section 4.5.4](#) for more information on ADT timing before randomisation.

Patients allocated to Arm L will go on to receive transdermal oestradiol in place of standard ADT methods.

The method of planned or current long-term standard-of-care ADT must be specified for each patient prior to randomisation.

The permitted methods of ADT are:

##### 6.1.1.A Bilateral Orchiectomy

Operations should be performed by appropriately trained surgeons. A total or sub-capsular orchiectomy may be performed. Patients having a bilateral orchiectomy are required to adhere to the same timelines as specified in [Section 4.5.4](#) unless there is a strong clinical reason not to do so.

##### 6.1.1.B LHRH Agonists

LHRH agonists used according to local practice. The prophylactic use of anti-androgens to prevent tumour “flare” is recommended.

##### 6.1.1.C LHRH Antagonists

LHRH antagonists used according to local practice. The use of prophylactic use of anti-androgens to prevent tumour “flare” is not necessary.

##### 6.1.1.D Dual Androgen Blockade

Long-term use of anti-androgens alongside LHRH agonists, according to local practice. Note this was previously referred to as maximum androgen blockade.

#### 6.1.1.E Others

Other methods of ADT should be discussed with the STAMPEDE trial team. The planned duration of ADT should be **at least 2 years**.

### 6.1.2 Standard-Of-Care (M0) RT

#### 6.1.2.A NOM0 Patients

Investigators should give standard radiotherapy (RT) to patients with node negative, non-metastatic disease (NOM0), in accordance with data from the PR07 and SPCG trials. If there is an intention to omit radiotherapy (e.g. RT is contraindicated for the patient) in patients with NOM0 disease this must be discussed with the STAMPEDE trial team before randomisation to confirm eligibility. See [Section 6.7](#) for further details of radiotherapy administration.

#### 6.1.2.B N+M0 Patients

The benefit of radiotherapy in this group is at present uncertain with no firm data to either support or refute its use. However, the PR07 trial included some node-positive patients as cross sectional imaging was not a part of the baseline assessment in this trial, which did include whole pelvis radiotherapy (11). For patients with node-positive, non-metastatic disease, radiotherapy is therefore recommended in suitable cases (12).

#### 6.1.2.C Planned use of SOC RT

Investigators will be asked to state their intention with regards to planned radiotherapy in this group at randomisation. Intention to give radiotherapy (or not) for **all** patients must be stated at randomisation to ensure that there is no bias towards particular combinations of systemic therapy with radiotherapy.

Standard-of-care radiotherapy is not a core part of the trial, therefore we intend to collect minimal data about the radiotherapy administered. It is accepted that some patients will develop progressive disease before radiotherapy can be administered and if this occurs the reasons for non-delivery of treatment must be recorded on the Radiotherapy Detail CRF.

Suitability for radiotherapy is assessed by the treating clinicians. Any patient who has had a previous, definite diagnosis of inflammatory bowel disease is at increased risk of disease re-activation following radiotherapy, and the risks of this must be balanced against the potential benefits of radiotherapy on an individual basis.

### 6.1.3 Standard-Of-Care Docetaxel

Investigators are strongly encouraged to consider giving docetaxel as part of the standard-of-care for patients with newly-diagnosed metastatic disease, based on the survival benefit demonstrated by both STAMPEDE in the primary analysis of the "original comparisons" and CHARTED (13) (14, 15). Investigators may also consider giving docetaxel to patients with high-risk locally-advanced disease, given both the significant improvement in failure-free-survival and consistency of effect for prostate cancer-specific survival shown by STAMPEDE.

The treating clinician and patient must have decided, prior to randomisation, if docetaxel is to be given. Chemotherapy treatment may have started when the patient is randomised. For patients allocated to receive transdermal oestradiol (Arm L) who have not already started docetaxel prior to randomisation, it is recommended that docetaxel commences around 4 weeks after starting research treatment (see [Section 6.2.8](#)). As with standard radiotherapy, minimum data collection will be required, however the start and end dates of docetaxel treatment are needed to ensure the appropriate timelines are met (see [Section 4.5.6](#)). A SOC Docetaxel Treatment CRF should be

completed for all patients confirming whether docetaxel was given or not, regardless of being planned.

Docetaxel is given according to local protocols as a standard non-trial treatment. The regime used previously within STAMPEDE was 75mg/m<sup>2</sup> Day 1 as 1hr IV infusion, plus prednisolone 5mg BID for 21 days repeated every 3 weeks for a maximum of 6 cycles. The STAMPEDE TMG would suggest prednisolone could be omitted and data on the use of co-prescribed steroid will be collected on the SOC Docetaxel Treatment CRF (please see [Table 19](#) for more details).

## 6.2 RESEARCH TREATMENTS

### 6.2.1 Required Timelines When Starting Research Treatment

Allocated treatment should start promptly after randomisation. Please refer to [Section 4.5.7](#) for more information on starting of research treatment.

### 6.2.2 Research Abiraterone + Prednisolone (relevant to arms G & J)

**Note:** recruitment has now closed to all research comparisons containing abiraterone; that is Arm G (SOC + abiraterone) and Arm J (SOC + enzalutamide + abiraterone).

Please note that for some patients treatment with abiraterone may continue until all categories of disease progression or up to a maximum duration of 2 years.

Arm G (SOC + abiraterone) patients who have now reached their maximum duration of 2 years on trial treatment include:

- All NOM0 patients
- All N+M0 patients receiving radical radiotherapy

Arm J (SOC + enzalutamide + abiraterone) patients who have now reached their maximum duration of 2 years on trial treatment include:

- NOM0 patients starting treatment over 2 years ago
- N+M0 patients receiving radical radiotherapy and starting treatment over 2 years ago

All such patients should have reported permanent stopping of research abiraterone on an End of Research Treatment CRF.

Please see sections below for more information.

**Abiraterone** will be administered as a single 1000mg daily oral dose (4 tablets to be taken together once a day) together with prednisolone or prednisone 5mg daily to prevent secondary mineralocorticoid excess. Abiraterone absorption is increased by food. The tablets should be taken at least 2 hours after food, swallowed whole with some water. No food should be eaten for 1 hour afterwards.

**Prednisolone** (prednisone in Switzerland) should be taken as a single dose with food in the morning.

Trial treatment must stop if other systemic treatments are initiated at any time for disease progression control (including chemotherapy, radium-223 etc).

Anti-androgens (e.g.. bicalutamide) should not be given in combination with abiraterone due to the risk of toxicity; as such patients on, or planned for dual androgen blockade (DAB), at randomisation should not continue with their anti-androgen use if allocated to receive abiraterone, additionally anti-androgens started whilst on abiraterone treatment should trigger abiraterone to be stopped. In patients with **M1 disease**, treatment with abiraterone will continue from randomisation until all categories of disease progression have occurred, consistent with the COU-AA-301 and COU-AA-302 trials (48, 49) i.e., abiraterone would be given for these patients until a composite of:

- PSA progression (as defined in [Section 7.1.3.A](#) )
- Radiological progression (appearance of new lesions or progression of existing lesions) **and**
- Clinical progression (defined as new cancer-related symptoms)

It is accepted that these flexible criteria for stopping treatment with abiraterone are open to the investigator's interpretation and discretion. Patients might continue treatment beyond the first failure-free survival (FFS) event; the first FFS event must be reported as per the other arms; all categories of disease progression (PSA, radiological and clinical) need to be reported once.

In patients **with NOM0 disease or N+M0 disease undergoing radical radiotherapy**, treatment would continue until the earliest of 2 years or all categories of disease progression as defined for M1 patients. ADT can be discontinued in this group at 2 years at the discretion of the local investigator (see [Section 6.1.1](#)).

For patients with **N+M0 disease not planned for radical radiotherapy**, or who do not receive planned prostate RT, treatment will continue as for patients with M1 disease until all categories of disease progression.

If a patient allocated to receive abiraterone develops only biochemical failure, the responsible clinician might switch from abiraterone + prednisolone 5mg od to abiraterone and dexamethasone 0.5mg od.

See [Section 7.1.3](#) for further information on the trial definition of progression.

### 6.2.3 Abiraterone + Prednisolone: Administration And Management Of Toxicities

**Abiraterone** absorption is increased by food therefore should be taken on an empty stomach without food. The tablets should be taken at least 2 hours after food, swallowed whole with some water. No food should be eaten for 1 hour afterwards.

**Prednisolone** (prednisone in Switzerland) should be taken as a single dose with food in the morning.

#### 6.2.3.A Abiraterone Contraindications

- Unusual or allergic reaction to past abiraterone acetate treatment
- Uncontrolled hypertension
- Uncontrolled heart failure
- Abnormal liver function or active or chronic liver disease

See [Table 16](#) for details on drugs that may interact with abiraterone.

### 6.2.3.B Abiraterone Special Warnings And Required Monitoring Whilst on Treatment

#### :: Hypokalaemia

Abiraterone may cause hypokalaemia due to secondary mineralocorticoid excess, this can be counteracted by co-prescription of prednisolone. Regular monitoring of serum potassium levels are required whilst receiving treatment with abiraterone. The Investigator Brochure states that monitoring should be performed 2-weekly for the first 12 weeks and then every month or as per protocol whilst receiving abiraterone (50).

When abiraterone is used routinely in the licenced setting (CRPC), it is common practice to prescribe the next course of treatment for 8 weeks to patients who have been on abiraterone for over 12 months with no abnormalities, having checked that the potassium is normal (or >3mmol/L and in line with previous results) prior to writing the prescription.

The STAMPEDE protocol requires continued monthly monitoring for patients who experience hypokalaemia related to research abiraterone. For patients who have been monitored appropriately with no evidence of hypokalaemia, the frequency of monitoring may be reviewed after 12 months on treatment and, at the discretion of the investigator, may be reduced to every 2 months if judged appropriate. This is consistent with the approach adopted in the LATITUDE trial in which abiraterone was evaluated in high-risk metastatic hormone-naïve prostate cancer (51). Treatment should always be interrupted presence of symptoms (constipation, palpitations, fatigue, muscle weakness or spasm, tingling or numbness), see [Table 8](#).

#### :: Hepatic Impairment

Abiraterone treatment can be associated with increased liver enzymes and hepatotoxicity therefore regular monitoring of liver function tests (LFTs) is required whilst on treatment. LFTs (ALT or AST and bilirubin). The Investigator Brochure states that monitoring should be performed 2-weekly for the first 12 weeks and then every month or as per protocol (50) whilst receiving abiraterone.

When abiraterone is used routinely, it is common practice to prescribe the next course of treatment for 8 weeks to patients who have been on abiraterone for over 12 months with no abnormalities, having checked that the liver function tests are normal, or no worse than grade 1 prior to writing the prescription (51). This is acceptable provided those with grade 1 abnormalities are monitored more frequently and treatment is interrupted if they increase to grade 2, see [Table 9](#). The STAMPEDE protocol requires monthly monitoring in the first 12-months on treatment with research abiraterone. For patients who have been monitored appropriately with no evidence of liver function abnormality, the frequency of monitoring may be reviewed after 12 months on treatment and, at the discretion of the investigator, reduced to every 2 months if judged appropriate, consistent with the approach adopted in the LATITUDE trial (51).

If clinical symptoms or signs suggestive of hepatotoxicity develop, serum transaminases, in particular serum alanine aminotransferase (ALT), should be measured immediately. See [Table 9](#) for the management of abiraterone induced hepatotoxicity.

#### :: Blood pressure management

Abiraterone may cause hypertension. Regular monitoring of blood pressure is required whilst receiving treatment. The Investigator Brochure states that monitoring should be performed 2-weekly for the first 12 weeks and then every month whilst receiving abiraterone. STAMPEDE Investigators will be required to ensure monthly blood pressure monitoring is performed and reviewed in the first 12-months on treatment, it is acceptable for this to be documented self-monitoring or via the GP providing this is reviewed at each follow-up. After 12 months on treatment, it is acceptable for blood pressure monitoring to be performed every 2-months and reviewed at each

follow-up visit, providing blood pressure has been well controlled. For the management of abiraterone induced hypertension see [Table 7](#).

#### **:: Cardiovascular history**

Abiraterone acetate should be used with caution in patients with a history of cardiovascular disease. The safety of abiraterone acetate in patients with left ventricular ejection fraction <50% or New York Heart Association (NYHA) Class III or IV heart failure has not been established. Before treatment with abiraterone acetate, hypertension must be controlled and hypokalaemia must be corrected.

Caution is required in treating patients whose underlying medical conditions might be compromised by increases in blood pressure, hypokalaemia, or fluid retention, e.g. those with heart failure, recent myocardial infarction, or ventricular arrhythmia.

#### **:: Renal Impairment**

No dose adjustments are required in renal impairment; however caution is advised if patients develop severe renal impairment as there is limited clinical data in this population. Systemic exposure to abiraterone after a single oral 1000mg dose did not increase in patients with end-stage renal disease on dialysis.

#### **6.2.3.C Abiraterone Undesirable Effects**

The most common adverse drug reactions observed in the integrated safety data for those patients who received 1000mg abiraterone acetate plus prednisone or prednisolone in clinical studies (n=1,070) were fatigue, arthralgia, peripheral oedema, back pain, bone pain, nausea, constipation, hypokalemia and anaemia.

The adverse events graded as 3 or 4 and which occurred in more than 5% of patients were fatigue, peripheral oedema, anaemia and back pain see [Appendix C](#).

#### **6.2.3.D Abiraterone Overdose**

Human experience of overdose with abiraterone is limited. There is no specific antidote to abiraterone acetate. In the event of an overdose, administration of abiraterone acetate should be stopped and general supportive measures undertaken, including monitoring for cardiac arrhythmias, liver function and electrolytes.

#### **6.2.3.E Management Of Specific Toxicities From Prednisolone**

The co-administration of prednisolone/prednisone 5mg once daily is required whilst receiving abiraterone to prevent secondary mineralocorticoid excess and 5 mg once daily is used in this trial.

Prednisolone/prednisone dose increase of up to 5mg BID is recommended to manage mineralocorticoid-related toxicities (e.g., hypokalaemia, hypertension, peripheral oedema) see [Table 7](#), [Table 8](#) and [Table 10](#).

Patients experiencing serious symptoms of Cushing's syndrome (e.g., weight gain, muscle loss) can decrease or discontinue (temporarily or permanently) steroids at the investigator's discretion but should be closely monitored for symptoms of secondary mineralocorticoid excess. It should be noted that weight gain and muscle loss are also associated with ADT.



**Table 7: Management of hypertension associated with abiraterone (given alone or in combination with enzalutamide)**

| TOXICITY EVENT | ACTION                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Grade 1-2      | Management as per investigator with anti-hypertensive treatment and increase frequency of blood pressure monitoring to at least weekly. Follow local guidance for selection of anti-hypertensives but avoid thiazide diuretics to minimise risk of serum potassium derangement. Calcium channel antagonists or beta blockers are often preferred.<br>As with other symptoms of mineralocorticoid excess, consider increasing prednisolone dose to 5mg BID. |
| Grade 3-4      | Withhold abiraterone and enzalutamide. Adjust or add anti-hypertensive medications to mitigate the toxicity. When hypertension resolves to Grade $\leq 1$ , resume both enzalutamide and abiraterone at full dose with prednisolone 5mg bid.                                                                                                                                                                                                               |

A cardiologist's opinion should be considered if blood pressure control is not achieved within 4 weeks.

**Table 8: Management of hypokalaemia associated with abiraterone (given alone or in combination with enzalutamide)**

| TOXICITY EVENT                                                                 | ACTION                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Grade 1<br>(LLN- 3.0mmol/L)                                                    | Supplement with oral potassium and monitor closely and increase prednisolone dose to 5mg BID.<br>Exclude and manage other causes of hypokalemia.                                                                                                                                                                                                                                  |
| Grade 2<br>( $<LLN - 3.0mmol/L$ and symptomatic)                               | <b>Pause abiraterone.</b><br>Supplement with oral potassium and monitor closely and increase prednisolone dose to 5mg BID.<br>Exclude and manage other causes of hypokalemia.<br>Re-start abiraterone with close monitoring, discontinue if recurs.                                                                                                                               |
| Grade 3<br>( $<3.0-2.5mmol/L$ )<br>or Grade 4 ( $<2.5mM$ )<br>life-threatening | Abiraterone will be <b>permanently discontinued</b> and the patients will be hospitalized for intravenous potassium replacement and cardiac monitoring. After the return of serum potassium to normal, prednisolone will be discontinued. The patient can continue on enzalutamide alone. If hypokalaemia persists consider a dose reduction of enzalutamide to 120mg once a day. |

**Table 9: Management of Abnormal Liver Function Tests (LFTs) associated with abiraterone (given alone or in combination with enzalutamide)**

| TOXICITY EVENT                                                                                                                                          | ACTION                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Grade 1</b> increases in AST, ALT or bilirubin (e.g. increase in AST or ALT from ULN to 2.5X ULN; increase in total bilirubin from ULN to 1.5X ULN)  | The frequency of LFT monitoring should be increased to at least weekly, if the investigator judges that the laboratory abnormalities are potentially related to study medication.<br>No dose reduction is required.<br>Providing LFTs are stable for 4 weeks, resume monthly checks                                                                                                                                                                     |
| <b>Grade 2</b> increases in AST, ALT or bilirubin (e.g. increase in AST or ALT to >2.5-5X ULN; increase in total bilirubin from >1.5-3X ULN)            | Withhold abiraterone, enzalutamide and all other concomitant medications that are potentially hepatotoxic.<br>The frequency of LFT monitoring should be increased to at least weekly until the liver function tests return to baseline value or grade 1 when all trial medication can be re-started.<br>No dose reduction is required after one episode providing this resolved within 4 weeks but should be considered if Grade 2 derangements recurs. |
| <b>Grade 3</b> increases in AST, ALT or bilirubin (e.g. increase in AST or ALT to >5X ULN; increase in total bilirubin to >3X ULN),                     | Withhold abiraterone and enzalutamide and all other concomitant medications that are potentially hepatotoxic.<br>At least weekly monitoring is required until the LFTs return to baseline value or Grade 1.<br>Enzalutamide can be re-started with no dose reduction. See below for abiraterone re-challenge.                                                                                                                                           |
| <b>Grade 4</b> increases in AST, ALT or bilirubin (e.g. increase in AST or ALT to >20x ULN; increase in total bilirubin to >10x ULN)                    | Patients must discontinue abiraterone and enzalutamide immediately.<br>At least weekly monitoring is required until the LFTs return to baseline value or grade 1 and then prednisone can be discontinued and the investigator can consider restarting enzalutamide.<br>Abiraterone should not be re-introduced.                                                                                                                                         |
| RE-CHALLENGE                                                                                                                                            | ACTION                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Recurrent grade 2 derangement                                                                                                                           | Reduce to 750mg once LFTs return to grade 1                                                                                                                                                                                                                                                                                                                                                                                                             |
| If study treatment resumption is considered for patients who have experienced Grade 3 increases in AST, ALT, or bilirubin                               | Resume study treatment with abiraterone dose reduction to 750mg when grade 3 toxicities resolve to grade 1 or baseline.                                                                                                                                                                                                                                                                                                                                 |
| If Grade 3 or higher increases in AST, ALT or bilirubin recur after the first dose reduction                                                            | Hold study medication and all other concomitant medications that are potentially hepatotoxic. At least weekly LFT monitoring is required, starting immediately regardless of study schedule and continued until a return to baseline values or Grade 1.                                                                                                                                                                                                 |
| If study treatment resumption is considered for patients who have experienced Grade 3 increases in AST, ALT, or bilirubin with the first dose reduction | Resume study treatment with abiraterone dose reduction to 500mg when AST, ALT or bilirubin returns to baseline value or grade 1.                                                                                                                                                                                                                                                                                                                        |

An opinion from a hepatologist should be considered if there are any concerns or liver function derangement shows no improvement within 2 weeks of discontinuation of abiraterone.

**Table 10: Management of fluid retention/oedema associated with abiraterone (given alone or in combination with enzalutamide)**

| TOXICITY EVENT | ACTION                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Grade 1-2      | Increase prednisolone dose to 5mg bid.                                                                                                                                                                                                                                                                                                                                                                                   |
| Grade 3-4      | Withhold abiraterone.<br>Consider addition of mineralocorticoid receptor antagonist eplerenone until resolution of symptoms. Enzalutamide can be continued. When fluid retention/oedema returns to baseline or resolves to ≤Grade 1, resume abiraterone at full dose with prednisone 5mg bid, if symptoms do not resolve abiraterone should not be re-started and enzalutamide should be dose reduced to 120 mg per day. |

**Table 11: Management of diarrhoea (associated with abiraterone or enzalutamide)**

| TOXICITY EVENT | ACTION                                                                                                                                                        |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Grade 1-2      | Symptomatic management.                                                                                                                                       |
| Grade 3-4      | Withhold abiraterone.<br>If no improvement reduce dose of enzalutamide to 120 mg per day. Once resolved to Grade 1, recommence abiraterone at 750 mg per day. |

## 6.2.4 Research Enzalutamide + Abiraterone + Prednisolone (Arm J)

Note: recruitment has now closed to Arm J (SOC + enzalutamide + abiraterone).

Please note that for some patients treatment with enzalutamide + abiraterone may continue until all categories of disease progression or up to a maximum duration of 2 years.

Arm J (SOC + enzalutamide + abiraterone) patients who have now reached their maximum duration of 2 years on trial treatment include:

- NOM0 patients starting treatment over 2 years ago
- N+M0 patients receiving radical radiotherapy and starting treatment over 2 years ago

Please see sections below for more information.

**Abiraterone** as described in [Section 6.2.2](#).

**Prednisolone/Prednisone** as described in [Section 6.2.2](#).

**Enzalutamide** will be administered as a 160mg oral dose (four capsules), taken together at the same time every day, with or without food.

Trial treatment must stop if other systemic treatments are initiated at any time for disease progression control (including chemotherapy, radium-223 etc).

Anti-androgens (i.e. bicalutamide) should not be given in combination with enzalutamide (as with abiraterone) due to the risk of toxicity; as such patients on, or planned for MAB, at randomisation should not continue with their anti-androgen use if allocated to receive enzalutamide + abiraterone, additionally anti-androgens started whilst on enzalutamide (+abiraterone) treatment should trigger

enzalutamide (+abiraterone) to be stopped. See [Table 16](#) and [Table 17](#) for further details on drugs that may interact with abiraterone and enzalutamide respectively.

In patients with **M1 disease**, treatment with both abiraterone and enzalutamide will continue until all categories of progression have occurred, consistent with the approach taken for abiraterone (see [Section 6.2.2](#)) i.e. abiraterone and enzalutamide will be given until a composite of:

- PSA progression (as defined in [Section 7.1.3.A](#))
- Radiological progression (appearance of new lesions or progression of existing lesions) **and**
- Clinical progression (defined as new cancer-related symptoms).

It is accepted that these flexible criteria for stopping treatment with abiraterone and enzalutamide are open to the investigator's interpretation and discretion. Patients may continue treatment beyond the first failure-free survival (FFS) event; the first FFS event must be reported as per the other arms.

In patients with **NOM0 disease or N+M0 disease undergoing radical radiotherapy**, treatment would continue for 2 years or all categories of disease progression as defined for M1 patients, whichever is the sooner. ADT can be discontinued in this group at 2 years at the discretion of the local investigator (see [Section 6.1.1](#)).

For patients with **N+M0 disease not planned for radical radiotherapy**, or who do not receive planned prostate RT, treatment will continue as for patients with M1 disease until all categories of disease progression.

If a patient develops PSA progression only whilst on abiraterone and enzalutamide, the local investigator might consider switching from from abiraterone + prednisolone 5mg od to abiraterone and dexamethasone 0.5mg OD.

See [Section 7.1.3](#) for further information on the definition of progression.

## **6.2.5 Enzalutamide: Administration, Dose Modification And Management Of Toxicities**

**Enzalutamide** can be taken with or without food.

### **6.2.5.A Enzalutamide Contraindications**

The full induction potential of enzalutamide may not occur until approximately 1 month after the start of treatment, when steady-state plasma concentrations of enzalutamide are reached, although some induction effects may be apparent earlier. Monitoring should continue for at least the first month of treatment and dose adjustments considered. Given the long half-life of enzalutamide (5.8 days), effects on enzymes may persist for one month or longer after stopping enzalutamide. A gradual dose reduction of the concomitant medicinal product may be necessary when stopping enzalutamide treatment. See [Table 17](#) for further details on specific drug interactions with enzalutamide.

### 6.2.5.B Enzalutamide Special Warnings And Precautions For Use

#### :: History of seizures

Caution should be used in administering enzalutamide to patients with a history of seizures or other predisposing factors including, but not limited to, underlying brain injury, stroke, primary brain tumours or brain metastases or alcoholism. In addition, the risk of seizure may be increased in patients receiving concomitant medications that may lower the seizure threshold. Enzalutamide should be **permanently discontinued** in patients who have a seizure while on treatment.

#### :: Hepatic impairment

A hepatic impairment study showed that the composite AUC of enzalutamide plus N-desmethyl enzalutamide after administration of a single dose of enzalutamide was similar in patients with baseline mild, moderate or severe hepatic impairment (Child-Pugh Class A, B or C respectively) relative to patients with normal hepatic function, and no starting dose adjustment is needed.

#### :: Renal impairment

No dose adjustment are required in renal impairment; however caution is advised if patients develop severe renal impairment as there is limited clinical data in this population.

### 6.2.5.C Enzalutamide Overdose

There is no antidote for enzalutamide. In the setting of an overdose, stop treatment with enzalutamide and initiate general supportive measures taking into consideration the  $t^{1/2}$  of 5.8 days. Patients may be at increased risk of seizures following an overdose.

### 6.2.5.D Management Of Specific Toxicities Due To Abiraterone And Enzalutamide

The safety monitoring and toxicity management plan described below takes into account AEs based on the reported clinical safety data of abiraterone and enzalutamide given separately. There are limited reported data on the safety and toxicity of the combination of enzalutamide and abiraterone however the recommendations summarised here have been updated in light of the experience gained in STAMPEDE as recommended by the STAMPEDE TMG.

#### :: Seizures

If any patient suffers a seizure whilst on treatment, enzalutamide should be permanently discontinued immediately. Abiraterone and prednisolone can be continued providing there are no abiraterone-specific toxicities.

**Table 12: Management of arthralgia & muscle pain (associated with enzalutamide)**

| TOXICITY EVENT | ACTION                                     |
|----------------|--------------------------------------------|
| Grade 1-2      | Symptomatic management                     |
| Grade 3-4      | Reduce dose of enzalutamide to 120 mg /day |

**Table 13: Management of fatigue (associated to enzalutamide)**

| TOXICITY EVENT | ACTION                                                                                                                                                                                     |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Grade 1-2      | Consider a dose reduction to 120 mg/day                                                                                                                                                    |
| Grade 3-4      | Pause enzalutamide for 1 week or until the toxicity grade improves to grade 2 or lower severity. Re-started at a reduced dose (120mg/day or 80mg/day) in consultation with the study team. |

### 6.2.6 Research Metformin (Arm K)

**Metformin** will be given as a daily dose in addition to standard-of-care treatment. The dose is 850mg OD. If tolerated, this should be increased to the target dose of 850mg BID after 4-6 weeks i.e. at the first follow-up visit.

In the case of **M0 patients**, if ADT is stopped after a minimum of 2 years, metformin should continue for a minimum of 3 years following randomisation and for a further 12 months after the administration of the last LHRH (whichever is longer). This is to allow for the delay in testosterone levels returning to normal following stopping ADT. If ADT is not stopped, then metformin should continue as it the case for M1 patients. In the event that ADT is stopped and then re-started for relapsed disease, if ADT is restarted whilst patients remain on metformin (i.e. within 12 months of the last administration of LHRH) then metformin should continue whilst on ADT. If metformin is stopped 12 months after the last administration of LHRH it should not be re-started in the event of relapse.

For **M1 patients** metformin should continue whilst on ADT. Treatment should continue post-progression providing it is judged to be in the patients best interest. Metformin can be given together with any additional treatments started for progression, excluding other IMPs i.e. investigators may choose to stop metformin treatment post-progression in order to enable patients to participate in another clinical trial evaluating treatments for CRPC.

In the event of stopping research treatment, unless a patient states otherwise, consent is assumed for continued recording trial data.

### 6.2.7 Metformin: Administration, Dose Modifications And Management Of Toxicities

The starting dose for metformin is 850mg once daily. If tolerated this should be increased to the target dose of 850mg twice daily after 4-6 weeks. Metformin should be taken around the same time each day and treatment tolerance is best if taken with or after food. For twice daily dosing, the minimum time between doses should be 8 hours, doses should not be taken closer together if forgotten or missed. If metformin is well tolerated and it is desirable to make a dose modification outside of the trial follow-up schedule, it is acceptable to conduct a telephone consultation. If metformin 850mg OD is not well tolerated, consider switching to 750mg SR OD or alternatively, reduce to 500mg OD.

## 6.2.7.A Metformin Special Warnings And Required Monitoring Whilst on Treatment

### :: Renal impairment

Metformin is not nephrotoxic, but is exclusively excreted by the kidneys. Therefore treatment should only be started in participants with stable renal function. From protocol version 16.0 the renal threshold has been revised in light of updated FDA guidance and published prescribing recommendations. Metformin should be only started when the  $GFR \geq 45 \text{ ml/min/1.73m}^2$ . Renal function should be monitored **at least every 6 months** in participants with stable renal function, whilst on metformin. Additional monitoring is required in any patient at risk of deteriorating renal function (see [Table 12](#)). In line with published prescribing recommendations, if the GFR falls to between 30-45  $\text{ml/min/1.73m}^2$  a **dose reduction** is required to ensure the maximum 24hr dose is 1000mg or less and monitoring of renal function is required at least 3 monthly (52). Metformin should be **permanently stopped** if the GFR falls to  $\leq 30 \text{ ml/min/1.73m}^2$ .

See [Table 14](#) for situations when metformin treatment should be paused due to the risk of deterioration in renal function.

**Table 14: Situations when metformin treatment should be paused**

| SITUATIONS                                                             | RISK FACTOR                                                                                                                                                                               |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Iodinated contrast agents                                              | If the $GFR < 60 \text{ ml/min/1.73m}^2$ metformin should be paused for 24 hours prior to receiving contrast and re-started 48 hours post-administration.                                 |
| Anaesthesia (peridural; spinal or general)                             | Pause metformin 48 hours prior to procedure and re-start no earlier than 48 hours following procedure, providing oral intake re-established and renal function is stable and at baseline. |
| Surgery                                                                | Pause metformin 48 hours prior to procedure and re-start no earlier than 48 hours following procedure, providing oral intake re-established and renal function is stable and at baseline. |
| Dehydration<br>e.g. nausea, vomiting or diarrhoea                      | Pause metformin and re-start only when oral intake is re-established and renal function is stable and at baseline.                                                                        |
| Obstructive uropathy<br>e.g. urinary retention or ureteric obstruction | Pause metformin and re-start only when renal function confirmed to be stable and at baseline.                                                                                             |

### :: Treatment breaks

It is anticipated that metformin treatment will be paused for approximately 72 hours around the time of contrast enhanced CT scans (see [Table 14](#)) and may need to be paused during episodes of inter-current illness. If metformin is paused for 6 days or less this information does not need to be recorded and no additional action is needed. Treatment pauses of 7 days or more should be recorded by updating the Metformin Treatment Log.

If metformin treatment is paused for more than 2 weeks, investigators may consider re-starting at 850mg once daily for the first 4 weeks before escalating to full dose providing tolerance is acceptable. It is suggested that, providing patients have a sufficient supply of labelled IMP metformin tablets, a telephone consultation may be sufficient to assess tolerance and advice regarding dose modification in order to limit hospital visits.

If treatment is paused for more than 3 months or >50% of doses are missed for any reason the trial team should be informed as metformin may need to be discontinued.

### 6.2.7.B Management Of Specific Toxicities From Metformin

#### :: Gastrointestinal disturbance

Gastrointestinal disturbances are very common with metformin and include nausea, vomiting, diarrhoea, abdominal pain and loss of appetite. These are most common when first starting treatment (occur in >1/10 individuals).

If toxicities occur we recommend a dose reduction and/or a switch to a sustained release (SR) preparation if available (see [Table 15](#)).

**Table 15: Management of metformin related gastrointestinal toxicity**

| TOXICITY EVENT    | ACTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Grade 1           | <ul style="list-style-type: none"> <li>Ensure metformin is taken with or after food.</li> <li>Consider switching to 750 mg BID SR preparation if available.</li> </ul> <p>Or, if unavailable consider:</p> <ul style="list-style-type: none"> <li>1 week treatment pause, re-start at reduced dose 850mg once daily. Attempt an escalation after 1 month but if necessary, remain at 850mg OD.</li> </ul> <p>And, if unable to tolerate 850mg OD or sustained release preparations are not available</p> <ul style="list-style-type: none"> <li>Consider dose reduction to 500mg OD (SR or IR if not available)</li> </ul> |
| Grade 2 or higher | <p>Reduce to 500mg OD SR (or IR if not available); re-attempt dose escalation after minimum of 1 week if symptoms improve, aiming to continue at the maximum tolerated dose</p> <p>If grade 2 toxicity persists:</p> <ul style="list-style-type: none"> <li>Pause treatment for 2 weeks, and re-start at 850mg sustained release or if not available 500mg OD.</li> <li>And re-attempt a dose escalation 2 months later.</li> <li>And continue at the maximum tolerated dose providing symptoms ≤ grade 1.</li> </ul>                                                                                                      |

If toxicities occur whilst on the initial starting dose of 850mg OD then the following dose modifications should be made:

- Switch to 750mg SR OD
- Alternatively, reduce to 500mg OD

If toxicities persist, consider a 1 week treatment pause, before re-starting at one of the dose-modified regimes and attempt an dose escalation after a minimum of 1 month.

Other possible metformin related toxicities included taste disturbance, skin reactions and B12 deficiency resulting in megaloblastic anaemia (see [Appendix C, Table 7](#)). Any patient who experiences anaemia whilst taking metformin should have haematinics including vitamin B12 measured and replaced if deficient.

#### :: Lactic acidosis

This is a very rare (3/100,000 patient years), but serious metabolic consequence. Reported cases have occurred primarily in diabetic patients with significant renal impairment who are also dehydrated. It is unclear whether this is due to the underlying diabetes or metformin. This is supported by a meta-analysis demonstrating similar rates of lactic acidosis in people with diabetes



taking metformin compared with diabetic patients not taking metformin (53). This evidence suggests this side effect may be a complication of diabetes and may not be associated with metformin treatment. The risk factors for lactic acidosis are: renal impairment, prolonged fasting or malnutrition, excessive alcohol intake, hepatic insufficiency or any condition associated with hypoxia e.g. cardiac or respiratory failure or circulatory shock due to any cause.

The risk of lactic acidosis should be considered in the event of non-specific symptoms such as muscle cramps, abdominal pain and/or severe weakness or lethargy. Any patient with a suspected metabolic acidosis requires immediate discontinuation of metformin and evaluation. Lactic acidosis is characterised by metabolic acidosis (decreased blood pH, high lactate above 5mmol/L and an increased anion gap and lactate/pyruvate ratio). The most effective way to remove lactate and metformin is haemodialysis.

#### :: Metformin overdose

Hypoglycaemia has not been reported with metformin doses of up to 85g although lactic acidosis has occurred in such circumstances. Patients should be urgently assessed in the event of an overdose and hospital admission considered. The management of metformin overdoses should be as per standard clinical care by the local team. The most effective way to remove lactate and metformin is haemodialysis.

### 6.2.8 Research Transdermal Oestradiol (Arm L)

Patients randomised to receive transdermal oestradiol may also receive SOC radiotherapy ([Section 6.1.2](#)) and SOC docetaxel ([Section 6.1.3](#)) as clinically appropriate, as has been done in the PATCH trial. It is recommended that patients commence SOC docetaxel (where planned) after they have been on transdermal oestradiol for around 4 weeks when most will have completed the induction period; radiotherapy, if used, would follow later.

Transdermal oestradiol is delivered as Progynova TS 100mcg/24 hours transcutaneous oestradiol patches according to the following dose regimen which has been shown within the PATCH trial to be sufficient for achieving castrate levels of testosterone.

#### 6.2.8.A Induction Regimen

**Four** Progynova TS 100 transdermal oestradiol patches to be changed twice weekly (e.g. Monday and Thursday) for four weeks. A confirmatory testosterone and oestradiol sample should be taken at 4 weeks with the sample drawn the **day before** the patches are changed.

#### 6.2.8.B Maintenance Regimen

If the patient has achieved a testosterone value of  $\leq 1.7\text{nmol/L}$  at 4 weeks then treatment is changed to a **maintenance regimen** of **three** patches changed twice weekly. The oestradiol level should also be monitored at the 4 week time point, with castrate levels of testosterone typically achieved with a plasma oestradiol level  $\geq 500\text{pmol/L}$ .

If a patient's testosterone is  $>1.7\text{nmol/L}$  at four weeks then they should remain on the induction regimen for another 4 week period, with monitoring of testosterone and oestradiol samples taken at around the week 8 time point, the day before patches are changed. Once the patient achieves a castrate level of testosterone  $\leq 1.7\text{nmol/L}$ , they can be reduced to the maintenance regimen.

#### 6.2.8.C Monitoring Hormone Levels

Oestradiol and testosterone levels should continue to be monitored throughout follow-up, while the patient remains on transdermal oestradiol treatment, to assess for evidence of compliance and to also ensure the patient is on the appropriate dose. See [Section 7.1.2](#) and [Table 1](#) for when these

values are required, noting also that the samples can be taken at the same time as scheduled PSA measurements.

A repeat blood test should be carried out within 4 weeks if, at any time, the patient's oestradiol level is found to be <300pmol/L or >2000pmol/L, with particular attention paid to the day that the patches are changed compared to when the blood sample is drawn (should be the day before changing patches). If the patient continues to have out of range oestradiol levels, and/or persistent testosterone >1.7nmol/L, then a member of the CTU team should be contacted for advice.

If the maintenance patch dose is changed at any time (for example, reducing from 3 to 2 patches changed twice weekly), then additional oestradiol and testosterone tests are required around 4 weeks after dose modification.

### **6.2.9 Transdermal Oestradiol: Administration, Dose Modifications And Management Of Toxicities**

Consecutive patches should be applied to different sites. It is recommended that patches are placed on dry, intact and hairless skin and on areas where little wrinkling occurs, at the following sites only:

- Shoulder girdle
- Back
- Upper arms
- Buttocks

Patches should not be placed on or near the breast area, or on areas of the body where there are large amounts of subcutaneous fat, particularly around the abdomen, as this could affect absorption. Please note that these recommendations are mainly based on studies in women using the patches.

To apply the patch, remove the protective liner and press on to the skin immediately, holding for at least 30 seconds to ensure proper adhesion. If necessary, tape can be used to fix the patch in place. If applied correctly, the patient can bath or shower as normal, however the patches might come off in very hot water or in a sauna.

Dermatitis can be a common side-effect of using the patches, especially in the induction period, which can usually be controlled by alternating the site of patch application. Patients should be advised that if patches become dislodged they should not put on extra patches, but apply their next set of patches when they are next due to be applied.

We expect patients to remain on the prescribed dose, and any potential dose modifications other than those indicated in [Section 6.2.8](#) should be discussed with the CTU team.

If a patient has a cardiovascular event (see [Section 7.1.4.B](#)), discontinuation of treatment with transdermal oestradiol may be considered at the discretion of the treating clinician.

## 6.3 CONCOMITANT MEDICATIONS AND DRUG INTERACTIONS

### 6.3.1 Abiraterone: Interaction With Medicinal Products And Other Forms Of Interaction

Details on drug interactions are described in [Appendix C](#) and [Table 16](#) provides a summary on the main interactions.

#### :: Anti-androgens

Abiraterone is steroid synthesis inhibitor and should **not** be given together with any other anti-androgens given the risk of toxicity. Cyproterone acetate should be discontinued 10 days and finasteride stopped 48 hours before commencing abiraterone. Concomitant use of dutasteride, bicalutamide, flutamide and tamoxifen are all **contraindicated**.

### 6.3.2 Enzalutamide: Interaction With Medicinal Products And Other Forms Of Interaction

Details on drug interactions are described in [Appendix C](#) and [Table 17](#) provides a summary on the main interactions.

#### :: Anti-androgens

Enzalutamide is potent androgen receptor antagonist and should **not** be given together with any other anti-androgens given the risk of toxicity.

Cyproterone acetate should be discontinued 10 days and finasteride stopped 48 hours before commencing enzalutamide. Concomitant use of dutasteride, bicalutamide, flutamide and tamoxifen are all **contraindicated**.

**Table 16: Drugs which may interact with abiraterone**

| DRUGS WHICH MAY INCREASE ABIRATERONE LEVELS            |                       |                                                                  |                                                                                                      |
|--------------------------------------------------------|-----------------------|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Substrate                                              | Clinical Use          | Drug                                                             | Recommendation                                                                                       |
| CYP3A4 inhibitors                                      | Macrolide antibiotics | Clarithromycin                                                   | Avoid or hold abiraterone if short term use unavoidable given increased risk of abiraterone toxicity |
|                                                        | Anti-fungals          | Ketoconazole<br>Itraconazole<br>Voriconazole                     | Avoid or hold abiraterone if short term use unavoidable given increased risk of abiraterone toxicity |
| DRUGS WHICH MAY REDUCE ABIRATERONE LEVELS              |                       |                                                                  |                                                                                                      |
| Substrate                                              | Clinical Use          | Drug                                                             | Recommendation                                                                                       |
| CYP3A4                                                 | Anti-epileptics*      | Phenytoin<br>Carbamazepine<br>Phenobarbital<br>Primadone         | Contraindicated                                                                                      |
|                                                        | Anti-depressants      | St Johns Wart                                                    | Contraindicated                                                                                      |
|                                                        | Anti-TB               | Rifampicin<br>Rifabutin                                          | Contraindicated                                                                                      |
|                                                        | Anti-retroviral       | Atazanavir<br>Saquinavir<br>Ritonavir<br>Indinavir<br>Nelfonavir | Contraindicated. Seek specialist advice and discuss with trial team                                  |
| DRUGS WHICH MAY ACCUMULATE WHEN GIVEN WITH ABIRATERONE |                       |                                                                  |                                                                                                      |
| Substrate                                              | Clinical Use          | Drug                                                             | Recommendation                                                                                       |
| CYP2D6                                                 | Cardiac               | Metoprolol<br>Propranolol<br>Propafenone<br>Flecainide           | Monitoring required as drug levels may increase with abiraterone use                                 |
|                                                        | Anti-depressants      | Desipramine<br>Venlafaxine<br>Citalopram                         | Monitoring required as drug levels may increase with abiraterone use                                 |
|                                                        | Anti-psychotics       | Haloperidol<br>Risperidone                                       | Monitoring required as drug levels may increase with abiraterone use                                 |
|                                                        | Analgesia             | Tramadol<br>Codeine<br>Oxycodone                                 | Monitoring required as drug levels may increase with abiraterone use                                 |
|                                                        | Alpha blockers        | Tamsulosin                                                       | Monitoring required as drug levels may increase with abiraterone use                                 |
|                                                        | Anti-diabetic         | Repaglinide                                                      | Monitoring required as drug levels may increase with abiraterone use                                 |

**Table 17: Drugs which may interact with enzalutamide**

| DRUGS WHICH MAY INCREASE ENZALUTAMIDE LEVELS            |                    |                                                                  |                                                                                                                             |
|---------------------------------------------------------|--------------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Substrate                                               | Clinical Use       | Drug                                                             | Recommendation                                                                                                              |
| CYP2C8 inhibitors                                       | Lipid-lowering     | Gemfibrozil                                                      | Avoid, if no alternatives, reduce enzalutamide dose to 80mg                                                                 |
| DRUGS WHICH MAY DECREASE ENZALUTAMIDE LEVELS            |                    |                                                                  |                                                                                                                             |
| Substrate                                               | Clinical Use       | Drug                                                             | Recommendation                                                                                                              |
| CYP2C8 inducers                                         | Anti-TB            | Rifampicin<br>Rifabutin                                          | Avoid and switch to an alternative if possible                                                                              |
| CYP3A4 inducers                                         | Anti-epileptics    | Phenytoin<br>Carbamazepine<br>Phenobarbital                      | Contraindicated                                                                                                             |
|                                                         | Anti-depressant    | St Johns Wart                                                    | Contraindicated                                                                                                             |
|                                                         | Anti-retrovirals   | Atazanavir<br>Saquinavir<br>Ritonavir<br>Indinavir<br>Nelfanavir | Contraindicated. Seek specialist advice and discuss with trial team                                                         |
| ENZALUTAMIDE MAY REDUCE DRUG LEVELS                     |                    |                                                                  |                                                                                                                             |
| Substrate                                               | Clinical Use       | Drug                                                             | Recommendation                                                                                                              |
| CYP2C19                                                 | Gastric protection | Omeprazole                                                       | Omeprazole AUC reduced by 70%<br>Consider increasing dose of omeprazole for same therapeutic effect                         |
| CYP3A4                                                  | Analgesia          | Fentanyl*<br>Alfentanil*<br>Tramadol                             | Monitor closely and consider alternatives                                                                                   |
|                                                         | Immunosuppressants | Sirolimus*<br>Tacrolimus*<br>Cyclosporine*                       | Monitor closely                                                                                                             |
|                                                         | Anti-migraine      | Ergotamine                                                       | Monitor closely                                                                                                             |
|                                                         | Cardiac            | Nifedipine<br>Ivabradine                                         | Monitor closely, consider alternatives as clinical effect may be reduced                                                    |
| CYP2C9                                                  | Anti-epileptics    | Phenytoin*                                                       | Contraindicated                                                                                                             |
|                                                         | Anti-coagulants    | Warfarin*                                                        | Warfarin AUC reduced by 56%<br>Consider switching to low molecular heparin, increase INR monitoring if this is not possible |
| DRUGS WHICH MAY ACCUMULATE WHEN GIVEN WITH ENZALUTAMIDE |                    |                                                                  |                                                                                                                             |
| Substrate                                               | Clinical Use       | Drug                                                             | Recommendation                                                                                                              |
| p-gp                                                    |                    | Colchicine*<br>Dabigatran*<br>Digoxin*                           | Monitor closely                                                                                                             |

\*narrow therapeutic index

### 6.3.3 Metformin: Interaction With Medicinal Products And Other Forms Of Interaction

Metformin does not interact with any of the other treatments for prostate cancer and can be continued during all further treatments started on progression.

Caution is needed however when initiating potential nephrotoxic drugs as metformin is renal excreted therefore may accumulate if renal function deteriorates, see [Table 18](#) for details.

As metformin is being given as an IMP in the context of a clinical trial, continued use will not be permitted if patients participate in other interventional clinical trials for prostate cancer (i.e. CRPC setting). Investigators should use their discretion and discuss discontinuing metformin with the trial team if it is felt to be in the patient's best interest.

**Table 18: Drugs which require additional monitoring of renal function**

| Clinical use                                 | Drug                                                                                      | Recommendation                                                                                                                                    |
|----------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Anti-hypertensives and other cardiac disease | ACE inhibitors/angiotension II receptor blockers<br>e.g. ramipril, lisinopril, Irbesartan | Monitor renal function until confirmed to be stable and providing GFR remains $>45\text{ml/min/m}^2$ . Repeat test if necessary                   |
|                                              | Diuretics<br>e.g. Frusemide, budesonide                                                   |                                                                                                                                                   |
| Antibiotics                                  | Aminoglycoside antibiotics<br>e.g. Gentamicin or amikacin                                 | Hold metformin during treatment and re-start providing renal function confirmed to be stable and GFR remains $>45\text{ml/min/m}^2$               |
| Analgesia                                    | NSAIDS<br>e.g. Ibuprofen, diclofenac, naproxen                                            | Avoid if possible<br>If no alternative increase renal monitoring to until confirmed to be stable and providing GFR remains $>45\text{ml/min/m}^2$ |

If the renal function declines to  $\text{GFR} < 45\text{ml/min/m}^2$  a dose reduction is required and the frequency of monitoring of renal function must increase. See [Section 6.2.7.A](#).

### 6.3.4 Transdermal Oestradiol: Drug Interactions

Tamoxifen should not be prescribed for patients receiving transdermal oestradiol.

The metabolism of oestrogens may be increased by concomitant use of substances known to induce drug metabolising enzymes, specifically cytochrome P450 enzymes, such as anticonvulsants (e.g. phenobarbital, phenytoin, carbamazepine) and anti-infectives (e.g. rifampicin, rifabutin, nevirapine, efavirenz). Ritonavir and nelfinavir, although known as strong inhibitors, by contrast exhibit inducing properties when used concomitantly with steroid hormones. Herbal preparations containing St. John's wort (*Hypericum Perforatum*) may induce the metabolism of oestrogens.

With transdermal administration, the first-pass effect in the liver is avoided and, thus, transdermally applied oestrogens might be less affected than oral hormones by enzyme inducers. Oestradiol levels are already monitored as part of trial follow-up while patients are on transdermal oestradiol. As a precaution, we recommend monitoring the drug levels of the above concomitant medications among patients receiving transdermal oestradiol.

## 6.4 TRIAL PRODUCTS

Details of the procedures for obtaining the drugs within the trial, dispensing and disposal of unused drug are given in [Appendix B](#). Arrangements for free or discounted drugs are given in the Finance section ([Section 15](#)).

## 6.5 TREATMENT DATA COLLECTION

Data will be recorded on case report forms (CRFs); the top copy/original should be sent to CTU for data entry and a copy kept at the local centre. Up-to-date versions of all CRFs can be found on the trial website (<http://www.stampedetrial.org/>) and centres will be notified of any changes throughout the course of the trial. The type of data to be recorded is detailed in the Assessments and Procedures section ([Section 7](#)).

## 6.6 MEASURES OF COMPLIANCE/ADHERENCE

Date of treatment, dose, delays and reasons for delays or dose modifications of all trial treatments will be recorded. The estimated number of abiraterone tablets and enzalutamide capsules taken in a given time period will also be recorded as well as any dose reductions.

Oestradiol levels will be collected for patients in the transdermal oestradiol arm and used to assess compliance to treatment (see [Section 6.2.8](#)).

Evidence of compliance with safety monitoring is required for patients on research abiraterone or metformin treatment, as described in [Section 6.2.3.B](#) and [Section 6.2.7.A](#) e.g. potassium, LFT and blood pressure monitoring for patients receiving abiraterone. Site investigators should document in the patient's medical records the date of the blood test or review of blood pressure measurements and confirmation that the results were known to be within acceptable limits and if not, the toxicity should be graded according to CTCAE and the action described. This should be available at on-site monitoring visits and used to verify the information provided on the follow-up CRF and treatment logs.

## 6.7 ADMINISTRATION OF STANDARD RADIOTHERAPY TO M0 PATIENTS

### 6.7.1 Treatment Details

Standard radiotherapy will be given to appropriate patients in each of the trial arms, following a period of neo-adjuvant ADT therapy, as is generally standard in UK practice. For patients with negative nodes on axial imaging, clinicians may choose between irradiating prostate and seminal vesicles alone or including the pelvic nodes in addition. Additional staging tests such as pelvic node sampling may be considered in making this decision. Conformal or intensity modulated radiotherapy should be used in all patients. Where patients have good clinical evidence that nodes are free of tumour or patients for whom nodal radiotherapy is contra-indicated (e.g. significant bowel disease), treatment may be given to the prostate gland and seminal vesicles only. The recommended dose is 74Gy in 37 fractions to the prostate and seminal vesicles or the equivalent using hypo-fractionated schedule, 60Gy in 20 fractions. Alternative dosing schedules are permitted but must be agreed with the STAMPEDE Trial Management Group.

#### **6.7.1.A Standard-Of-Care RT Timing In M0 patients**

If receiving docetaxel as part the standard-of-care (permitted from Protocol version 14.0), the patient must have sufficiently recovered from any docetaxel toxicity before RT can begin. In all other patients not receiving SOC docetaxel, SOC RT may be started sooner (2-6 months post-randomisation) consistent with the data from the MRC PR07 trial (11).



## 7 ASSESSMENTS AND PROCEDURES

### 7.1 SCHEDULE FOR ASSESSMENTS

#### 7.1.1 Follow-Up Schedules

An individualised form with a follow-up schedule will be provided for each randomised patient. A detailed follow-up schedule is given in [Table 1](#) and [Table 20](#).

Note that for the transdermal oestradiol arm, the first follow-up visit post-randomisation can be scheduled at 4 instead of 6 weeks to coincide with the 4-week hormone tests (see [Section 6.2.8A](#)).

See [Table 1](#) for a summary of required investigations at each follow-up visit.

#### 7.1.2 PSA, Testosterone And Oestradiol Measurements

All patients should have PSA measured prior to starting ADT and at every subsequent trial follow-up visit, regardless of allocated treatment arm. For patients who do not have a scheduled hospital visit, it is acceptable for arrangements to be made for blood samples to be drawn at their GP surgery.

For arm L patients, oestradiol and testosterone levels should continue to be monitored while the patient remains on transdermal oestradiol treatment; see [Table 1](#) for when these measurements should be obtained. These samples could be taken at the same time as the PSA tests, unless additional tests are required as detailed in [Sections 6.2.8.B](#) and [6.2.8.C](#). It is also preferable for samples to be taken the day before the oestradiol patches are changed, to allow consistent measurements of testosterone and oestradiol with respect to the pharmacokinetic profile of the patches.

### 7.1.3 Assessment Of Treatment Failure (Definition Of Progression)

It is not proposed to routinely assess patients for response. However, in order that objective progression can be assessed, it is necessary to have imaging taken at time of best response as judged by the treating clinician.

All patients should have baseline radiological examinations as detailed in [Section 4.5.2](#). In addition, it is recommended all patients should have scans or X-rays repeated at 24 weeks (and whenever clinically appropriate) if they were abnormal at baseline, particularly if they have a low PSA value on entry in to the trial making biochemical assessment of treatment failure difficult.

The following outcomes should be reported on the Progression log:

- Biochemical failure
- Local progression
- Lymph node progression
- Progression or development of new distant metastases, defined as lymph nodes outside the pelvis, bone or organ involvement
- Skeletal-related events confirmed as progression (see below)

#### 7.1.3.A Biochemical Failure

For the purposes of the STAMPEDE trial, a unique threshold PSA value for biochemical failure is calculated, referred to as the **PSA progression value**.

This value is derived for each patient based on their **PSA nadir**, defined as the lowest PSA value reported between randomisation and 24 weeks on trial. Please refer to the PSA progression value calculator on the STAMPEDE website.

The exact method for deriving the progression value for a patient depends on the value of their PSA nadir, and how this compares to their pre-treatment PSA value (i.e. the extent of the fall in PSA from the starting point).

The PSA progression value is calculated in one of three ways:

- A. If the lowest recorded PSA value in the 24 weeks following randomisation is more than 4ng/ml and more than 50% of the pre-treatment PSA level then the patient fulfils the criteria for immediate treatment failure.
- B. For patients whose PSA nadir in the 24 weeks following randomisation is less than or equal to 50% of the pre-treatment PSA level but remains above 4ng/ml, biochemical failure will be defined as a rise of 50% above the nadir level.
- C. For patient whose PSA nadir is less than or equal to 4ng/ml, biochemical failure is defined as at least a 50% rise above the nadir value that is also above 4ng/ml.

**Confirming biochemical failure:** the timing of assessments needs to be considered because spurious rises in PSA can occur e.g. following procedures involving the urinary tract. For this reason, any isolated rise in PSA should be confirmed before reporting biochemical failure.

In the case that the raised PSA value reaches the progression value, a confirmatory PSA test should be performed between one week and 3 months later. Biochemical failure is confirmed if the second

value is around the same level or higher i.e. the trend is confirmed. The date of PSA progression should be provided as the date of the **first** raised PSA that fulfilled the trial definition of progression. Only the first instance of biochemical failure needs to be reported.

A confirmatory PSA is not required if there are other signs of progression e.g. progression of cancer related symptoms (clinical progression) or new radiological progression.

Second line treatment commenced specifically for biochemical failure should not start until the trial definition for biochemical failure has been met. However, if second line treatment does start before the trial definition is met then report the closest PSA value prior to the treatment start date as the progression value. This is not required if second line treatment is being started for other signs of progression e.g. clinical or radiological.

**Testosterone levels:** are only required when reporting biochemical progression whilst receiving hormone treatment to confirm the diagnosis of castrate resistant prostate cancer. Testosterone levels are not required when reporting biochemical progression in patients not receiving hormone therapy e.g. patients who presented with non-metastatic disease have relapsed following completion of treatment.

See [Appendix E](#) for further details on the trial definition of biochemical failure.

#### 7.1.3.B Local, Lymph Node And Metastatic Failure

For each of local, lymph node and distant metastases progression, **both** the following should be reported:

1. Date of first clinical/symptomatic progression
2. Date of first objective/radiological progression

#### 7.1.3.C Skeletal-related Events

Skeletal-related events (SREs) are defined as:

- Pathological Fracture
- Spinal cord compression
- Requirement for RT to bone (e.g. for pain or impending fracture)
- Requirement for surgery (e.g. for prevention or management of fracture)

SREs are a secondary outcome measure and a disease event of interest. SREs may represent disease progression but can also occur due to treatment-related effects e.g. osteoporotic fracture due to treatment-related bone-mineral density loss. From Protocol version 15.0 information regarding SREs will be collected at each follow-up visit. All SREs should be investigated further to establish whether or not the patient has progressed and, if confirmed as progression, a Progression Log should be completed to record this along with an Additional Treatment Log to give details of any treatment received (e.g. palliative RT or surgery)

The summary of timing of Case Report Forms can be viewed in [Table 19](#).

#### 7.1.4 Additional Metabolic And Cardiovascular Outcomes

A number of metabolic and cardiovascular (CVS) outcomes are being assessed in the “metformin comparison” and “transdermal oestradiol comparison” as outlined below. From protocol version 17.0 onwards, a metabolic profile (lipids, glucose and HbA1c) will be measured for all patients randomised from Sept-5-2016 onwards to capture data on metabolic and cardiovascular outcomes

for both comparisons. See [Table 1](#) for a schedule of assessments, please note it is permitted to obtain these measurements within 12 weeks of the scheduled follow-up visit.

The summary of timing of Case Report Forms can be viewed in [Table 19](#).

#### 7.1.4.A Cardiovascular Outcomes: Transdermal Oestradiol Comparison

Cardiovascular morbidity and mortality was the primary outcome measure for the first stage in the PATCH trial (completed in 2010), which showed similar rates of CVS events in patients receiving transdermal oestradiol compared to those receiving LHRH injections(43). These results have been confirmed by longer-term data within the trial (see [Appendix I](#)). Continued monitoring of CVS outcomes will be undertaken by the PATCH IDMC for both the PATCH trial, as well as for the patients in STAMPEDE allocated to transdermal oestradiol together with their contemporaneous controls.

While Arm L patients are undergoing treatment with transdermal oestradiol, the majority of these CVS events will fall under the definitions of Serious Adverse Events (see [Section 11](#)). Once a patient has a cardiovascular event, the discontinuation of treatment with transdermal oestradiol may be considered at the discretion of the treating clinician and the patient switched to standard of care hormone therapy.

An increased risk of venous thromboembolism has been observed when docetaxel is used in combination with certain agents for the treatment of prostate cancer. As yet, there are limited safety data available on docetaxel use in combination with transdermal oestradiol from the PATCH trial. Therefore, the rate of CVS events will be closely monitored among patients within Arm L who are receiving docetaxel as part of their first-line treatment. For more details see [Appendix I](#).

#### 7.1.5 Additional Safety Assessments

Medical review and PSA measurements are repeated for all patients across all research arms (including the control arm) and follow the trial FU schedule. Patients have FU assessments every 6 weeks for 6 months (apart from the transdermal oestradiol arm where the first assessment is at 4 weeks instead of 6 to coincide with the hormone tests), every 12 weeks up to 2 years, six-monthly up to 5 years and annually thereafter). In addition, there are arm-specific assessments as outlined below and summarised in [Table 1](#). The summary of the timing of Case Report Forms also can be viewed in [Table 19](#).

##### 7.1.5.A Additional Safety Assessment: Abiraterone

Due to the risk of liver toxicity and secondary hyperaldosteronism with abiraterone, all patients require regular monitoring of **potassium, liver function tests and blood pressure** whilst receiving research abiraterone. Monitoring should be performed 2-weekly in the first 12 weeks of treatment, then **monthly until 12-months on treatment** and then, providing treatment is well tolerated, **2-monthly thereafter** whilst treatment with research abiraterone continues, see [Section 6.2.3.B](#).

Investigators are responsible for ensuring blood tests are performed at the required frequency and need to review and document the results. It is acceptable for blood pressure to be self-monitored at the required frequency by trial participants or via the GP, providing this is reviewed at each follow-up by investigators.

Confirmation that potassium and liver functions test have been performed regularly and blood pressure control reviewed will be required at each follow-up visit. Any abnormalities should be graded according to CTCAE version 4.0 and recorded on the toxicity section of the follow-up CRF; any

abnormalities fulfilling the criteria for a SAE (e.g. requiring hospital admission) should also be reported on a SAE CRF (see [Section 11](#)).

#### 7.1.5.B Additional Safety Assessment: Metformin

Patients with normal and stable renal function receiving metformin require monitoring of **renal function (U&Es) every 6 months** whilst on treatment. More frequent monitoring is required in patients with declining renal function, or when initiating new potentially nephrotoxic medications or at times of intercurrent illness (see [Section 6.2.7.A](#)). Changes in renal function (eGFR, graded according to CTCAEv.4) are recorded on the Follow-Up CRF. It is acceptable for bloods sampling to be arranged via the GP at the patient's home or local hospital.

#### 7.1.5.C Additional Safety Assessment: Transdermal Oestradiol

Hormone levels are monitored while patients are on transdermal oestradiol, and if oestradiol levels are found to be >2000pmol/L with confirmed repeat test, please contact CTU for advice (see [Section 6.2.8.C](#)).

## 7.2 DATA COLLECTION PROCEDURES

Treatment-related data are collected on Treatment Specific Forms or Logs. It is important that any treatment given for progressive disease is recorded on the Additional Treatment Log. This should be updated with any subsequent changes e.g. treatment for CRPC. The summary of timing of Case Report Forms can be viewed in [Table 19](#).

### 7.2.1 Data Collection For SOC Hormone Therapy

Information relating to SOC hormone therapy is recorded on the SOC Hormone Therapy Log unless it is a treatment change for disease progression. The SOC Hormone Therapy Log should be updated with any changes in long-term hormone therapy e.g. if anti-androgens are being added to LHRH for dual androgen blockade in the **absence of progression**. If however, anti-androgens are being added as an additional treatment for progressive disease, then this should be recorded on the Additional Treatment Log. SOC hormone therapy only refers to LHRH or anti-androgens; if second-generation AR-targeted treatments such as abiraterone or enzalutamide are used as second-line treatments for progression this should only be recorded on the Additional Treatment Log. Corresponding details of the progression event should be reported on the Progression Log.

If a patient allocated to receive transdermal oestradiol switches to receiving SOC Hormone Therapy i.e. LHRH in the **absence of progression**, then this information should be recorded on the SOC Hormone Therapy Log. However, any changes in hormone therapy initiated to treat disease progression should be recorded on the Additional Treatment Log e.g. switching from transdermal oestradiol to LHRH due to progressive disease.

### 7.2.2 Data Collection For Standard Docetaxel

The decision to use docetaxel as part of the standard-of-care must be made before randomisation and should be recorded on the Randomisation CRF. The date of the first cycle should be recorded at the time of randomisation; this can be a planned date when randomisation occurs prior to docetaxel commencing but must be within 12 weeks of starting ADT (see [Section 6.2.7](#)). All further details should be recorded on the SOC Docetaxel Treatment CRF upon completion of the final cycle.

If a patient does not receive the planned docetaxel, this must also be recorded on the SOC Docetaxel Treatment CRF, together with the reason why.

### **7.2.3 Data Collection And Non-Administration Of Standard Radiotherapy**

There are CRFs to be completed for ALL patients regardless of being planned for, or subsequently receiving, primary radiotherapy. Where radiotherapy is not received a reason should be provided on the Radiotherapy Detail CRF whether this is standard-of-care radiotherapy for patients (on any research arm) or research RT to the prostate for Arm H patients.

All radiotherapy and acute side effects details should be recorded on the Radiotherapy Detail and Radiotherapy Acute Toxicity CRFs upon completion of the RT schedule; any RT late side effects should be recorded on the Follow-Up CRF under the section for RTOG Toxicities.

If RT is not given, this should be stated on the Radiotherapy Detail CRF together with the reason for non-administration of the treatment in those instances where RT was planned and not given (for example, due to early metastatic progression or patient refusal).

### **7.2.4 Data Collection For Palliative Radiotherapy**

Details of any radiotherapy given for progressive disease should be recorded on the Additional Treatment Log.

This includes palliative RT for SREs e.g. bone pain and spinal cord compression (note that these should also be reported as SREs on the Follow-Up CRF and only reported on the Progression Log and Additional Treatment Log if confirmed as progression), as well as salvage RT to the prostate.

### **7.2.5 Data Collection for Research (M1) Radiotherapy**

Arm H only: all radiotherapy and acute side-effects details will be recorded on the Radiotherapy Detail and Radiotherapy Acute Toxicity CRFs upon completion of the RT schedule; any RT late side effects will be recorded on the Follow-Up CRF under the section for RTOG Toxicities.

In those cases where RT is not given (for example, due to early metastatic progression or patient refusal), this should be stated on the Radiotherapy Detail CRF together with the reason for non-administration of the treatment.

### **7.2.6 Data Collection for Additional Treatments Given for Disease Progression**

All treatments given for disease progression are recorded on the Additional Treatment Log. This should be updated with all subsequent changes to treatment. Only treatments for progressive disease need to be recorded; details of supportive treatments such as pain killers or bone-strengthening agents e.g. zoledronic acid, given to relieve symptoms, does not need to be provided.

In some scenarios, SOC hormone therapies such as LHRH or anti-androgens may be given as a treatment for progressive disease. For example, LHRH may be re-started on relapse for patients with M0 disease who discontinued hormone therapy and commenced surveillance. In addition, patients allocated to transdermal oestradiol may switch to LHRH on progression. Historically, some patients progressing on LHRH will have commenced anti-androgens (dual androgen blockade) as a treatment for progression. In all cases, if treatment is being started for disease progression, treatment data are collected on the Additional Treatment Log and the details of the progression event recorded on the Progression Log.

Please note that any change in ADT which are solely a change in the patient's long-term hormone therapy, and not for disease progression, should be reported on the SOC Hormone Therapy Log only and not on the Additional Treatment Log.

### 7.3 FOLLOW-UP PROCEDURE

Every effort should be made to follow-up all patients who have been randomised. Patients should, if possible, remain under the care of an oncologist or urologist for the duration of the trial. If care of a patient is returned to the GP, it is the responsibility of the responsible clinician who obtained the patient's consent to participate in the trial to ensure that all relevant data collection forms are completed. Nurse-led follow-up is permitted and should be conducted in line with local practice and procedures.

If the patient moves away from the local area, arrangements should be made for trial follow-up to be undertaken by their new local centre. Details of other participating centres can be obtained from the STAMPEDE Trial Team. Information on patient transfer procedures is detailed in [Section 8.2](#). If the responsible clinician moves, appropriate arrangements should be made to arrange for trial follow-up to continue at the centre.

All efforts should be made to preserve the initial patient's consent for long-term survival information to be flagged through national registries, for example NHS Digital (previously the Health and Social Care Information Centre); Office of National Statistics (ONS) in England/Wales; General Register Office in Scotland; Hospital Episode Statistics (HES) or Public Health England.

Please see [Section 8](#) for more information on early stopping of follow-up.

#### 7.3.1 Follow-Up Telephone Consultations

In certain circumstances it may be appropriate to replace hospital visits with telephone consultations providing that it is still possible to collect all the necessary follow-up information. Situations where this may be considered include at the point where patients would normally be discharged from oncology or urology services. In these instances, it is acceptable to alternate appointments with telephone consultations providing the required blood results are available to the research team. All necessary information required to complete the Follow-Up CRF is still required. All details on the telephone consultation must be recorded in the patients' notes as per in person assessments.

Other circumstances where it may be appropriate to use telephone consultations is when assessing treatment tolerance and advising regarding dose modifications, providing that all the required safety monitoring has been adhered to. For example, when re-commencing metformin after a treatment break it may be appropriate to confirm tolerance and advise regarding dose escalation over the phone.

### 7.4 TRIAL CLOSURE

For the purpose of complying with UK Clinical Regulations introduced in May-2004, each comparison will only be considered 'closed' when follow-up has ceased. This will be reviewed for each comparison separately after the point of the primary analysis and, if appropriate, a later, updated analysis. Longer term outcome data may be sought via site research teams and/or through linkage with national registers where possible and adequate consent has been obtained.

**Table 19: Summary of timing of case report forms**

| CASE REPORT FORMS                          | TIMING OF ASSESSMENT AND CRF                                                                                                                                                                                                                                                       |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Registration*</b>                       |                                                                                                                                                                                                                                                                                    |
| Registration                               | At registration. For patients participating in the Biomarker-Screening Pilot, when FFPE tumour sample has been retrieved and is ready to be sent to the Sponsor's designated laboratory.                                                                                           |
| Biomarker Test Request Form                | As soon as possible after registration.                                                                                                                                                                                                                                            |
| Blood Form                                 | As soon as possible after registration.                                                                                                                                                                                                                                            |
| Saliva Pathology                           | As soon as possible after registration.                                                                                                                                                                                                                                            |
| <b>Baseline</b>                            |                                                                                                                                                                                                                                                                                    |
| Randomisation                              | At randomisation                                                                                                                                                                                                                                                                   |
| Baseline                                   | At randomisation                                                                                                                                                                                                                                                                   |
| Cardiovascular Assessment                  | At randomisation                                                                                                                                                                                                                                                                   |
| Bone Density Risk Factor                   | At randomisation                                                                                                                                                                                                                                                                   |
| Saliva Pathology                           | At randomisation or any point on trial. When saliva sample has been taken and sent to Sponsor's designated laboratory.                                                                                                                                                             |
| <b>Treatment</b>                           |                                                                                                                                                                                                                                                                                    |
| SOC Docetaxel Treatment                    | To be completed for all patients 20 weeks after randomisation.                                                                                                                                                                                                                     |
| SOC Hormone Therapy Log                    | To be completed every time there is a change in SOC hormone therapy to report (including when Arm L patients switch to SOC HT pre-progression). To be sent in with the corresponding Follow-Up CRF.                                                                                |
| Abiraterone and Enzalutamide Treatment Log | To be completed when treatment is first started and subsequently every time there is a dose change, treatment pause and re-start. To be sent in with the corresponding Follow-Up CRF.                                                                                              |
| Metformin Treatment                        | To be completed when treatment is first started and subsequently every time there is a dose change, treatment pause and re-start. To be sent in with the corresponding Follow-Up CRF.                                                                                              |
| Transdermal Oestradiol Treatment Log**     | To be completed when treatment is first started and subsequently when reporting change in dose or type of patch.                                                                                                                                                                   |
| RT Detail                                  | To be completed for all patients:<br>Upon completion of SOC RT<br>If planned RT is no longer to be given (at 10 months after randomisation)<br>Arm H patients when research RT completed<br>Arm A patients with newly-diagnosed M1 disease at 3 months to confirm RT was not given |
| RT Acute Toxicity                          | For all patients who receive primary RT                                                                                                                                                                                                                                            |
| <b>Assessments</b>                         |                                                                                                                                                                                                                                                                                    |
| Follow-Up                                  | To be completed every 6 weeks for 6 months, then every 12 weeks until 2 years, then every 6 months until 5 years and annually thereafter*. (See <a href="#">Table 1</a> for more information)                                                                                      |



| CASE REPORT FORMS                                    | TIMING OF ASSESSMENT AND CRF                                                                                                                                                                                                                                              |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Toxicity                                             | Required at each follow-up and in the event that treatment is changed due to toxicity.                                                                                                                                                                                    |
| Transdermal Oestradiol Treatment Hormone Results Log | To be completed whenever there are testosterone and oestradiol test results while arm L patients on transdermal oestradiol.                                                                                                                                               |
| End of Research Treatment                            | To be completed when (each) allocated research treatment is permanently stopped or in the event that allocated research treatment is never started (in each case a reason for stopping/never starting should be provided).                                                |
| Progression Log                                      | To be completed at the occurrence of each progression event (PSA, local, nodal, distant metastases) and for each method of detection (clinical/symptomatic and objective/radiological).<br>Skeletal-related events confirmed as progression should also be reported here. |
| Additional Treatment Log                             | To be completed each time a patient who has progressed starts or completes any additional treatment for progression.                                                                                                                                                      |
| Serious Adverse Event                                | To be completed following any Serious Adverse Event having confirmed none of the trial specific exemptions are met                                                                                                                                                        |
| Death                                                | At Death                                                                                                                                                                                                                                                                  |
| <b>Administration</b>                                |                                                                                                                                                                                                                                                                           |
| Patient Transfer Confirmation Form                   | To be completed when a patient is transferred to a different hospital for the administration of trial treatment and follow-up                                                                                                                                             |
| Co-enrolment                                         | To be completed when a patient is co-enrolled in any other clinical trial. Please see <a href="#">Section 5.2</a> for more information                                                                                                                                    |

\* For centres participating in the biomarker-screening only

\*\* For the transdermal oestradiol arm, the 6-week follow-up form can be completed at the same time as the 4-week visit for the hormone tests (see Section 6.2.8.A)

**Table 20: Schedule for completion of treatment and outcome forms by arm**

| TIMING FROM RANDOMISATION |        |       | TREATMENT LOG <sup>§</sup> |
|---------------------------|--------|-------|----------------------------|
| YEARS                     | MONTHS | WEEKS | (IF REQUIRED)              |
| <b>6-Weekly</b>           |        |       |                            |
| -                         | -      | 6*    | G, J, K, L                 |
| -                         | -      | 12    | G, J, K, L                 |
| -                         | -      | 18    | G, J, K, L                 |
| -                         | 6      | 24    | G, J, K, L                 |
| <b>12-Weekly</b>          |        |       |                            |
| -                         | 9      | 36    | G, J, K, L                 |
| 1                         | 12     | 48    | G, J, K, L                 |
| -                         | 15     | 60    | G, J, K, L                 |
| -                         | 18     | 72    | G, J, K, L                 |
| -                         | 21     | 84    | G, J, K, L                 |
| -                         | -      | 96    | G, J, K, L                 |
| <b>6-Monthly</b>          |        |       |                            |
| 2                         | 24     | 104   | G, J, K, L                 |
|                           | 30     | 130   | G, J, K, L                 |
| 3                         | 36     | 156   | G, J, K, L                 |
|                           | 42     | 182   | G, J, K, L                 |
| 4                         | 48     | 208   | G, J, K, L                 |
|                           | 54     | 234   | G, J, K, L                 |
| 5                         | 60     | 260   | G, J, K, L                 |
| <b>Annual</b>             |        |       |                            |
| 6                         | 72     | -     | G, J, K, L                 |
| 7                         | 84     | -     | G, J, K, L                 |
| Etc.                      | -      | -     | G, J, K, L                 |

**Key:**

G = SOC + abiraterone  
J = SOC + enzalutamide + abiraterone  
K = SOC + metformin  
L = Transdermal oestradiol ± RT ± docetaxel

**Notes:**

\* For the transdermal oestradiol arm, the 6-week follow-up form can be completed at the same time as the 4-week visit for the hormone tests (see Section 6.2.8.A)  
§ For patients in Arm L on transdermal oestradiol, the hormone tests results are to be reported on the Transdermal Oestradiol Treatment Hormone Results Log

## 8 STOPPING OF TREATMENT OR FOLLOW-UP

Patients should be given every encouragement to adhere to their allocated protocol treatment and follow-up schedule, in order to reduce bias. However, a patient has the right to withdraw consent for participation in any aspect of this trial at any time.

### 8.1 STOPPING RESEARCH INTERVENTIONS

A patient may stop **any trial treatment** for the following reasons:

- Unacceptable toxicity
- Intercurrent illness which prevents further treatment
- Patient refusal
- Any alteration in the patient's condition which justifies the discontinuation of treatment in the clinician's opinion

#### 8.1.1 Stopping Trial Treatment: Abiraterone, Enzalutamide + Abiraterone

For **patients randomised to Arm G or J**, trial treatment should also be discontinued for the following reasons:

- Disease progression whilst on therapy (please refer to [Section 7.1.3](#))
- Intention to commence a new anti-cancer treatment due to evidence of relapse

As detailed in [Section 7.1.3](#), the disease event for stopping treatment may be after the first reportable Failure-Free Survival event.

In all cases reason for permanent stopping of research treatment should be recorded on the End of Research Treatment CRF.

#### 8.1.2 Stopping Trial Treatment: Metformin

For **patients randomised to Arm K**, treatment duration is detailed in [Section 6.2.6](#).

Reasons for early stopping of metformin can be:

- Decline in renal function (metformin must be stopped if  $GFR < 30 \text{ ml/min/1.73m}^2$ , see [Section 6.2.7.A](#))
- Decline in performance status (WHO PS >2)
- Unacceptable toxicity
- Patient refusal
- Intercurrent illness preventing continued metformin treatment
- Investigator decision e.g. administration of IMP within a CTIMP in CRPC setting

If metformin is paused for more than 3 months or >50% of doses are missed please discuss with the trial team as treatment is likely to need to be stopped.

#### 8.1.3 Stopping Trial Treatment: Transdermal Oestradiol

For **patients randomised to Arm L**, treatment with transdermal oestradiol may be discontinued for the following main reasons:

- Unacceptable toxicity
- Patient refusal

- Intercurrent illness
- Investigator decision
- Cardiovascular event (see [Section 7.1.4B](#))

In addition, upon evidence of disease progression and at the investigator's discretion, a switch to LHRH analogues is appropriate to facilitate the addition of further therapies where concurrent treatment with transdermal oestradiol is untested. For patients who are on the 3 patch maintenance dose and have castrate levels of testosterone, there is currently no evidence that increasing the number of patches further once the patient has progressed would be beneficial and is therefore not recommended.

In the event of stopping research treatment, unless a patient states otherwise, consent is assumed for continued recording of trial data.

## 8.2 PATIENT TRANSFERS

For patients moving away from the area and planning to transfer care, every effort should be made for the patient to be followed-up at another participating trial centre. The patient will need to sign a new consent form. Once this has been done, the new trial centre will take over responsibility for the patient ongoing participation in the trial, until this has been done, responsibility for the patient lies with the original trial centre.

To document the transfer process the main contact person at both the current and receiving hospitals should complete and sign the Patient Transfer Confirmation form.

A fully completed form must be returned to CTU prior to the patient transfer and any outstanding data queries for the patient should be completed prior to transfer.

On receipt of the completed transfer form a member of the STAMPEDE team will confirm the database has been updated and request confirmation of the name of the patient's new clinician. Photocopies of the following documents may then be sent to the new hospital to complete the transfer and copies must be also retained at the original site for monitoring purposes:

- Consent form
- Completed CRFs
- Any documentation relating to the patient's participation in STAMPEDE (patient names must be removed from any documentation).

## 8.3 EARLY CESSATION OF TRIAL PARTICIPATION

If a patient explicitly withdraws consent to have any further trial data recorded their decision must be respected and CTU must be informed in writing. All communication surrounding the early cessation of trial participation should be noted in the patient's records. Please note data for the patient prior to this decision will still be required.

In the majority of cases, patients continue to give permission for their data and information on their health to continue to be collected via clinical notes and national registries. Any information on the follow-up status, however minimal, would be helpful. Investigators are encouraged to facilitate

ongoing collection of follow-up data for example, through considering telephone consultations (see [Section 7.3.1](#)).

Early cessation of trial participation should not be undertaken lightly and the site must consider the implications for the trial and the patient in reaching such a decision. Without long-term data, the efficacy of trial treatments would be less reliable and could lead to inconclusive results. The early stopping of trial treatment should not lead to the early cessation of trial participation and in such cases follow-up assessments should be continued as per trial protocol.

Patients can change their minds about withdrawal at any time and re-consent to participate in the trial. Follow-Up data should be collected only from the point of when consent was re-instated.

## 9 STATISTICAL CONSIDERATIONS

### 9.1 METHOD OF RANDOMISATION

Patients will be randomised centrally using a computerised algorithm developed and maintained by CTU. Randomisation will be performed using the method of minimisation over a number of clinically important stratification factors with an additional random element. To decrease determinability, the factors are not listed here but can be found in the Statistical Analysis Plan.

Protocol version 16.0 introduced a new allocation, Arm L: transdermal oestradiol. Allocation to Arm K remains available only to non-diabetic patients with no contraindication to metformin.

See [Appendix H](#) for the allocation weighting of each arm by previous protocol version; this also shows allocation weighting for previously closed research arms.

### 9.2 OUTCOME MEASURES

The overall, definitive primary outcome measure for each comparison in the trial is overall survival (all-cause mortality), unless otherwise stated. The design of the trial is such that it is important to have additional intermediate primary outcome measures to assess activity in each research arm as the trial progresses.

For comparisons involving research arms B to J the intermediate primary outcome measure is failure-free survival; this and other outcome measures are listed in [Table 21](#).

**Table 21: Trial Outcome Measures by Comparison Stage (Arms B-J)**

| COMPARISON STAGE     | PRIMARY OUTCOME MEASURE                  | SECONDARY OUTCOME MEASURES                                                                                                   |
|----------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Pilot phase          | Safety*                                  | Feasibility                                                                                                                  |
| Activity Stages (AS) | Failure-free survival (FFS) <sup>†</sup> | Overall survival (OS)<br>Toxicity<br>Symptomatic skeletal events (SSE)                                                       |
| Efficacy Stage (ES)  | Overall survival                         | Quality-of-life<br>Cost effectiveness<br>Failure-free survival <sup>†</sup><br>Toxicity<br>Symptomatic skeletal events (SSE) |

\*Based on toxicity

<sup>†</sup>Including biochemical failure (see Section 7.1.3 and Appendix E)

For the “metformin comparison” the intermediate and definitive primary outcome measure are the same, being overall survival; see [Table 24](#) for full details of all outcome measures for that comparison.

For the “transdermal oestradiol comparison”, overall survival and progression-free survival are the definitive co-primary outcome measures, and the intermediate primary outcome measure is

progression-free survival (PFS); see [Table 26](#). The rationale for choosing progression-free survival rather than failure-free survival as the outcome measure for this comparison is outlined in [Section 9.8.2](#).

The reasons for different emphases in each recruitment stage are explained in [Section 9.3](#).

### 9.3 SAMPLE SIZE: PRINCIPLES

The design is a multi-arm multi-stage, multi-centre, platform, randomised controlled trial. There are a number of stages for each research arm: a Pilot/Feasibility/Safety Phase, Activity Stages and a final Efficacy Stage. Full details of the methodology underlying the trial design are given by Royston et al. (54, 55) The original sample size calculations were performed using the stage2 (version 1.2.0, Mar-2002) and stagen (version 1.1.1, May-2004) programs, both implemented in Stata (Stata Corp, TX) and updated using the later nstage program (version 1.0.3, Jun-2007; version 2.1.0, Jun-2009; version 3.0.1, Sep-2014). (56)

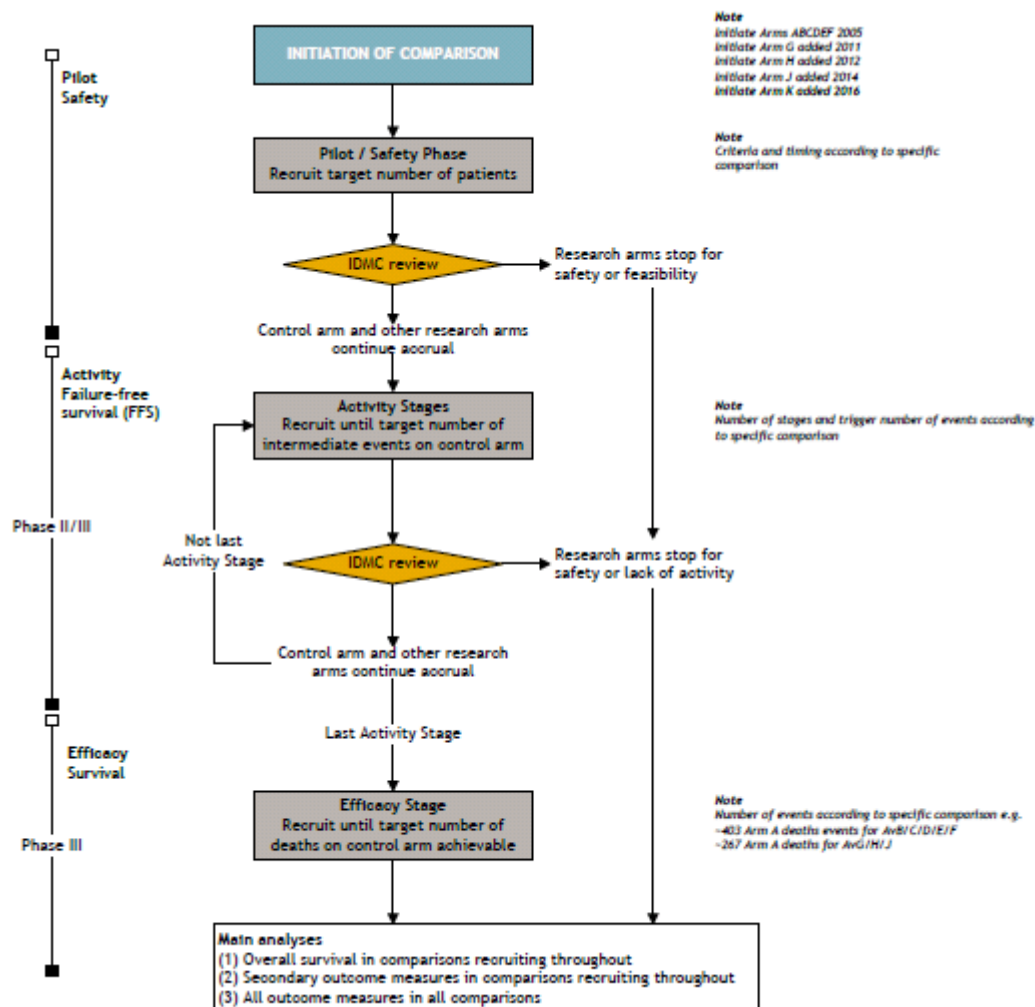
For each of the comparisons, other than transdermal oestradiol, we have adequately powered each comparison to detect an appropriate improvement in overall survival at the final Efficacy Stage, with high power at each of the planned interim Activity Stages on the intermediate primary outcome. For example, in a cohort with 2 years median FFS and 4 years median survival a target HR of 0.75 for research arm relative to control would translate into an absolute improvement in FFS of 10%, from approximately 50% to 60% at two years and OS of 10%, from approximately 50% to 60% at four years.

The “transdermal oestradiol comparison” is powered only for contributing to a meta-analysis of patients from the STAMPEDE “transdermal oestradiol comparison” and the PATCH trial. It will assess non-inferiority of transdermal oestradiol in terms of overall and progression-free survival which are co-primary outcome measures.

As each comparison is powered to detect a difference in relative improvement, the analyses will be performed when the pre-planned number of events has been reported in the control arm, rather than after a certain number of patients have been recruited to the comparison or a certain amount of time has elapsed. Further details of the sample size calculations and varying assumptions for each research comparison are summarised in the relevant [Sections 9.4-9.8](#) and detailed in a separate Statistical Design Document which are available on request.

As with all trials, changes in both the standard-of-care and second-line therapies over time are possible which improve outcomes and thus will affect the observed control arm event rates and associated reporting timelines. In particular, from Protocol version 8.0, standard-of-care RT was mandated for all patients with N0 M0 disease and no RT contraindication (this is likely to improve outcomes for this subgroup) and docetaxel permitted from Protocol version 14.0. Further agents are starting to be licensed for patients with castrate-refractory disease which may also improve survival rates. Improved FFS rates would delay the intermediate analyses, for comparisons where FFS is the intermediate primary outcome measure; whilst improved survival rates would delay the definitive analyses. Similarly, improved PFS rates could delay both the time of intermediate and definitive analysis for the “transdermal oestradiol comparison”. For each comparison event rates are estimated based on data which are publicly available at the time of design. The Statistical Design Document for arms A-K includes models where median survival is varied around such estimated rates.

Figure 4: Schema of progress of STAMPEDE through the trial\*



Key  
FFS: Failure-free survival  
HR: Hazard ratio  
IDMC: Independent Data Monitoring Committee  
Pts: Patients

Notes  
Exact accrual depends on many factors including  
accrual rate, event rate and arms recruiting in  
each stage

\* Except for the “transdermal oestradiol comparison”



## 9.4 SAMPLE SIZE ISSUES AND TRIAL STAGES: ADDITIONAL RESEARCH ARM G

This is the “abiraterone comparison” and includes patients allocated to research Arm G (SOC+abiraterone) and patients contemporaneously allocated to the control Arm A (SOC).

### 9.4.1 Pilot Phase: Additional Research Arm G

The IDMC reviewed safety data, in the context of data from the control arm, when the first 30 patients allocated to Arm G had been on trial for at least 18 weeks.

Furthermore, an additional review of safety was performed when 30 patients with newly-diagnosed non-metastatic disease, allocated to Arm G, had been on trial for at least 18 weeks. Both of these milestones were successfully completed.

### 9.4.2 Activity Stages I-III: Additional Research Arm G

The same principles were applied to this new comparison as to the original research comparisons. The notable difference was in the accrual rate to this comparison which was anticipated to be higher. There were two reasons for this. First, STAMPEDE initially started recruitment slowly in only a limited number of pilot sites. As more sites have been activated, including internationally, accrual has increased. At the time of adding Arm G (Protocol version 8.0), monthly accrual to the trial was averaging around 60 patients/month (over 700 patients/year). Second, there was an equal allocation ratio for the abiraterone arm compared to the control arm. It was this different allocation ratio which meant that the number of control arm events required to trigger the intermediate analyses was lower for the assessment of abiraterone than the assessment of the original research arms. This is shown in the table below.

**Table 22: Guidelines for stopping accrual to additional research Arms G and H**

| ACTIVITY STAGE | SIG LEVEL | POWER | TARGETED HR | NUMBER OF CONTROL ARM EVENTS | CONSIDER DISCONTINUATION IF HR(OBSERVED) IS... |
|----------------|-----------|-------|-------------|------------------------------|------------------------------------------------|
| I              | 0.50      | 95%   | 0.75        | ~75                          | >1.00                                          |
| II             | 0.25      | 95%   | 0.75        | ~142                         | >0.92                                          |
| III            | 0.10      | 95%   | 0.75        | ~221                         | >0.89                                          |

### 9.4.3 Efficacy Stage IV: Additional Research Arm G

The analysis of Efficacy Stage IV for this comparison will be performed when around 267 deaths have been observed in the control arm. This will give 90% power to detect the targeted hazard ratio of 0.75 at a one-sided significance level of 0.025.

### 9.4.4 Sample Size For Additional Research Arm G

Up to around 1,800 patients were targeted join the abiraterone comparison, with half allocated to the research arm G; the observed allocation was 1,917. Consideration was given to ceasing further randomisations to Arm G if it was not showing sufficient evidence of activity at the interim analyses.

The original plan intended for accrual to be halted either when 1,500 patients had been recruited or after 3 years, whichever was the sooner, providing the accrual rate remained above 50 pts/m.

The total number of patients joining this comparison depended not just on observed accrual and event rates, but also the length of time that the original research arms co-recruited alongside this additional research arm; it was originally assumed that this would be for approximately 1 year, but it was closer to 1.5 years. The sample size calculations and projected durations are fairly robust to changes in the length of co-recruitment with the original research arms and future co-recruitment with any further research arms which the Trial Management Group may introduce. Many scenarios are detailed in the Statistical Design Document.

In Protocol version 11.0 in Sep-2013, the target sample size for the "abiraterone comparison" was increased from around 1,500 patients to around 1,800 patients, with note that the efficacy analysis remains unchanged and is still to be triggered by around 267 control arm deaths. This increase in sample size was primarily because of an increase in the proportion of non-metastatic patients joining the comparison; this related to the activation of Arm H which only recruits patients with newly-diagnosed metastatic disease and thereby reduces the numbers of metastatic patients randomised to the "abiraterone comparison". Non-metastatic patients have a lower event rate than the metastatic patients and maintaining the same overall sample size would lead to a delay in time to the primary analysis. The increase in sample size was achievable because recruitment rates to the trial had been substantially higher than the anticipated 50 patients/month for the 6 months preceding the increase.

## 9.5 SAMPLE SIZE ISSUES AND TRIAL STAGES: ADDITIONAL RESEARCH ARM H

This is the "M1|RT comparison" and includes patients allocated to research Arm H (SOC+RT) and newly-diagnosed M1 patients with no contraindication to RT allocated to the control Arm A whilst Arm H was open to recruitment. Suitability for allocation to the comparison was assessed before randomisation to ensure comparability with contemporaneous control arm patients.

### 9.5.1 Pilot Phase: Additional Research Arm H

The IDMC reviewed safety data, in the context of data from the control arm, when the first 30 patients allocated to Arm H had been on trial for around six months.

### 9.5.2 Activity Stages I-III: Additional Research Arm H

The same principles were applied to this new comparison as to previous comparisons and an equal allocation ratio of control arm patients to patients allocated to Arm H was employed; as for Arm G. The number of control arm events required to trigger the intermediate analyses are the same as for the "abiraterone comparison" (see [Table 22](#)).

### 9.5.3 Efficacy Stage IV: Additional Research Arm H

The analysis of Efficacy Stage IV for this comparison will be performed when around 267 deaths have been observed in the relevant control arm patients. This will give 90% power to detect the targeted hazard ratio of 0.75 at one-sided significance level of 0.025.

### 9.5.4 Sample Size For Additional Research Arm H

Consideration was given to ceasing further randomisations to Arm H if it was not showing sufficient evidence of activity on the intermediate primary outcome measure (FFS), just as for the other research arms. This research comparison is relevant to around 60% of patients joining STAMPEDE. At the point of the scientific approval, accrual was averaging around 80 patients per month to the trial. If accrual to the trial was slower at 70 patients per month, then accrual to this comparison could be

between 18 and 42 patients per month, depending on which other trial arms are open to recruitment at the time.

We are targeting a 25% relative improvement in overall survival following local radiotherapy to the prostate in this patient group. This is the same size of effect targeted with the other research arms in STAMPEDE. This relative improvement can be further justified in the light of MRC PR07 which demonstrated an improvement of this magnitude for adding radiotherapy to ADT in locally-advanced disease, with a hazard ratio for overall survival of 0.77 (95% CI 0.61 to 0.98). In that trial, fewer than half of the deaths were from prostate cancer, whereas in newly-diagnosed metastatic patients nearly all men will die of their disease. Therefore, it is relevant to note the relative benefit of radiotherapy in PR07 in terms of prostate cancer-specific survival, where the hazard ratio was 0.46 (95% CI 0.34 to 0.61) after a median follow-up time of 8 years (57).

We anticipated that around 1250 patients were required over 4 years to observe 267 control arm deaths after 5.25 years. This assumed that (i) recruitment was constantly 70 pts/m to the trial overall; (ii) the original research arms stopped accrual within 6 months after activation of the RT arm; (iii) the abiraterone arm stops accrual around 24 months after activation of the RT arm; and (iv) a further new research arm with an equal allocation ratio was introduced 18 months after activation of the RT arm. In Protocol version 13.0, we reflected on these four points: (i) recruitment to the trial has been faster; (ii) the original research arms completed accrual 2 months after activation of the RT arm; (iii) the abiraterone arm stopped accrual 12 months after activation of the RT arm; and (iv) Arm J was activated 18 months after activation of the RT arm, Arm H.

Of patients joining STAMPEDE during this time, 60% have been eligible for the “M1|RT comparison”. Prior to randomisation, a RT schedule must be nominated: Weekly or Daily. We have observed that around half of patients in the comparison are nominated for RT with the Daily schedule and half for the Weekly schedule, primarily chosen by trial site with patient groups nominated for each schedule observed to be comparable at baseline. There will likely be interest to know the effect of each RT schedule when the main results are reported. This will be explored by “within schedule” comparisons of patients randomised to research vs control (arms H vs A) within each nominated RT schedule.

Therefore, in Protocol version 13.0, the target sample size was increased from 1,250 patients up to around 1,800 patients, resulting in an approximate increase in the split by planned RT schedule from 625 to 900 in each “within schedule” analysis. A FFS analysis “within schedule” will be carried out at the time of the “main analysis”; predicted to have ~300 control arm FFS events by schedule (FFS “within schedule” analysis parameters: target HR=0.75, power 90%, 1-sided  $\alpha=0.015$ ). For either of the RT schedules showing evidence of an effect on FFS, a comparative “within schedule” analysis will be carried out on survival when ~199 control arm deaths are observed in that schedule comparison. This is a closed test with OS only formally compared within schedule if there is an advantage in FFS for that RT schedule at the main analysis. Thus, extending recruitment enables a secondary analysis of the impact of RT on survival by planned “RT schedule” to happen within around 18 months from the first main analysis.

All sample scenarios are documented in the Trial Master File.

All patients joining the trial will be starting long-term ADT for the first time. The focus of this comparison will be on the newly-diagnosed, metastatic patients (with no contraindications to RT), which is the largest subgroup of patients in the trial and the group of patients at highest risk of death from prostate cancer. Patients with non-metastatic disease will be excluded from this particular comparison as there are already randomised data demonstrating the survival benefit from

radiotherapy in patients with locally-advanced disease. Radiotherapy is now mandatory in node negative patients; it is also recommended in the node-positive, non-metastatic (N+ M0) group. Relapsing patients are also excluded from this comparison.

For the control arm of the whole trial, we constructed sample size scenarios median failure-free survival being 18, 24 or 30 months and constructed sample size scenarios around each of these options; the event rate would depend on the patient mix. We now know that around 60% of patients have M1 disease at trial entry and we have reported that FFS at 24 months is 51% across the whole of the control arm.(58)

For the updated sample size calculation for this comparison, we based our estimates on the subgroup of patients with newly-diagnosed M1 disease in the control arm. Therefore, we estimate median FFS to be 1 year and estimate that median overall survival will be around 3.5 years.

## 9.6 SAMPLE SIZE ISSUES AND TRIAL STAGES: ADDITIONAL RESEARCH ARM J

This is the “enzalutamide + abiraterone comparison” and included patients allocated to research Arm J (SOC + enzalutamide + abiraterone) and patients contemporaneously allocated to the control Arm A.

### 9.6.1 Pilot Phase: Additional Research Arm J

The IDMC first reviewed safety data for this combination when the first 50 patients allocated to Arm J had been on trial around 6 weeks (i.e. to the first follow-up visit).

Furthermore, an additional review of safety was performed when these 50 Arm J patients had been on trial for around 6 months. Safety is routinely reviewed at regular intervals and additional safety reviews will be performed if the IDMC raises any concerns.

Direct comparison will be available with contemporaneously randomised patients on Arm A (hormones alone). Contextual data will be provided from Arm G (hormones plus abiraterone). Indicative safety data may also be available on the combination from other studies in CRPC.

### 9.6.2 Activity Stages I-II: Additional Research Arm J

The principles of intermediate analyses were applied to this new comparison as to previous comparisons, but some of the details were different, and an equal allocation ratio of control arm patients to patients allocated to Arm J was employed; as for Arms G and H. Owing to the expected accrual rate to the trial (>100 pts/m) and the expected slower event rate, only two activity stages were planned before accrual completed. These are set out in [Table 23](#).

**Table 23: Guidelines for stopping accrual to the additional research Arm J**

| ACTIVITY STAGE | SIG LEVEL | POWER | TARGETED HR | NUMBER OF CONTROL ARM EVENTS | CONSIDER DISCONTINUATION IF HRJ(OBSERVED) IS... |
|----------------|-----------|-------|-------------|------------------------------|-------------------------------------------------|
| I              | 0.40      | 95%   | 0.70        | ~66                          | >0.957                                          |
| II             | 0.12      | 95%   | 0.70        | ~139                         | >0.869                                          |

### 9.6.3 Efficacy Stage III: Additional Research Arm J

The analysis of the final Efficacy Stage for this comparison will be performed when around 267 deaths have been observed in the control arm. This would give 90% power to detect the targeted hazard ratio of 0.75 at a one-sided significance level of 0.025.

### 9.6.4 Sample Size For Additional Research Arm J

Consideration was given to ceasing further randomisations to Arm J if it was not showing sufficient evidence of activity on the intermediate primary outcome measure (FFS), just as for the other research arms.

The patient mix for this comparison is likely to represent a more favourable prognosis on average than in the original research trial's other arms, due to concurrent recruitment of M1 but not M0 patients, to Arm H.

We anticipate that around 1,800 patients are required within 3.5 years to observe ~267 control arm deaths within 6 years. This time will be dependent on the observed overall survival. The default scenario assumes that (i) recruitment is constantly 70pts/m to the trial overall, (ii) the M1|RT arm H accrues throughout and (iii) a further new research arm with an equal allocation ratio is introduced 18 months after activation of Arm J. The stopping date for Arm G is no longer an assumption.

Variations on these factors are documented in a Statistical Design Document. If accrual rates to the trial are at 150pts/m (as observed during Summer 2013), accrual of around 1,800 patients to the comparison could be achieved within 2 years. These sample scenarios will also be documented in the Trial Master File.

Updating the standard-of-care to include docetaxel has minimal impact on the projected time to maturity of the "enzalutamide + abiraterone comparison".

### 9.6.5 Further Sample Size Issues For Additional Research Arm J

Careful consideration will be given to the implications of any emerging data from the "abiraterone comparison". This has no effect on recruitment to the "enzalutamide + abiraterone comparison" because the recruitment target was reached before any data are available from the "abiraterone comparison".

Indirect comparisons to understand the contribution from each agent may be possible if this research arm is demonstrably superior to the standard-of-care. These plans will be developed and documented elsewhere, but a higher number of patients will help with the power to the indirect comparison.

## 9.7 SAMPLE SIZE ISSUES AND TRIAL STAGES: ADDITIONAL RESEARCH ARM K

This is the "metformin comparison" and includes patients allocated to research Arm K (SOC + metformin) and the equivalent non-diabetic patients with no contraindication to metformin contemporaneously allocated to the control Arm A whilst Arm K is open to recruitment. Suitability for allocation to the comparison is assessed before randomisation to ensure comparability with contemporaneous control arm patients

### 9.7.1 Implementation: Additional Research Arm K

The implementation of the MAMS principles are different in this comparison for the following reasons:

- Although all non-diabetic patients will be eligible for allocation to the “metformin comparison”, the timing of the analyses will be driven only by the M1 patients.
- Failure-free survival will not be used as the intermediate primary outcome measure; overall survival will be used as both the intermediate and definitive primary outcome measure. This is because we are not convinced that any comment on metformin’s usefulness should be determined from an ability to act on a PSA-driven outcome measure. Furthermore, treatment with metformin is intended to continue throughout long-term hormone therapy which may include going well beyond an FFS event, particularly in M1 patients.
- The target HR is 0.80 for overall survival (a 20% relative improvement). This is a smaller relative improvement in survival than targeted for previous comparisons because of metformin’s known low toxicity profile, the low cost of the drug and the potential positive effects on metabolic parameters and morbidity; a smaller impact may still have clinical benefit.

### 9.7.2 Outcome Measures: Additional Research Arm K

**Table 24** lists the outcome measures for this comparison and can be compared with the outcome measures for the other comparisons in **Table 21**.

**Table 24: Trial outcome measures by stage for the “metformin comparison”**

| COMPARISON STAGE       | PRIMARY OUTCOME MEASURES | SECONDARY OUTCOME MEASURES                                                                                                                                                                                                                        |
|------------------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pilot phase            | Safety*                  | Feasibility<br>Metabolic effects including:<br>:: Changes in BMI<br>:: Changes in haemoglobin A1c (HbA1c)<br>:: Changes in waist circumference<br>:: New diagnosis of diabetes mellitus<br>:: Cardiovascular event: major adverse cardiac events‡ |
| Activity Stage (AS) I  | Overall survival         | Metabolic effects<br>Toxicity<br>Symptomatic skeletal events (SSE)<br>Failure-free survival† (FFS)                                                                                                                                                |
| Efficacy Stage (ES) II | Overall survival         | Metabolic effects<br>Toxicity<br>Symptomatic skeletal events (SSE)<br>Failure-free survival† (FFS)<br>Quality-of-life<br>Cost effectiveness<br>Correlative outcomes <sup>¶</sup>                                                                  |

\*Based on toxicity

‡MACE; nonfatal MI, nonfatal stroke, & death from CV causes

†Including biochemical failure (see Section 6.1.2 and Appendix J)

¶Plasma lipid and fasting triglyceride levels, fasting plasma glucose

Sarcopenia and/or radiological progression free survival (rPFS)

Plasma insulin

AMP Kinase

**Note:** All arms are unblinded so primary outcome measures for this comparison are objectively measured with caution to be taken around interpretation of more subjective secondary outcome measures such as symptomatic skeletal events

### 9.7.3 Pilot Phase: Additional Research Arm K

The IDMC will review safety data for this comparisons when the first 50 patients allocated to Arm K have been on trial around 12 months. Furthermore, analyses will be conducted on metabolic parameters (see [Table 24](#)). If there is harm observed in metabolic effects, or any serious concerns regarding the toxicity profile, recruitment would be stopped; there are no formal criteria to guide this.

Safety is routinely reviewed at regular intervals and additional safety reviews will be performed if the IDMC raises any concerns.

### 9.7.4 Activity Stage I: Additional Research Arm K

The principles of intermediate analyses will be applied to this new comparison as to previous comparisons, but some of the details will be different, and an equal allocation ratio of control arm patients to patients allocated to Arm K is employed; as for Arms G, H and J. Owing to the expected accrual rate to the trial overall (>100 pts/m) and the interim primary outcome being overall survival, only one intermediate activity stage is planned before accrual is completed; this is set out in the [Table 25](#).

Although analyses are triggered by events in M1 patients, they will include all patients in the “metformin comparison”; this will have high power. A separate subgroup analysis in M1 patients (conventionally-powered) and M0 patients (limited power) will then look at consistency of effect; there will be few deaths in M0 patients at this time. The IDMC recommendation will be based on the totality of the available data, including safety, metabolic and compliance data.

**Table 25: Guidelines for stopping accrual to the additional research Arm K**

| ACTIVITY STAGE | SIG LEVEL | POWER | TARGETED HR | NUMBER OF CONTROL ARM EVENTS | CONSIDER DISCONTINUATION IF $HR_K$ (OBSERVED) IS... |
|----------------|-----------|-------|-------------|------------------------------|-----------------------------------------------------|
| I              | 0.40      | 90%   | 0.80        | ~104 M1 deaths               | >0.965                                              |

### 9.7.5 Efficacy Stage II: Additional Research Arm K

The analysis of the final Efficacy Stage for this comparison will be performed when around 374 deaths have been observed for M1 patients randomised contemporaneously to the control arm. This would give 85% power to detect the targeted hazard ratio of 0.80 at a one-sided significance level of 0.025.

As with the intermediate activity, this analysis will include all patients in the comparison, with a separate subgroup analysis in M1 and M0 patients looking at consistency of effect. At this time point we predict <60 control arm M0 deaths will be observed. Further subgroup analyses, defined by the stratification factors, are planned to check for consistency of effect at intermediate and final analyses.

### 9.7.6 Sample Size For Additional Research Arm K

Consideration will be given to ceasing further randomisations to Arm K if it is not showing sufficient evidence of improvement on overall survival at the intermediate analysis.

We anticipate that around 1,800 patients, including around 1,100 M1 patients, are required over 3 years to observe ~374 control arm M1 deaths over around 8 years. This number and time will be dependent on the observed overall survival. The default scenario assumes (i) recruitment is constantly 100pts/m to the trial overall, (ii) co-recruitment throughout of the equivalent of one other research arm, and (iii) the majority of metastatic patients will also have docetaxel but non-metastatic patients will not.

Variations on these factors are documented in a Statistical Design Document. If accrual rates to the trial are at 150pts/m (as observed during summer 2013), accrual of around 1,800 patients to the comparison could be achieved within 2 years. These sample scenarios will also be documented in the Trial Master File.

Updating the standard-of-care to permit first-line use of docetaxel was assumed within the sample size scenarios and is reflected in the projected time to maturity of the “metformin comparison”.

### 9.7.7 Further Sample Size Issues For Additional Research Arm K

Careful consideration will be given to the emerging data from the “abiraterone comparison” when this reports in 2017.

Analyses for the “metformin comparison” will be timed from randomisation. The point of randomisation compared to the start of hormone therapy may differ, depending on the planned use of docetaxel. This practical information will be reviewed by the TMG and IDMC.

## 9.8 SAMPLE SIZE ISSUES AND TRIAL STAGES: ADDITIONAL RESEARCH ARM L

This is the “transdermal oestradiol comparison” and includes patients allocated to research Arm L (transdermal oestradiol  $\pm$  RT  $\pm$  docetaxel) and the equivalent, eligible patients contemporaneously allocated to the control Arm A (SOC).

The phase III evaluation of the clinical efficacy of transdermal oestradiol will ultimately be based on the relevant data from this comparison within STAMPEDE and the PATCH trial, combined using an individual patient data meta-analysis. The overall evaluation is based on a non-inferiority design.

### 9.8.1 Implementation And Outcome Measures: Additional Research Arm L

The transdermal oestradiol evaluation is based on the following approach.

#### 9.8.1.A Earlier Stages In The PATCH Trial

- The early stages of the PATCH trial already demonstrated the safety and early activity of transdermal oestradiol in comparison to LHRH therapy (see [Appendix I](#))(43). The pilot phase (completed in 2010, n=254) showed the rates of cardiovascular events in the transdermal oestradiol and LHRH arms were similar, and the castration rates were equivalent. These results were confirmed by longer-term data including nearly 900 patients enrolled up to Oct-2015.
- A pre-planned, confidential interim analysis undertaken in Jun-2013, based on progression-free survival, at the end of the Phase II component of the PATCH trial, led the PATCH IDMC to recommend further recruitment for an extension to Phase III. That analysis included 638 patients with 206 PFS events, and reviewed data against a pre-specified non-inferiority margin hazard ratio of 1.25 with a 1-sided alpha 0.25.



### 9.8.1.B STAMPEDE And PATCH Meta-analysis

- To assess the clinical efficacy of transdermal oestradiol, the relevant data from the STAMPEDE “transdermal oestradiol comparison” will be combined with that data from all patients recruited into PATCH; the data from STAMPEDE will not be analysed alone.
- As the eligibility criteria with respect to the timing of start of ADT differs between the STAMPEDE “transdermal oestradiol comparison” and the PATCH trial (see [Section 4.4.2](#)), the “transdermal oestradiol comparison” will undergo an initial Pilot Phase to assess castration rates and safety among those patients on Arm L. This will also include a safety review of patients receiving transdermal oestradiol in combination with docetaxel. The data will be reviewed by the PATCH IDMC when there are 30 patients in Arm L who have been followed up for at least 18 weeks. A feasibility review will also be performed at the same time.
- There will be an additional safety review for transdermal oestradiol used in combination with docetaxel, based in the first instance on data from the PATCH trial. This analysis will primarily assess cardiovascular events, but will also review other toxicities including neutropenia. This will be carried out when there are 30 research patients from the PATCH trial who have received both transdermal oestradiol and docetaxel as part of their first-line treatment and been followed up for at least 12 weeks (expected date around May-2017). These results will be made available to the STAMPEDE TMG and TSC, pending approval by the PATCH IDMC, and will inform whether an additional safety review of patients on transdermal oestradiol with docetaxel is required within both STAMPEDE and PATCH.
- The pre-planned Activity Stage II, on intermediate primary outcome measure progression-free survival, will take place based on combined data from the STAMPEDE “transdermal oestradiol comparison” patients and PATCH patients.
- The same approach will be used at the final Efficacy Stage, with progression-free and overall survival as definitive co-primary outcome measures (see PATCH Protocol v10.0 for further details). The rationale for choosing progression-free survival as both the intermediate primary outcome measure and as part of the definitive co-primary outcome measure for the “transdermal oestradiol comparison” is outlined in [Section 9.8.3](#).

**Table 26** summarises the outcome measures for each stage of this research comparison. The target sample size for the meta-analysis of the “transdermal oestradiol comparison” is approximately 2,000 patients, with around 500 to be recruited through the STAMPEDE “transdermal oestradiol comparison”. By Feb-2017, around 1,200 patients had been recruited directly to the PATCH trial.

**Table 26: Trial outcome measures by stage for the “transdermal oestradiol comparison”**

| COMPARISON STAGE                     | DATA SOURCE(S)            | PRIMARY OUTCOME MEASURES                       | SECONDARY OUTCOME MEASURES                                                                |
|--------------------------------------|---------------------------|------------------------------------------------|-------------------------------------------------------------------------------------------|
| Pilot phase<br>(completed 2010)      | PATCH trial               | Cardiovascular morbidity and mortality         | Castration rates<br>Other toxicities<br>Metabolic effects                                 |
| Activity Stage I<br>(completed 2013) | PATCH trial               | Progression-Free Survival*                     | Cardiovascular and other toxicities<br>Castration rates<br>Metabolic effects              |
| Activity Stage II <sup>‡</sup>       | PATCH and STAMPEDE trials | Progression-Free Survival*                     | Cardiovascular & other toxicities                                                         |
| Efficacy Stage III <sup>§</sup>      | PATCH and STAMPEDE trials | Progression-Free Survival*<br>Overall survival | Cardiovascular & other toxicities<br>Prostate cancer specific survival<br>Quality-of-life |

\* Defined as the earliest among biochemical failure, clinical progression (local progression, lymph node progression, distant metastases), or death from any cause (see Section 9.8.3).

‡ In addition, there is Pilot Phase to assess castration rates and safety among Arm L patients within STAMPEDE, since the eligibility criteria with respect to timing of start of ADT differs between the transdermal oestradiol comparison within STAMPEDE and the PATCH trial (see Section 4.4.2).

§ The timing of these analyses is determined by when a pre-specified number of events for the primary outcome measure have been observed in the control arms for the PATCH and STAMPEDE trials combined. Please see the PATCH Protocol v10.0 for further details.

## 9.8.2 Additional Use of Outcome Data from the “transdermal oestradiol comparison”

Patients allocated to the “transdermal oestradiol comparison” may provide additional consent to participate in translational sub-studies, see [Section 4.6](#) for details. Subsequent correlative analysis using outcome data from these patients will be undertaken by the STAMPEDE team and collaborators, overseen by the STAMPEDE BRG and other STAMPEDE oversight committees.

## 9.8.3 Definition of PFS and Use As Co-primary OM: Additional Research Arm L

Note that the definition of progression-free survival (PFS) used within the “transdermal oestradiol comparison” analyses differs slightly to that of failure-free survival used for other research comparisons within STAMPEDE. This is because it includes death from any cause as an event- i.e. both PCa deaths and non-PCa deaths (see [Appendix D](#) for further details of the definition of progression). Progression-free survival is hence defined as time from randomisation to the first of: biochemical failure, clinical progression or death from any cause.

The use of PFS rather than FFS for the “transdermal oestradiol comparison” has no practical impact on STAMPEDE. The rationale for choosing PFS as part of the co-primary outcome measure for the “transdermal oestradiol comparison” is to capture any potential effects on survival due to the different toxicity profiles between transdermal oestradiol and LHRH.

Although PFS and survival are co-primary endpoints, their respective primary analyses will be triggered at different timepoints particularly because PFS is likely to contain a relatively low proportion of deaths as the contributing first PFS event.

## 9.9 FURTHER NOTES ON TRIAL DESIGN

### 9.9.1 Overall Sample Size

Given the adaptive nature of the study, there is no formal overall sample size target, but the numbers of patients required for each comparison are detailed in [Sections 9.4-9.8](#). To date, more than 8,000 patients have been recruited overall and at least 10,000 patients will join the trial.

### 9.9.2 Factorial Design

We note here that we have not employed a factorial design in the original design of this trial because we anticipate the possibility of synergy between SOC, zoledronic acid and docetaxel and between SOC, zoledronic acid and celecoxib.

It would not be possible to assess any such interactions reliably in a factorial trial (see the Statistical Design Document for further details).

## 9.10 INTERIM MONITORING AND ANALYSES

The accumulating data will be reviewed at regular intervals (approximately annually) by an Independent Data Monitoring Committee (IDMC), including pre-specified formal intermediate analyses of activity data (see also [Section 16](#)). These analyses will be performed by the trial team at CTU. Only patients randomised contemporaneously, and eligible for that comparison, will be included in the comparison of each research arm against control e.g. patients allocated to the control arm prior to Protocol version 12.0 will not contribute to the "enzalutamide + abiraterone comparison" (Arm A vs Arm J). For the "transdermal oestradiol comparison", the relevant STAMPEDE data will only be analysed as a meta-analysis in combination with the PATCH trial. Therefore, interim data from this comparison will be reviewed by the PATCH IDMC.

The IDMC will be asked to give advice on whether the accumulating data from the trial justifies continuing recruitment of further patients or further follow-up; guidelines for discontinuation of accrual for the relevant Activity Stages, together with results from any other relevant trials will aid them in this. A decision to discontinue recruitment, either in all patients or in selected subgroups, will be made only if the result is likely to convince a broad range of clinicians including those entering patients into the trial and the general clinical community. The intermediate stopping guidelines apply to the intermediate primary outcome measure.

To stop accrual early for benefit in any comparison would require convincing data in terms of the definitive primary outcome measure, overall survival. For example, this could be  $p < 0.001$  as proposed by Haybittle-Peto.<sup>(59, 60)</sup> The use of such a guideline for stopping for benefit has a minimal impact on the operating characteristics.

If a decision is made to continue without change, the IDMC will advise on the frequency of future reviews of the data on the basis of accrual and event rates. The IDMC will make recommendations to the Trial Steering Committee (TSC, see [Section 16](#)) as to whether the trial should continue in its present form. While the trial is ongoing the accumulating data will generally remain confidential, unless the TSC and IDMC agree that the data should be made public.

## 9.11 OUTLINE ANALYSIS PLAN

Analyses will be performed on an intention-to-treat basis. The standard unadjusted log-rank approach will be applied to analyses of intermediate and definitive primary outcome measures. The impact of potential confounders including the stratification factors used at randomisation will be considered in a Cox proportional hazard model. Flexible parametric models will be used to calculate the absolute differences between the arms to show treatment differences over time and to estimate restricted mean “survival” times (RMST). The estimated difference in restricted means survival time (RMST) will be used preferentially to compare treatment arms if the proportional hazards assumptions required for hazard ratios cannot be supported. The  $\chi^2$  test or Mann-Whitney test will be implemented for categorical data comparisons, including toxicity, as appropriate. Where relevant the primary outcome measure(s) (see [Section 9.2](#)) will be considered for all arms of the trial at each phase, but the main emphasis will be placed on the comparison of the research arms that have continued to recruit throughout the trial.

In the “transdermal oestradiol comparison,” a meta-analysis approach will be used to combine data from the STAMPEDE and PATCH trials. The analysis will also take into account the change in randomisation ratio partway through the PATCH trial (from 2:1 for transdermal oestradiol versus LHRH before Feb-2011, to 1:1 thereafter). In addition, as the comparison uses a non-inferiority design, sensitivity analyses will be conducted based on a number of pre-defined definitions for the per-protocol population.

### 9.11.1 Pilot / Safety Phases

The Pilot Phase randomised patients between all the trial arms so that the results from these patients can be included in the main trial. Feasibility is considered in terms of acceptability of the trial randomisation and reported toxicities and adherence to trial medication. Centres participating in the Pilot Phase for the original research arms were required to keep an anonymised log of all patients assessed for trial eligibility (see Protocol version 2.0) so that the number of patients who did not participate in the study and the number of eligible patients who chose to not participate in the study could be summarised (reasons for non-participation were collected where the patients was willing). The anonymised logs are no longer needed for new research arms (since Protocol version 8.0).

For the patients who are randomised, we shall describe the incidence of expected and unexpected severe toxicities and adverse events/reactions (see [Section 11](#)) to decide whether to continue with research arms beyond the Pilot Phase.

### 9.11.2 Activity And Efficacy Stages

The approach to analysis of these stages is summarised within the sample size calculations (see earlier subsections of [Section 9](#)). Each research arm will be compared in a pairwise fashion against the contemporaneously recruited control arm.

Full details are available in the Statistical Analysis Plan. See [Figure 4](#) for an overview of the schema of progress.

## 10 MONITORING AND QUALITY ASSURANCE

### 10.1 MONITORING AT CTU

Data provided to the CTU will be checked for missing or unusual values (range checks) and consistency over time. If missing or questionable data are identified, staff at the CTU will request that the data be clarified. The exact procedures for data clarification and the amendment of CRFs will be described in the trial Data Management Plan and instructions will be sent to all STAMPEDE institutions as soon as they have been approved to participate in the trial. The CTU will also send reminders for any overdue data.

#### 10.1.1 Central monitoring of consent

Anonymised copies of the initial patient's consent form (including the additional research consent) and any subsequent re-consent should be sent to the STAMPEDE team at the CTU as soon as possible to enable central monitoring of consent. The dates and signatures should be visible on the copies sent to the CTU however the name of the patient in block capitals must be omitted. Any queries resulting after central monitoring will be redirected to sites for clarification.

The original un-anonymised consent forms should be kept at site in the ISF.

### 10.2 DIRECT ACCESS TO DATA

Collaborating institutions should be aware that direct access to patient data by CTU staff may be required for trial-related monitoring or audit. Patient consent for this will be obtained as part of the general trial consent process.

### 10.3 VISITS TO INVESTIGATOR SITES

A selection of institutions will be visited at least once during the course of the STAMPEDE trial. The CTU will give the responsible investigator adequate notice of the monitoring visit to allow adequate time, space and staff for these visits. The standard operating procedures (SOPs) for monitoring are available from the CTU.

After the monitoring visit the monitor will complete a site visit report. This report may be circulated to the TMT for comment. Once the TMT have reviewed the report and agreed on any recommendations the monitor will finalise the report and send a copy to the Principal Investigator (PI) at the site. A copy will be kept in the CTU STAMPEDE Trial Master File.

### 10.4 CONFIDENTIALITY

All information collected during the course of the research will be kept strictly confidential. In addition, all procedures for handling, processing, storage and destruction of data are compliant with the Data Protection Act 1998. No individual patients will be identified when results from the trial are published.

Patients will be asked for permission for information about their health status to be obtained from the Office of National Statistics (ONS) or via NHS Digital (formerly HSCIC) or similar or national equivalent by CTU, if necessary. In addition, patients will be asked for permission to inform their GP of their involvement in the STAMPEDE trial.

## 11 SAFETY REPORTING

The principles of GCP require that both investigators and sponsors follow specific procedures when reporting adverse events/reactions in clinical trials. These procedures are described in this section of the protocol and in [Section 7](#). Further information on the expected toxicities for the investigational medicinal products (IMPs) (LHRH analogues, docetaxel, zoledronic acid, abiraterone, enzalutamide, metformin and transdermal oestradiol) can be found in [Appendix C](#).

### 11.1 SAFETY REPORTING DEFINITIONS

The definitions of the EU Directive 2001/20/EC Article 2 based on the principles of GCP apply to this trial protocol. These definitions are given in [Table 27](#).

**Table 27: Event Terms and Definitions**

| TERM                                                                                                                   | DEFINITION                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adverse Event (AE)                                                                                                     | Any untoward medical occurrence in a patient or clinical trial patient to whom a medicinal product has been administered including occurrences which are not necessarily caused by or related to that product.                                                                                                                                                                                                                                                    |
| Adverse Reaction (AR)                                                                                                  | Any untoward and unintended response to an investigational medicinal product related to any dose administered.                                                                                                                                                                                                                                                                                                                                                    |
| Unexpected Adverse Reaction (UAR)                                                                                      | An adverse reaction, the nature or severity of which is not consistent with the information about the medicinal product in question set out in the summary of product characteristics (or Investigator brochure) for that product.                                                                                                                                                                                                                                |
| Serious Adverse Event (SAE) or Serious Adverse Reaction (SAR) or Suspected Unexpected Serious Adverse Reaction (SUSAR) | Respectively any adverse event, adverse reaction or unexpected adverse reaction that: <ul style="list-style-type: none"> <li>• results in death</li> <li>• is life-threatening*</li> <li>• requires hospitalisation or prolongation of existing hospitalisation**</li> <li>• results in persistent or significant disability or incapacity</li> <li>• consists of a congenital anomaly or birth defect</li> <li>• Other important medical condition***</li> </ul> |

#### Clarifications and Exceptions

\*The term 'life-threatening' in the definition of 'serious' refers to an event in which the patient was at risk of death at the time of the event; it does not refer to an event which hypothetically might have caused death if it were more severe.

\*\*Hospitalisation is defined as an inpatient admission, regardless of length of stay, even if the hospitalisation is a precautionary measure for continued observation. Hospitalisations for a pre-existing condition (including elective procedures that have not worsened) do not constitute an SAE.

\*\*\*Medical judgement should be exercised in deciding whether an AE/AR is serious in other situations. Important AE/ARs that are not immediately life-threatening or do not result in death or hospitalisation but may jeopardise the patient or may require intervention to prevent one of the other outcomes listed in the definition above, should also be considered serious.

STAMPEDE is an adaptive platform protocol in which research treatments are given in addition to standard-of-care (SOC) therapies, or as alternatives in the case of transdermal oestradiol.

### 11.1.1 Defining treatment for the purposes of safety reporting

- **Protocol research treatments** are the IMPs under investigation in STAMPEDE i.e. the additional or alternative treatments patients allocated to research arms (B-L) receive as part of the STAMPEDE protocol:
  - Arm B: zoledronic acid
  - Arm C: docetaxel
  - Arm D: celecoxib
  - Arm E: docetaxel + zoledronic acid
  - Arm F: zoledronic acid + celecoxib
  - Arm G: abiraterone
  - Arm J: abiraterone & enzalutamide
  - Arm K: metformin
  - Arm L: transdermal oestradiol

Note, the research treatment in arm H (prostate RT) is not an IMP, but safety reporting requirements to the CTU are the same.

- **Protocol SOC treatments** are standard forms of background treatment permitted as part of the STAMPEDE protocol
  - Licenced ADT (e.g. LHRH analogues) given in the setting of castrate-sensitive prostate cancer.
  - Docetaxel given in castrate-sensitive prostate cancer

Please note standard forms of ADT e.g. LHRHa given in the setting of CRPC is not considered protocol treatment

- **Non-protocol treatments** are all prostate cancer treatments given following disease progression in the management of CRPC. This may include a “protocol research treatment” which is provided to a participant following disease progression.

### 11.1.2 Safety data collection in STAMPEDE

Safety data is collected in three ways:

- **Adverse Events** (AEs) are collected systematically for all patients and recorded on the Toxicity CRF required at each follow-up and in the event that treatment is changed due to toxicity. Adverse events are collected whilst trial treatment is ongoing, please refer to the CRF completion guidance available on the STAMPEDE website for further details. The purpose of safety data collected in this way is to enable comparative analysis of toxicity between the control and research group in each comparison therefore all toxicity data needs to be captured on this form.

- **Serious Adverse Event (SAE)** are AEs that fulfil the definition of serious as detailed in [Table 27](#). SAEs are reported using the SAE CRF. There are two types of SAEs: these are defined based on their causal relationship to treatment. If the event is judged to be possibly, probably or definitely related to treatment, it is categorised as a Serious Adverse Reaction (SARs). If the event is judged unlikely or unrelated to treatment, it is classed as an unrelated SAE, see [Table 28](#). The purpose of expedited SAE data collection is to meet regulatory requirements and to enable central monitoring.
- **Other important medical conditions** including pregnancy occurring in a STAMPEDE patient's partner during the patient's participation in the trial. This must be reported to the CTU within the same timelines as an SAE and the outcome of a pregnancy should be followed up carefully and any abnormal outcome to the mother or child should be reported. Patients who develop any new primary carcinomas should have the event reported on a SAE CRF with the exception of non-melanoma skin cancer (e.g. basal cell carcinomas and squamous cell carcinomas) which do not require reporting.

## 11.2 SAFETY PROCESSES: EXEMPTIONS

### 11.2.1 Exemptions: Trial-specific SAE reporting exemptions

The following events which may fulfil the definition of "serious" are exempted from regulatory reporting and therefore do not require an SAE CRF to be completed. They may still require reporting as an AE on the toxicity form, or using an alternative CRF e.g. progression log as appropriate.

- **Serious adverse events** unrelated to STAMPEDE research treatment i.e. unrelated SAEs occurring more than 30 days after stopping STAMPEDE research treatment
- **Serious adverse events** unrelated to protocol SOC ADT i.e. unrelated SAEs occurring more than 30 days after the last exposure to ADT (please note this is assumed to be 30 days after the expiration date of a depot preparation)
- **Serious adverse events** unrelated to protocol treatment (research or SOC) occurring after disease progression, providing research treatment has stopped more than 30 days previously.
- **Non-fatal progression events:** events that fulfil the definition of serious e.g. result in hospital admission, but are due to disease progression are exempt from reporting as an SAE, instead details should be provided on the progression log.
- **Death as a result of disease progression** is also not considered to be a SAE. Do not complete an SAE CRF, instead details should be reported on the STAMPEDE Death Form.
- **Elective hospitalisation** and surgery for treatment of locally-advanced or metastatic prostate cancer or its complications. Instead, record this as a non-trial inpatient admission on the follow-up form under Non-Trial visits.
- **Elective hospitalisation** to simplify treatment or procedures. If related to prostate cancer, record until non-trial inpatient admission on the follow-up form, if unrelated e.g. pre-existing conditions that have not been exacerbated by trial treatment, do not report.



## 11.3 SITE INVESTIGATOR RESPONSIBILITIES

All toxicities experienced, both non-serious and serious, occurring whilst protocol treatment is ongoing and up to 30 days after discontinuation, should be recorded in the participants medical notes and on the Toxicity CRF linked to the Follow-Up CRF. The toxicity CRF should be sent to the CTU within one month of the corresponding Follow-Up CRF being due.

In addition, if an AE meeting the definition of serious (see [Table 27](#)) occurs then it is the responsibility of the site investigator to determine if it requires reporting to the CTU, see [Figure 5](#) for guidance on when it is necessary to complete a SAE CRF.

### 11.3.1 Investigator Assessment

#### 11.3.1.A Seriousness

When an adverse event occurs the investigator or delegates must first assess whether the event is serious using the definitions given in [Table 27](#). If the event is serious and does not meet any of the exemption criteria then it must be reported using the SAE CRF and submitted to the CTU; see [Section 11.2.1](#) for details on exemptions.

#### 11.3.1.B Grading severity of adverse event

The severity (i.e. intensity) of all AEs/ARs (serious and non-serious) in this trial should be graded using Common Terminology Criteria for Adverse Events (CTCAE) v4.0<sup>4</sup>.

The complete CTCAE v4.0 can be found at: [http://evs.nci.nih.gov/ftp1/CTCAE/CTCAE\\_4.03\\_2010-06-14\\_QuickReference\\_5x7.pdf](http://evs.nci.nih.gov/ftp1/CTCAE/CTCAE_4.03_2010-06-14_QuickReference_5x7.pdf). Any questions concerning this process should be directed to the CTU team in the first instance.

Please note, prior to September 2016 all events were graded according to CTCAE version 3.0. Please ensure the correct version of the CTCAE grading system is referred to and the correct corresponding version of the CRF.

#### 11.3.1.C Causality

The Investigator must assess the causality of all SAEs in relation to protocol treatment using the definitions in [Table 28](#). There are 5 categories: unrelated, unlikely, possibly, probably and definitely related. If the causality assessment is unrelated or unlikely to be related the event is classified as unrelated, therefore an unrelated SAE. If the causality is assessed as either possibly, probably or definitely related then the event is classified as related and therefore a SAR. The assessment of causality determines whether the event requires notification to meet regulatory requirements, see [Section 11.3.2.A](#) for details.

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<sup>4</sup> [ctep.cancer.gov/reporting/index.html](http://ctep.cancer.gov/reporting/index.html)

**Table 28: Assigning type of SAE through assessment of causality and expectedness**

| CAUSAL<br>RELATIONSHIP | DESCRIPTION                                                                                                                                                                                                                                                                                                    | EVENT TYPE                                                |                                                     |            |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------|------------|
|                        |                                                                                                                                                                                                                                                                                                                | N/A                                                       | EXPECTED                                            | UNEXPECTED |
| Unrelated              | There is no evidence of any causal relationship                                                                                                                                                                                                                                                                | Unrelated<br>SAE                                          | No assessment required as<br>unrelated to treatment |            |
| Unlikely               | There is little evidence to suggest there is a causal relationship (e.g. the event did not occur within a reasonable time after administration of the trial medication). There is another reasonable explanation for the event (e.g. the patient's clinical condition, other concomitant treatment).           | Unrelated<br>SAE                                          |                                                     |            |
| Possibly               | There is some evidence to suggest a causal relationship (e.g. because the event occurs within a reasonable time after administration of the trial medication). However, the influence of other factors may have contributed to the event (e.g. the patient's clinical condition, other concomitant treatments) | Must assess<br>expectedness<br>to categorise<br>reactions | SAR                                                 | SUSAR      |
| Probably               | There is evidence to suggest a causal relationship and the influence of other factors is unlikely.                                                                                                                                                                                                             |                                                           | SAR                                                 | SUSAR      |
| Definitely             | There is clear evidence to suggest a causal relationship and other possible contributing factors can be ruled out.                                                                                                                                                                                             |                                                           | SAR                                                 | SUSAR      |

#### 11.3.1.D Expectedness

If there is at least a possible involvement of the protocol research treatment the investigator must assess the expectedness of the event, i.e. expectedness must be assessed for all SARs. An unexpected adverse reaction is one not previously reported or one that is more frequent or more severe than previously reported in the reference safety information. The definition of an unexpected adverse reaction (UAR) is given in [Table 27](#). If a SAR is assessed as being unexpected, it becomes a SUSAR.

The reference safety information will either be the current Summary of Product Characteristics (SPC) or Investigator Brochure. The reference safety information for all protocol research treatment and common forms of SOC ADT are summarised in [Appendix C](#).

### 11.3.2 Notification responsibilities

STAMPEDE investigators are responsible for notifying the CTU of all serious events that do not fulfil any of the trial-specific exemptions (see [Section 11.2.1](#)). All unrelated SAE, SARs and SUSARs must be notified within 24 hours of the investigator being made aware of the event. The notification period differs according to whether the event is classified as an unrelated SAE, SAR or SUSAR. See [Table 28](#) for details on event classification.

#### 11.3.2.A Unrelated SAEs

All unrelated SAEs must be reported up until STAMPEDE protocol treatment has been discontinued for 30 days. In the case of LHRH analogue this is assumed to be 30 days after the depot expiration date e.g. up to 8 weeks after a 4-week depot or 16 weeks after the last administration of a 12-week depot. Providing research treatment has been stopped, unrelated SAEs are also exempt from reporting **after disease progression**, see [Section 11.2.1](#) for details on exemptions.

### 11.3.2.B SARs and SUSARs

All reactions judged by the investigator as possibly, probably or definitely related to protocol research or SOC treatment are reportable whilst follow-up continues.

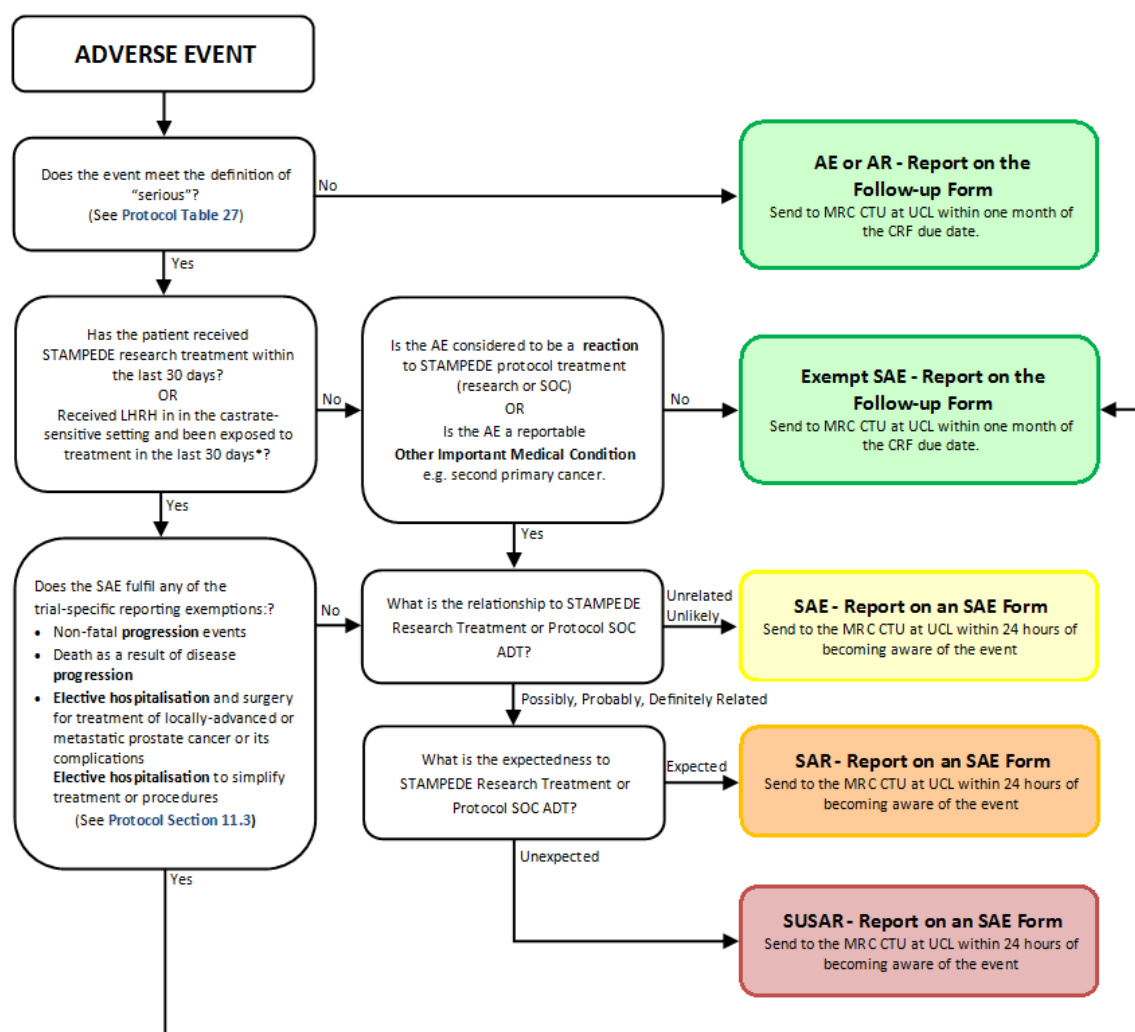
### 11.3.2.C Other important medical conditions

All other important medical conditions are reportable regardless of when they occur following randomisation whilst follow-up continues.

It should be noted that docetaxel, abiraterone and enzalutamide may be given as non-trial treatments in the management of CRPC. It is not necessary to report safety data to the STAMPEDE trial team relating to this non-trial use, instead the yellow card system should be used to notify the regulatory authorities of adverse drug reactions if appropriate (<https://yellowcard.mhra.gov.uk/>)

See **Figure 5** which provides an overview of when SAE, SAR and SUSARs are required to be reported to the STAMPEDE Trial team.

**Figure 5: SAE reporting flowchart**



*\*Exposure to LHRHa is assumed to be until the depot expiration date, therefore unrelated SAEs are reportable up until 8 weeks after the administration of a 4-week depot or 16 weeks after the administration of a 12-week depot.*

### Box 1: SAE report notification checklist

Before sending the SAE CRF please check that the event does not meet any of the exemption criteria, see [Section 11.2.1](#). Once confirmed, please ensure that the information provided meets **all** of the following minimum criteria required for initial processing and review:

1. At least **two** patient identifiers
2. Indication of why the event was **serious**
3. **Grade** severity of event/reaction according to CTCAE version 4.0
4. Assessment of **causality** in relation to research protocol treatment and SOC protocol treatment – *confirm that this is a reportable event*
5. Assessment of **expectedness if possibly related to treatment**. Please refer to list of expected toxicities in [Appendix C](#)
6. Provide the **date of last administration** for all trial treatments (minimum month/year) – *if reporting an unrelated SAE confirm that this is still reportable*
7. **Signature** (if not by a clinician, by a site trial team member in the first instance)

#### SAE REPORTING

**Fax** to 020 7670 4818 within 24 hours of becoming aware of the event  
Or send via **encrypted** email to [mrcctu.stampede@ucl.ac.uk](mailto:mrcctu.stampede@ucl.ac.uk)

### 11.3.3 Event Follow Up

Patients must be followed-up until clinical recovery is complete or stabilised. Follow-up should continue after completion of protocol treatment if necessary. Follow-Up information can be updated on the original SAE CRF by ticking the box marked “follow-up” and faxing to the CTU as information becomes available. Extra, annotated information and/or copies of test results may be provided separately. The patient must be identified by trial number, date of birth and initials only. The patient’s name should not be used on any correspondence.

## 11.4 CTU RESPONSIBILITIES

The STAMPEDE trial team will confirm receipt of the SAE report to the main point of contact via email. Please contact the STAMPEDE trial team if receipt is not received within 24 hours.

At least one medically qualified person at the CTU, or comparison chief-investigator or another appropriate TMG member will review all SAE reports received. The causality assessment given by the

local Investigator at the hospital cannot be overruled and in the case of disagreement, both opinions will be provided in any subsequent reports.

The CTU is undertaking the duties of trial sponsor and is responsible for the reporting of SUSARs and other SARs to the regulatory authorities (through the MHRA to competent authorities in other European member states) and the UK research ethics committees; and in addition has sponsor oversight for reporting in other countries in which the trial is taking place.

The CTU will also keep all investigators informed of any safety issues that arise during the course of the trial.

## 12 ETHICAL CONSIDERATIONS AND APPROVAL

### 12.1 ETHICAL CONSIDERATIONS

#### 12.1.1.A Randomisation

This is a randomised trial therefore neither the patients nor their physicians will be able to choose the patients' treatment. Treatment will be allocated randomly using a computer-based algorithm. This is to ensure that the groups of patients receiving each of the different treatments are as similar as possible.

All patients, with the exception of those allocated to transdermal oestradiol (Arm L), will receive standard hormone treatment. All patients, including those allocated to Arm L, may also receive other standard-of-care treatments which may include prostate radiotherapy and/or docetaxel. Use of radiotherapy and/or docetaxel will be unaffected by trial participation and is left to the discretion of the treating clinician and patient. Patients may be randomised to receive additional treatment (metformin) given with standard-of-care treatments, or an alternative form of hormone treatment (transdermal oestradiol). An even allocation ratio is being currently being used which means all eligible patients have an equal chance of being randomised to the control or research arms.

Through the introduction of a "transdermal oestradiol comparison" into the STAMPEDE trial platform, sufficient data will be collected to evaluate this treatment approach more rapidly. By undertaking a meta-analysis using data collected in both PATCH and STAMPEDE, fewer patients overall are allocated the control arm i.e. more patients gain access to novel treatments.

#### 12.1.1.B Evaluation of Novel Therapeutic Strategies

The newer treatment options are being assessed in a detailed and systematic fashion in this trial. There is some evidence to suggest that the newer treatment options may have advantages over standard treatment alone with regards to clinical outcome, but this is not confirmed and toxicity may be increased. This trial will follow a large group of men who have been randomly allocated to either the standard treatment(s) or the novel treatment strategies in order to measure the benefits of these approaches. The patients will also be followed-up for toxicity and safety issues, so that any benefits can be weighed against any negative aspects including the impact treatments have on other aspects of medical health e.g. cardiovascular disease, as well as quality-of-life and value for money (health economic analysis).

#### 12.1.1.C Additional Tests and Hospital Visits

Patients participating in the trial will have some additional hospital visits and some extra blood samples taken compared to patients who are not participating in the trial, with the amount varying according to the allocated treatment and stage of disease. Sometimes the blood samples can be taken when the patient is attending hospital for treatment anyway. On some of the trial arms, the patient may have to make additional visits to the hospital for the blood sample to be taken, although in some cases it may be possible for the blood sample to be taken in the GP's surgery.

#### 12.1.1.D Facilitating Participant Feedback From Investigations and Additional Analyses

For participants who choose to take part in additional sub-studies, biological samples including blood, saliva and remaining stored FFPE tumour samples will be used in research projects. These projects will enable the study of genetic factors and other biomarkers that can help identify individuals who serve to benefit most from the treatments tested in STAMPEDE and to further understand why and how treatment resistance develops. All samples will be link-anonymised and only made accessible to approved collaborators granted access by the STAMPEDE oversight

committees. We will make every effort to protect the confidentiality of this information and make sure personal identities are protected.

From Protocol version 16.0, patients may opt to receive feedback regarding genetic results that may arise from analyses of research analyses of genetic material extracted from any of the biological samples collected as part of the trial e.g. saliva, FFPE tumour blocks or circulating tumour DNA extracted from blood. Only results which are of established clinical relevance and for which testing would be available under standard NHS genetic testing guidelines will be fed back. Any genetic analysis undertaken as part of additional research associated with STAMPEDE does not replace clinically indicated investigations as only a proportion of STAMPEDE patient will undergo prospective testing and therefore it cannot be guaranteed that results will be fed back in a timely fashion.

This change has been made in response to emerging data that demonstrates a small proportion of men may have genetic faults in genes such as BRCA2. This has implications for both patients and potentially their biological relatives. For patients and their treating clinician, knowledge of this information may facilitate access into further clinical trials and may potentially impact on the choice of treatment following progression i.e. metastatic castrate-resistant prostate cancer.

Any patient who consents to receive feedback and in whom a known pathogenic mutation of clinical significance is detected on testing of research samples collected as part of STAMPEDE will be told of this. Patients will be recommended to undergo genetic counselling accessed via clinical genetics services and consider confirmatory testing. This is necessary to determine if the defect is germline (inherited) and ensures access to appropriate ongoing support. If confirmed as a germline (inherited) abnormality, this will enable biological relatives to also access appropriate genetic counselling and testing if they wish.

However, patients and STAMPEDE investigators are informed that any genetic analysis undertaken as part of additional research associated with STAMPEDE does not replace clinically indicated investigations as only a proportion of STAMPEDE patient will undergo prospective testing and therefore it cannot be guaranteed that results will be fed back in a timely fashion. Therefore participation in the additional research conducted on biological samples collected as part of STAMPEDE should not impact on a clinician's decision to recommend genetic screening.

The introduction of the "metformin comparison" means that all patients, not known to be diabetic, will be screened for diabetes prior to trial entry. This is to enable the effect of metformin to be studied in non-diabetic patients. All patients in whom screening bloods are abnormal will be referred for confirmatory tests and further management according to local guidelines e.g. via their GP. Screening is expected to lead to a small proportion of potential trial participants receiving a new diagnosis of diabetes but will ensure appropriate management of both conditions.

#### **12.1.1.E Considering the Impact of Emerging Data**

If new information emerges during the course of the trial which may affect the treatment or follow-up of patients who have joined the trial, information will be provided through by the trial team to all Principal Investigators. PIs therefore have the duty to inform the patients in their care of any new information emerging using any appropriate channel (e.g. letter, communication at follow-up clinic, etc.)

## 12.2 ETHICAL APPROVAL

The protocol has a Favourable Opinion from an appropriate Research Ethics Committee, according to national guidelines. Additionally, each site must also obtain management permission for research (Local R&D approval or equivalent) from the relevant host organisations before patients can be entered into the trial. The patient's informed consent to participate in the trial should be obtained after a full explanation has been given of the treatment options, including the conventional and generally accepted methods of treatment. Patient information sheets and patient consent forms are available on the STAMPEDE website ([www.stampedetrial.org](http://www.stampedetrial.org)).

The right of the patient to refuse to participate in the trial without giving reasons must be respected. After the patient has entered the trial, the clinician must remain free to give alternative treatment to that specified in the protocol, at any stage, if he feels it to be in the best interest of the patient. However, the reason for doing so should be recorded and the patient will remain within the trial for the purpose of follow-up and data analysis according to the treatment option to which he has been allocated. Similarly, the patient must remain free to withdraw at any time from the protocol treatment without giving reasons and without prejudicing his further treatment.

A statement of MRC policy on ethical considerations in clinical trials of cancer therapy, including the question of informed consent, is available from the MRC Head Office web site (<http://www.mrc.ac.uk>). In addition, the MRC and the Wellcome Trust framework on the feedback of health-related findings in research is readily available (<https://www.mrc.ac.uk/documents/pdf/mrc-wellcome-trust-framework-on-the-feedback-of-health-related-findings-in-researchpdf/>) and has been used when developing the trial specific processes.



## 13 REGULATORY APPROVAL

This trial has been approved in the UK by the MHRA and will be conducted under a CTA (Ref: 00316/0026/001-0001) in the UK.

The trial has been approved in Switzerland by Swissmedic (Ref: 2009 DR 3235).

## 14 INDEMNITY

University College London holds insurance against claims from participants for injury caused by their participation in this clinical trial. Participants may be able to claim compensation if they can prove that UCL has been negligent. However, as this clinical trial is being carried out in a hospital, the hospital continues to have a duty of care to the participant of the clinical trial. University College London does not accept liability for any breach in the hospital's duty of care, or any negligence on the part of hospital employees. This applies whether the hospital is an NHS Trust or otherwise.

Participants may also be able to claim compensation for injury caused by participation in this clinical trial without the need to prove negligence on the part of University College London or another party. Participants who sustain injury and wish to make a claim for compensation should do so in writing in the first instance to the Chief Investigator, who will pass the claim to the managing organisation's Insurers, via the managing organisation's office.

Hospitals selected to participate in this clinical trial must provide clinical negligence insurance cover for harm caused by their employees and a copy of the relevant insurance policy or summary can be provided on request.

## 15 FINANCE

STAMPEDE is funded by Cancer Research UK's Clinical Research Committee (formerly the Clinical Trials Advisory Awards Committee; CTAAC). It is also funded by the MRC through the MRC Clinical Trials Unit at UCL. The trial has National Institute for Health Research Clinical Research Network (NIHR CRN) approval and, therefore, local NCRN funds may be available at each centre to support entry of patients into this trial.

Funding arrangements for research arms now closed to recruitment can be found in [Protocol version 13.0](#)

**Standard therapies** including **ADT**, **prostate radiotherapy** and **docetaxel** will be administered as per routine clinical care using local NHS supplies.

**Abiraterone** is manufactured by Janssen Pharma PV (pharmaceutical companies of Johnson & Johnson). They have agreed to provide free drug and funds to distribute drug to participating sites and to help support the conduct and management of the trial.

**Enzalutamide** is manufactured by Astellas Pharma. They have agreed to provide free drug and funds to distribute drug to participating sites and to help support the conduct and management of the trial.

**Metformin** will be administered using local NHS supplies.

**Transdermal oestradiol** will be administered as Progynova TS 100 patches, manufactured by Bayer who have agreed to supply these patches at a trial-specific discounted price. All accredited STAMPEDE centres will be able to order Progynova patches through AAH Pharmaceuticals Ltd wholesalers at the discounted rate.

**Biomarker-Screening Pilot** will be funded by Clovis Oncology who will fund all sample analysis and help support the coordination of sample retrieval including site reimbursement through an educational grant to the MRC CTU at UCL.

## 16 TRIAL COMMITTEES

### 16.1 TRIAL MANAGEMENT GROUP (TMG)

A Trial Management Group (TMG) has been formed comprising the Chief Investigator, other co-investigators and members of the CTU. The membership of the TMG may be expanded if other groups of trialists wish to participate. It will also be amended during the trial if other circumstances require e.g. retirement.

The TMG will be responsible for the day-to-day running and management of the trial. They will meet by teleconference at least 3-monthly and in person as needed. The TMG members are detailed in [Appendix F](#).

Further details of TMG functioning are provided in the TMG charter (available on request).

### 16.2 TRIAL STEERING COMMITTEE (TSC)

A Trial Steering Committee (TSC) has been formed to provide overall supervision for the trial and provide advice through its independent chair. The ultimate decision for the continuation of the trial lies with the TSC. The TSC will meet regularly.

Further details of TSC functioning are provided in the TSC charter (available on request).

### 16.3 INDEPENDENT DATA MONITORING COMMITTEE (IDMC)

An Independent Data Monitoring Committee (IDMC) has been formed. The IDMC will be the only group who sees the confidential, accumulating data to the trial. Reports to the IDMC will be produced by the CTU. The IDMC will meet within 6 months of the trial opening with the frequency of meetings dictated by the IDMC. The IDMC will consider data in accordance with the analysis plan (see [Section 9](#)) and will be advisory to the TSC. The IDMC can recommend premature closure or reporting of the trial, or that recruitment to any research arm is discontinued.

From Protocol version 8.0 onwards, any recommendation from the IDMC to stop recruitment to one or more trial arms will be acted upon immediately, pending ratification from the TSC. As this period between meetings should be very short, sites would not be notified until after the TSC have made a decision. IDMC recommendations based on emerging safety issues will be discussed with sites promptly.

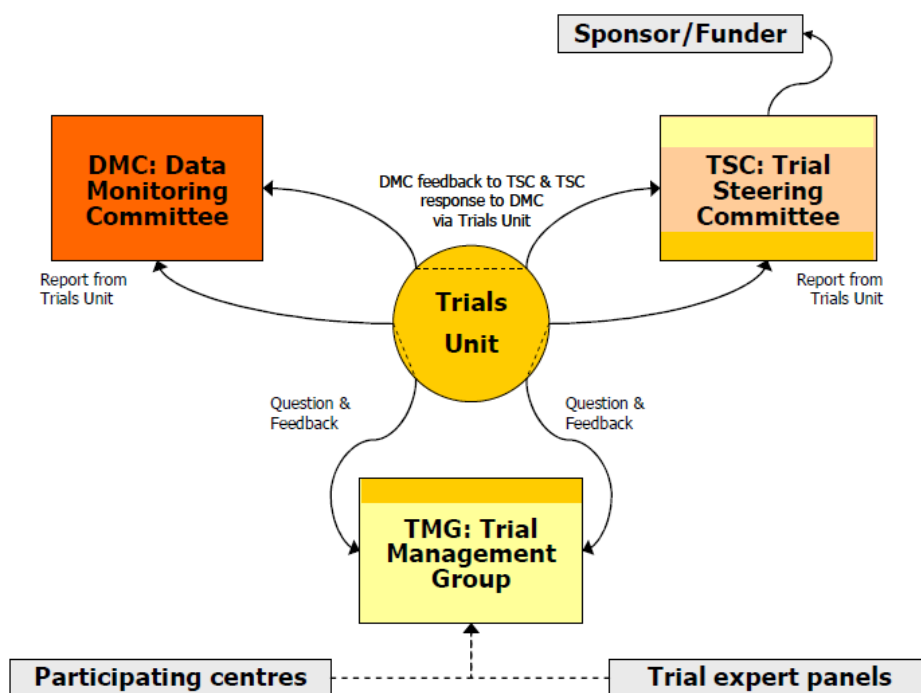
Further details of IDMC functioning and the procedures for interim analysis and monitoring are provided in the IDMC charter (available on request).

Data from the “transdermal oestradiol comparison” are viewed by the PATCH IDMC, in meta-analysis with PATCH, rather than by the STAMPEDE IDMC. Recommendations of any actions relating to STAMPEDE would be made to the STAMPEDE TSC.

## 16.4 TRIAL EXPERT PANELS

The trial has two established translational expert groups chaired by TMG members the Biological Research Group (BRG) and the Metabolic Translational Group (MTG). Both groups input and provide expert oversight of relevant translational aspects of the trial and associated sub-studies.

**Figure 6: Diagram of relationships between trial committees**



## 17 ANCILLARY STUDIES

### 17.1 QUALITY-OF-LIFE

A quality-of-life (QL) study is being performed to assess the impact of each treatment arm on the quality of patient's lives. Initial participation in this study was limited to the first 700 patients recruited (this was reached in Sep-2008) patients. The QL study re-opened from the implementation of Protocol version 8.0.

The EORTC QLQ-C30 with the prostate-specific module QLQ PR25 will be used. Key items for assessment are pain reduction for patients with metastatic disease and urinary symptoms for patients with locally-advanced disease. In addition specific hypotheses will be generated for each of the research arms. The EuroQol (EQ-5D) will be used in the study as a generic measure of health-related quality-of-life which can be linked to public preferences. These data will be used to calculate quality-adjusted life-years as part of the economic evaluation (see [Section 17.2](#)). **Patients recruited into the QL study, should continue to complete QL data for five years after randomisation or until progression, whichever is sooner.** Questionnaires should be self-administered, although it is recommended that a key person (e.g. research nurse) at each centre be responsible for the data collection to optimise compliance and completeness of the data.

The QL and the HE questionnaires should be completed by the patient without conferring with friends or relatives and all questions should be answered even if the patient feels them to be irrelevant.

The responsible person should check each questionnaire for its completeness, ensuring that the correct date of completion and patient identifiers are present. The research nurse should approach patients at appropriate clinical visits to complete a questionnaire. If no clinical visit is scheduled for the patient (with a window of 4 weeks around the expected date) the nurse should organise the completion of the questionnaire, by post or by a visit to the patient at home (or in a hospice).

### 17.2 HEALTH ECONOMICS

A health economics (HE) sub-study will be performed. Core resource use information will be collected, using CRFs on days in hospital (by speciality) and outpatient visits. Data collected on concomitant medication will also be used in the economic analysis. Information on patients' use of primary care and community-based services will be collected as additional questions in the QL questionnaire. Costs will be calculated on the basis of representative UK unit costs at the point of analysis. Health outcomes will be assessed in terms of quality-adjusted life years (QALYs). Quality adjustments will be based on patients' responses to the EQ-5D health status measure which will be administered at baseline and each point of follow-up as part of the QL questionnaire. A cost-effectiveness analysis will compare all regimens that continue to recruit into their final Efficacy Stage. For further details please refer to [Appendix G](#).

### 17.3 TRANSLATIONAL SUB-STUDIES

Samples obtained from consenting STAMPEDE participants are analysed as part of separate translational sub-studies. These are conducted through collaborations with other academic and

industry partners. All applications for collaboration and sample accesss are reviewed by the STAMPEDE oversight committees and are overseen by the STAMPEDE BRG. For further details of each ongoing substudy please see [Section 4.6](#). For details regarding sample collection please refer to the [Sample collection and handling manual](#) available via the website.

## **17.4 DISEASE VOLUMETRIC ANALYSIS SUB-STUDY**

Baseline imaging obtained from STAMPEDE participants are accessed and analysed as part of the trial data collection. Collection and analysis will be undertaken in collaboration with partners on the TMG, initially, in ordered to determine disease volume. For details partaking to retrospective imaging centralisation and image handling, please refer to the individual sub study Working practices available from CTU. All subsequent applications for collaboration and imaging access are reviewed by the STAMPEDE oversight committees following the usual processes.

## 18 PUBLICATIONS

The results from different centres will be analysed together and published as soon as possible. Individual clinicians must not publish data concerning their patients that are directly relevant to questions posed by the study until the TMG has published its report. The TMG together with the STAMPEDE collaborators will form the basis of the writing committee and decide on the nature of publications. For the “transdermal oestradiol comparison”, as the efficacy analyses will be based on relevant data from the STAMPEDE and PATCH trials, TMGs for the two studies will form the writing committee. Any release, of efficacy or safety data, presentation or publication will be agreed with the TSC according to the terms of their charter.

All publications will acknowledge the participating centres and clinicians, and these will be detailed in an appendix to the main report. Papers will have named authors determined by the TMG according to the following principles:

- To be as inclusive as possible where this is practicable
- To ensure that there is justification for anyone to be named as an author

Reasons for nomination for authorship may include: trial design; grant holding; day-to-day trial oversight (TMG membership); analysis; discussion and interpretation of data; representation for key groups; active participation at large recruiting sites.

It should be accepted that the people qualifying for authorship will vary over time. In addition, key positions will vary depending on the nature of the publication: clinical lead for clinical papers, statistician lead for methodology papers, translational papers may be led by authors not on the main TMG if appropriate (e.g., the bone sub-study). In the event of any dispute related to authorship or data release, the TSC will be responsible for making the executive decision.

In the manuscript, a full list of sites and the number of patients recruited will be provided. In the presentations, this list of sites will also be shown. The term “the STAMPEDE investigators” will clearly be stated and relevant names included in the presentation credits.

A detailed **Publication Plan** is documented separately.



## 19 PROTOCOL AMENDMENTS

### 19.1 PROTOCOL

#### 19.1.1 Amendments Made To Sections In Protocol Version 1.0 (May-2004)

Administrative changes such as typos, word change etc.

Name additions/changes to:

TMG members

TSC members

IDMC members

'General Information' Section – additional information re. Abridged version of protocol

Section 1.2 – Figure 1, Celecoxib duration amended

Section 1.3 – Figure 2, addition of cardiovascular assessment form, name and timings amended

Section 2.3 – Docetaxel information updated

Section 2.4 – Additional text re dose and duration justification for Celecoxib use.

Section 3 – Title change and content updated

Section 4.2 – New exclusion criteria added

Section 4.3.1 – New investigations added and additional text re testosterone measurements and additional text re. prior celecoxib treatment

Section 6.1.4 – Celecoxib duration amended

Section 6.1.5 – Additional text re. Co-administration of docetaxel and bisphosphonates

Section 6.1.6 – Celecoxib duration amended

Section 6.2.2 – additional docetaxel information

Section 6.2.3 – addition of CV event history

Section 11 – Safety reporting updated

Section 12.1 – Additional text re. the collection of blood for genetic and serum marker studies

Section 15 – Additional information re. Central Subvention for docetaxel arms

#### 19.1.2 Amendments Made To Sections In Protocol Version 1.1 (May-2005)

Section 6.2 Administration and Dose Modifications, subsection 6.2.1 Zoledronic Acid

#### 19.1.3 Amendments Made To Sections In Protocol Version 2.0 (Jun-2005)

General Information section – SAE reporting fax number and timeframe added.

Section 1.2 – Addition of anti-androgen use for M0 patients as a method of HT

Section 1.2 – Increase in amount of blood needed & addition tissue sample request.

Section 1.3 Trial Documentation updated to include new table detailing trial documentation ahead of accreditation, the inclusion of the radiotherapy forms and correct case report form timings

Section 2.1 – Addition of anti-androgen use for M0 patients as a method of HT

Section 4.1.3 – Inclusion criteria Vii "Normal testosterone prior to hormone treatment" removed.

Section 4.1.3 - note has been omitted and moved to section 4.2 (see number 8)

Section 4.2 – Exclusion criteria added to exclude patients with active peptic ulceration, gastrointestinal bleeding and inflammatory bowel disease.

Section 4.2 – Exclusion Criteria added to exclude patients with planned major dental work

Section 4.3.1 - All blood test timelines changed from 14 days to 28 days.

Section 4.3.1 – Hormone Therapy pre-randomisation deadline extended from 4 weeks to 12 weeks.

Section 4.3.1 – Additional information regarding the use of NSAIDs and cox-2-inhibitors before coming on to the STAMPEDE study and once commenced on study treatment

Section 4.3.2 – Updated to ask for all vitamins and minerals the patient is taking to be recorded.

Section 4.3.3 – Updated to include the extra blood required and the request for consent of patients' tissue samples.

Section 6.1.1 – Addition of anti-androgen use for M0 patients as a method of HT

Section 6.1.6 – Addition of the calcium & vitamin name "calcichew".

Section 6.6.2 – asking also to collect vitamins and minerals under concomitant medication.

Section 6.6.3 – New section to inform investigators that patient's, who they wish to give radiotherapy to, are also eligible for STAMPEDE

Section 6.6.4 – New section to detail what data is being collected on the radiotherapy given to patients.

Section 7.1; figure 4 – Addition of radiotherapy form and in note, addition of AA alone

Section 7.1.2 – omission of repeated scans and x-rays at 24 weeks, also omitted in note under figure 4.

Chapter 11 – Safety reporting section updated

Section 17.3 – Increase in amount of blood needed & additional tissue sample request.

### **19.1.4 Amendments Made To Section In Protocol Version 3.0 (Jul-2006)**

Front Cover - NCRN logo added for accuracy

Front Cover - Clarification that protocol developed with NCRI rather than on behalf of

Front Cover - Clarification that it is a 6 arm trial

General Information section - MRC CTU staff section updated

Section 1.2 – Statistics section updated.

Section 1.2 - Additional research paragraph updated to reflect additional studies and for clarification of terms

Section 1.2 - Blood collection volume changed to reflect new technique used

Section 1.3 (figure 3) - Table showing case report form schedule updated to reflect clarification of follow-up schedule and addition of new CRF (End of Treatment)

Section 2.2 - AS changed to HT (clarification of terms)

Section 2.3 - Updated in information in regard to use of docetaxel added to reflect up to date practice

Section 2.5 - Sub-headings numbered for consistency

Section 3.0 - Information in regard to the Pilot Phase now written in past tense as Pilot Phase has now been completed

Section 4.1.1 - Inclusion criteria extended so that patients who fulfil 2 out of the three of the first inclusion criteria can be eligible.

Section 4.3.1 - Change in time scales by which baseline investigations need to be completed.

Section 4.3.1 - Clarification that chest X-ray is only required if chest is not included in the CT

Section 4.3.1 - Removal of 12 week timeline for baseline PSA test to be performed. (Stipulation that it must be performed before start of HT)

Section 4.3.2 – Information added in regard to time allowed from randomisation to start of treatment

Section 4.3.3 - Additional research paragraph updated to reflect additional studies and for clarification of terms

Section 4.3.3 - Blood collection volume changed to reflect new technique used

Sections 6.1.2-6.1.6 - Androgen Suppression replaced with hormone therapy for consistency of terms

Section 6.2.2 - '(Taxotere)' Removed for consistency

Section 6.2.2 \_ information added in regard to the need to closely monitor liver function prior to docetaxel administration

Section 7.1 - Page number reference updated

Section 7.1.1 - PSA measurement timings updated to accurately reflect follow-up schedule

Section 7.3 (Table 4) - Table and key updated to accurately reflect follow-up schedule and to include information about new CRFs and removal of withdrawal CRF

Section 8 - Rewording for clarification of definition of trial withdrawal

Section 8.1 - Instruction that withdrawal from trial treatment should be recorded on End of Treatment Form rather than withdrawal form

Section 8.1 - Information updated to emphasise that trial treatment must be discontinued following a progression

Section 8.2- Information added in regard to patient transfers

Section 8.3 - Instruction that withdrawal from trial completely must be notified in writing to the MRC CTU rather than included on withdrawal form

Section 9 and Summary – Target event numbers updated to reflect the slightly revised numbers obtained by using –nstage- which is the new, recommended program for MAMS trials

Sections 11.1 and 11.2 - Form numbers removed to allow for future changes in numbering

Section 11.2 – Reference to toxicity grading website added

Section 11.2.1 - Reference to table in appendix G added

Section 12.2 - 'Suggested' removed from 'Suggested patient information sheets'

Section 13 - CTA reference added

Section 17.3 - Information added to reflect new blood collection method for DNA analysis and in regard to additional translational studies for which funding has recently been approved

### **19.1.5 Amendments Made To Protocol Version 4.0 (Dec-2007)**

General Information Section - Randomisation and SAE reporting details sections clarified

Section 1.2 and throughout protocol - Efficacy Stages 1-111 renamed to Activity Stages 1-111 for accuracy and clarity

Section 1.2 - Follow schedule corrected

Section 4.1.2 - Inclusion criteria widened to include high risk relapsing patients, that would not have met the previous PSA based criteria

Section 4.1.3 - Note added to reference location of WHO performance status definitions

Section 4.2 - Notes added to reference locations of toxicity gradings and NYHA classifications

Section 4.3.1 - Timings of baseline scan information changed to accurately reflect most common current practice

Section 6.1.1 - Information about use of LHRH antagonists to ensure that the protocol accurately reflects current and future practice

Section 6.1.1 - Information about suggested duration of hormone therapy added to ensure that the protocol accurately reflects current practice

Section 6.2.2 - Additional information added about the timing of liver function tests prior to docetaxel administration added for clarity

Section 6.6.4 - Information on radiotherapy data collection added

Section 7.1.1 - Erroneous information about the timing of PSA measurements removed

Figure 3 - Moved to new section in protocol for clarity and extended to include current information on data collection

Figure 3b - Added to describe how extent of data collection during follow-up should change, post treatment and post progression

Figure 4 - Notes added to explain the changes in data collected at follow-up and to information that the quality-of-life study will be applicable to the first 700 patients randomised only

Figure 4 - Note added to include palliative radiotherapy CRF

Section 11.3 - SAE reporting information updated

Section 19 - Protocol amendments list updated

### **19.1.6 Amendments Made To Protocol Version 5.0 (Aug-2008)**

1. General Information Section – Randomisation phone line number updated – non UK extension added
2. Section 3 – Information about QL study removed to reflect closure of QL study after first 700 patients
3. Section 4.2 – Exclusion criteria clarified to explain that only patients with severe poor cardiovascular history should be excluded
4. Section 4.3.1 – Information on co-administration of NSAIDS with celecoxib changed based on clinical advice.
5. Section 5 - Randomisation phone line number updated – non UK extension added
6. Section 6.2.1. – Information added to clarify that patients who develop an osteonecrosis of the jaw should stop zoledronic acid treatment
7. Section 6.2.3 – ‘severe’ text added to accurately reflect which patients should be excluded based on their cardiovascular history

8. Section 7.1.2 – Definition of disease progression extended for clarity
9. Figure 3 – Updated to include reference to newly created skeletal related event form
10. Figure 4 – Previous error in table amended to show that the 4th Zoledronic Acid form that is submitted contains information about 3 cycles rather than 2 as previously indicated
11. Table 4 – ‘Other important medical condition’ added to definition of serious in the SAE section, to accurately reflect SAE form and current practice
12. Section 11.1 – Information added on reporting or pregnancies
13. Section 17 - Information about QL study removed to reflect closure of QL study after first 700 patients

### **19.1.7 Amendments Made To Protocol Version 6.0 (Jul-2009)**

1. General Information Section – Trial Pharmacist removed and changes of:

Co-Investigator

Patient Representatives

Trial Manager

Data Manager

General Information Section - Coordinating Centre – address change

General Information Section – change of Sponsor address

Section 1.1 – ratio of patients randomised to the investigational arms updated

Section 1.2 – figure 1b added to clarify trial design from Apr-2011 onwards

Section 1.2 – paragraph added to explain trial changes after the second activity analysis

Section 1.2 – wording added to clarify that QL data only collected for first 700 patients randomised

Section 1.3 – SSA Favourable Opinion removed from list of trial documentation required ahead of site accreditation

Section 2.1 – Amount of men diagnosed with prostate cancer annually updated

Section 2.4 – note added to explain completion of recruitment to celecoxib- containing arms

Section 2.5.2 - note added to explain completion of recruitment to celecoxib- containing arms

Section 3 – SSA Favourable Opinion removed

Section 4.2 – Exclusion criterion xiii greyed out

Section 4.3.1 – paragraph removed regarding potential randomisation to celecoxib-containing arms

Section 5 – Randomisation instructions expanded to exclude public holidays or dates when notice has been given by the CTU

Section 6.1.4 – formatting changed to grey font to reflect recruitment completion for arm D

Section 6.1.6 - formatting changed to grey font to reflect recruitment completion for arm F

Section 6.2.3 – recruitment note added

Section 6.6.3 – radiotherapy statement changed to reflect data from recent trials

Section 7.1.2 – removal of reference to SRE- specific CRF

Section 7.3 – Figure 3 - Addition of Bone Density Risk Factor Form and BMD sub-study assessment forms to summary of timing table

Section 7.3 – Figure 4 – Weeks added to timings of assessments post 2 years

Section 7.3- Figure 4 – note added to explain recruitment completion for arms D and F

Section 12.1 – Wording changed to reflect change to randomisation allocation ratio

Section 12.1 – Addition of statement regarding new information emerging during the trial

Section 12.2 – Reference to SSA removed

Section 16.3 – Statement added regarding actioning IDMC recommendation ahead of TSC ratification

### **19.1.8 Amendments Made To Protocol Version 7.0 (Jul--2011)**

1. General Information Section- SAE reporting fax number corrected

2. Section 11- SAE reporting fax number corrected

### **19.1.9 Amendments Made To Protocol Version 7.1 (Jul-2011)**

Throughout protocol – numbering has been updated in some sections new accommodate new information that has been added.

General Information Section – contact details updated

General Information Section – Funding information updated to include involvement from additional company

General Information Section – Wording on compliance and regulations updated to reflect current MRC CTU standard wording

General Information Section – Abbreviations list updated

Section 1.1 – The number of investigational agents being studied updated from three to four

Section 1.1 – Information regarding celecoxib updated to reflect that recruitment to these arms was discontinued in Apr-2011

Section 1.1 – Information about new IMP, Abiraterone inserted

Section 1.1 – Sample size and trial duration information updated to reflect changes brought about by additional trial arm

Section 1.2 – Summary information updated to reflect the discontinuation of recruitment to celecoxib arms and the addition of abiraterone

Figures 1a, b and c - Updated to reflect the discontinuation of recruitment to celecoxib arms and the addition of abiraterone

Section 1.2 – Information on trial stages updated to reflect changes brought about by additional trial arm

Section 1.2 – Information updated regarding the re-opening of the quality-of-life sub-study from implementation of protocol version 8.0

Section 2.1 – Wording related to hormone therapy updated for clarity

Section 2.1 - Updated to reflect the discontinuation of recruitment to celecoxib arms and the addition of abiraterone

Section 2.2 – Updated references added

Section 2.3 – Updated references added

Section 2.5 – Section added to give background information on new IMP, abiraterone

Section 2.6.1 – Updated references added

Section 2.7 – Section added to give information regarding radiotherapy which is to be given as part of standard care following recently published trial data.

Section 3 – Wording updated regarding selection of investigators to reflect current MRC CTU practice

Section 4.1 – Inclusion criteria updated with new criterion regarding radiotherapy use

Section 4.1 - Inclusion criteria updated with new criterion regarding contraceptive use

Section 4.1 – Wording of inclusion and exclusion criteria updated for clarity

Section 4.1 – Exclusion criteria updated with new criterion regarding acceptable liver function for trial entry

Section 4.1 – Exclusion criteria updated with specifics related to blood pressure levels

Section 4.1 - Exclusion criteria updated with new criterion regarding concomitant medications

Section 4.1 - Exclusion criteria updated with new criterion regarding prior treatment with abiraterone

Section 4.1 - Exclusion criteria updated with new criterion regarding prior treatment with chemotherapy

Section 4.1 - Exclusion criteria updated with new criterion regarding prior treatment with zoledronic acid

Section 4.3 – Wording updated to reflect that patients who initially fail screening can be re-screened at a later date

Section 4.3.2 – Wording updated regarding prior anti-androgen and LHRH use updated for clarity

Section 5.1 – Co-enrolment guidelines information updated to describe newly created co-enrolment CRF

Section 6.1 – Trial treatment information updated to reflect the fact that anti-androgens alone will be no longer permitted as hormone therapy

Section 6.1.1 – Updated to describe patients for whom radiotherapy should be given as standard practice

Section 6.1.1 a and b - Sections added to give information regarding radiotherapy treatment

Section 6.1.1-6.1.6 – References to further sections updated

Section 6.1.7 – Section added to describe abiraterone treatment

Section 6.2.4 - Section added to describe abiraterone treatment

Section 6.6 - Section added to give information regarding radiotherapy treatment

Section 7.1.1 – Reference to blood being taken at patient's home removed as this does not occur in practice

Section 7.1.2 – Wording updated regarding the reporting of biochemical failures for clarity

Section 7.1.2 – Wording updated regarding skeletal-related events for clarity

Section 7.1.3 – Section added to describe additional assessments required related to abiraterone treatment

Section 7.1.4 – Section added to provide information on when treatment should commence

Figure 4 – Updated for clarity regarding return of BMD sub-study forms, the addition the co-enrolment CRF and the description of the re-opening of the QoL Sub-study.

Figure 5 – Updated with reference to abiraterone and co-enrolment form

Section 7.3 - Wording on trial closure updated to reflect current MRC CTU standard wording

Section 8.1 – Additional criteria for definition of progression added for clarity

Section 8.1 – Definition of progression for abiraterone patients added.

Section 9 – Statistical information updated to describe the addition of the new trial arm

Section 11 – Safety reporting wording updated for clarity

Section 11 – SAE reporting fax number updated

Section 12 – Ethical information updated to describe the unequal randomisation allocation ratio

Section 12 – Ethical information updated to describe that the visit schedule will vary according to trial arm

Section 12.2 – Wording updated to reflect international participation in the trial

Section 13 – Wording updated to reflect international participation in the trial

Section 14 – Wording updated to reflect international participation in the trial

Section 15 - Updated to reflect the discontinuation of recruitment to celecoxib arms and the addition of abiraterone

Section 16 – Reference to trial committee charters added for information

Section 17.1 – Information added to reflect re-opening of quality-of-life sub-study

Section 17.2 – Timing of health economics analysis updated to previous error

Section 18 – Information on publication policy expanded for clarity

Section 19 – Information regarding amendments to protocol appendices moved to the separate appendices document

Section 20 – References extensively updated

### **19.1.10 Amendments Made To Protocol Version 8.0 (Sep-2011)**

Throughout protocol – numbering, sections headings, tables, figures and bibliographical references have been updated in some sections to accommodate new information that has been added

Throughout protocol – Androgen Deprivation Therapy has replaced Hormone Therapy as deemed more representative of the type of hormone therapy used in the study

General Information Section – New staff members of the MRC CTU and Co-Investigators added and contact details updated

General Information Section – Abbreviations list updated

Section 1.1 – Information regarding the new research radiotherapy treatment inserted

Section 1.1 – Information regarding docetaxel updated

Section 1.2 – Wording updated to reflect the addition of the new research comparison arm

Section 1.3 – Additional criteria for the re-accreditation of participating centres (for protocol version 9.0 only)

Section 2.1.1 – Wording updated to clarify the use of anti-androgen in trial patients

Section 2.1.2 – Information added to describe the rationale for the RT comparison arm

Section 2.8 – Information added to describe research RT treatment to prostate for patients with newly diagnosed metastatic disease

Section 3.1 – Information added to describe RT Quality Assurance procedures and centre accreditation

Section 4.1.1 to 4.1.3 – Wording updated to clarify inclusion criteria for all patients groups (newly diagnosed non-metastatic, metastatic and relapsing patients)

Section 4.2 – Clarification added on cardiovascular exclusion criteria

Section 4.2 – New exclusion criterion added concerning patients with prior exposure to hormone therapy

Section 4.2 – New exclusion criterion added to reflect the addition of the new RT comparison arm

Section 4.4.1 – Clarification added regarding pre-randomisation checks

Section 4.4.2 – Clarification added regarding permissible hormone therapy duration prior to randomisation

Section 4.4.5 – Information added regarding starting research radiotherapy treatment

Section 4.4.6 – Information updated on concomitant medications

Section 5 – Clarification regarding randomisation allocation added to reflect the addition of the new RT research arm

Section 6.1.8 – Information added to describe the administration of research radiotherapy

Section 6.2.1 – Clarification added regarding the measurement of serum creatinine levels prior to the administration of zoledronic acid

Section 6.2.3 – Clarification regarding the completion of recruitment to the celecoxib containing arms

Section 6.25 – Information added regarding the administration of research radiotherapy treatment

Section 6.6 – Clarification incorporated to describe the administration of standard-of-care radiotherapy

Section 7.1.4 – Information added regarding data collection and non-administration of standard radiotherapy

Section 7.2 – Section updated to include new treatment specific CRFs and timing of CRFs

Section 8.1 – Clarification added for the criteria to stop treatment for patients randomised to arm G

Section 8.2 – Section expanded to include additional details on study patient transfer to different centres

Section 8.3 – Additional sentence inserted to reinforce the importance of compliance with follow-up assessments

Section 9.1 – Additional paragraph inserted to clarify the method of randomisation and allocation distribution in the light of the introduction of the new RT arm

Section 9.4 – Wording updated to clarify the assessment of safety data

Section 9.5.4 – Wording updated concerning the end of randomisations to arm G

Section 9.6 to 9.6.4 – Section added describing sample size issues and trial stages for arm H

Section 9.8 – Clarification on intermediate stopping guidelines

Section 9.9 – Clarification on the outline analysis plan

Section 11 – Information on safety reporting updated to reflect the addition of the research RT comparison arm

Section 11 – Clarification added regarding arm A safety reporting timelines

Section 12.1 – Clarification added regarding the Principal Investigator's responsibilities

Section 14 – Indemnity section updated to reflect current MRC policy

Section 16 – Clarification regarding TMG membership

Section 17.3 – Section on Bone Mineral Density sub-study removed

Section 19 – Information regarding amendments to protocol appendices moved to the separate appendices document

Section 20 – References updated

### **19.1.11 Amendments Made To Protocol Version 9.0 (Oct-2012)**

Throughout protocol – numbering, sections headings, tables, figures and bibliographical references have been updated in some sections to accommodate the completion of recruitment to original research arms B, C and E.

Throughout protocol – Tenses have been changed to reflect activities that were in the future and which have now been passed.

Section 1 – Figure added and clarifications added to each figure

Section 2 – Previous reference 8 removed

Section 4 – Clarification of acceptable alternatives to bone scans

Section 6.2.5 – Correction of an error defining the PTV: the wording has been reordered

Table 4 – Dose-volume objectives corrected: order swapped

Table 5- Correction CRFs names

Section 17.3.2 – Clarification that DNA may be extracted

### **19.1.12 Amendments Made To Protocol Version 10.0 (Apr-2013)**

Throughout protocol – numbering, sections headings, tables, figures and bibliographical references have been updated in some sections

Throughout protocol – typos have been corrected

Section 4 – Clarification of exclusion criteria V (now V and VI)

Section 6 – Timing of orchidectomy prior to randomisation extended to 12 weeks

Section 6 – Clarification of hypokalaemia, blood pressure and fluid retention management

Section 9 – Statistical considerations amended in light of the recruitment extension for the abiraterone comparison

Section 14 - Section updated to reflect the changes in the structure of the MRC CTU (now MRC CTU at UCL) and indemnity arrangements

### **19.1.13 Amendments Made To Protocol Version 11.0 (Sep-2013)**

Throughout protocol – numbering, sections headings, tables, figures and bibliographical references have been updated in some sections

Throughout protocol – typos have been corrected

Co-investigators list updated to reflect the addition of the “enzalutamide + abiraterone comparison” lead

Section 1.2 – Enzalutamide added as trial treatment

Section 1.2 – Protocol version 12.0 added to the list of amendments

Section 2.10 – Rationale for the combination of enzalutamide and abiraterone

Section 4.2 – Eligibility criteria amended to reflect the addition of enzalutamide + abiraterone arm

Section 4.4.2 – Wording clarified

Section 6.8 – Clarification regarding end of trial treatment after starting trial therapy

Section 6.10 – Section added to describe enzalutamide and abiraterone treatment for the new research arm (Arm J)

Section 6.11.4.A – Section added to describe the management of toxicities from trial abiraterone

Section 6.11.4.B - Section added to describe the management of toxicities from trial enzalutamide

Section 9.1.4 – Section added to describe the statistical considerations concerning the introduction of Arm J

Section 9.3 – Principles and assumption for the introduction of Arm J added

Section 9.7 and sub-sections – Sample size issues and trial stages for Arm J

Section 9.9 – Details on interim monitoring and analyses for Arm J added

Section 11.2.1.D – Wording clarified regarding safety reporting requirements for control arm

Section 12.1 – Wording clarified

Section 15 – Details on funding for the “enzalutamide + abiraterone comparison” added

Section 19 - Amendments made to protocol updated

Reference list updated

### **19.1.14 Amendments Made To Protocol Version 12.0 (Jan-2014)**

Throughout protocol – typos have been corrected

Section 4.4.2. Wording clarified

Section 4.3. Wording clarified for eligibility to M1|RT comparison

Section 6.10. Addition of use of dexamethasone post-biochemical progression for Arm J patients

Section 6.11.4.A. Correction of CTCAE version

Section 6.11.4.C. Clarification on enzalutamide dose modification to be in line with current SmPC

Section 9.6. Sample size increase for M1|RT comparison

Section 11. Correction of safety reporting timelines for Arm A patients



Section 17. Addition of saliva samples collection for DNA analysis

Table 4, 5 and 6. Clarification on Case Report Forms and Follow-up schedule

### **19.1.15 Amendments Made To Protocol Version 13.0 (Feb-2015)**

Throughout protocol – typos have been corrected

Throughout protocol – clarification on the new definition of standard-of-care

Table of contents updated to reflect any changes to the protocol

Section 1.1. Wording added throughout section to include reference to survival results from “original comparisons”

Section 2.1.1. Section improved to include reference to survival results from “original research comparisons”

Section 2.1.2. Section improved to include reference to survival results from “original research comparisons”.

Section 2.1.3. Additional section added to describe the role of docetaxel for men with M0 or M1 disease

Section 2.9. Clarification on treatment completion and primary results for “original research comparisons”

Section 4.2. Clarification of Exclusion criteria XIII and XVI

Section 4.4.2. Clarification on HT prior to randomisation

Section 4.4.3. New section to clarify standard-of-care docetaxel treatment prior to randomisation

Section 4.4.7. Clarification on concomitant medication and contra-indicated concomitant medications

Section 4.5. Clarification provided on tissue block collection

Section 6. Inclusion of docetaxel into the standard-of-care

Section 6.2.3 New section to describe standard-of-care docetaxel administration

Section 6.11. Improvement throughout sections and sub-sections for abiraterone and enzalutamide-related toxicity management

Section 6.12. Section improved throughout to incorporate clearer details on concomitant medications and drug-to-drug interactions

Section 7.1.4. New section to describe data collection for standard-of-care docetaxel

Section 9.7.4. Clarification provided about implications for “enzalutamide+ abiraterone comparison” following change of standard-of-care treatment

Section 11.2.1.D Clarification on SAE notification timelines to reflect change in standard-of-care treatments (addition of docetaxel)

Figure 1. Figure updated to reflect change in standard-of-care

Figure 2. Figure updated to reflect trial history and recruitment over time

Figure 3. Figure updated to reflect changes in standard-of-care and recruiting arms

Table 1. Table updated to remove repetition

Table 13. Table updated to include new CRF to report standard-of-care docetaxel treatment

Table 15. Table updated to include only active trial treatments

### **19.1.16 Amendments Made To Protocol Version 14.0 (Oct-2015)**

Throughout protocol – typos have been corrected

Throughout protocol – clarification on the new definition of standard-of-care

Table of contents updated to reflect any changes to the protocol

Section 1. Wording added throughout section to include reference “metformin comparison”

Section 2. Section updated to include reference “metformin comparison”

Section 4.2. Exclusion criteria review to reflect Arm J closure and instruction of “metformin comparison”

Section 4.3. Clarification of comparison specific eligibility (M1|RT and metformin)

Section 4.5.7. Clarification on concomitant medication and contra-indicated concomitant medications

Section 6. Treatment sections improved throughout

Section 6.11. Section updated to include details on metformin treatment

Section 6.12. Amendment throughout sections and sub-sections for metformin treatment

Section 6.13. Amendment throughout sections and sub-sections for metformin treatment

Section 6.13. Improvement throughout sections and sub-sections for abiraterone and enzalutamide treatment

Section 7.0. Amendment throughout sections and sub-sections to include assessment and procedures specific to “metformin comparison”

Section 9.0. Section updated and streamlined to capture statistical considerations on each comparison

Section 9.0. Details on “metformin comparison” added

Section 11. Safety processes updated and clarified

Section 16.0 Membership to oversight groups updated

Section 11.2.1.D Clarification on SAE notification timelines to reflect change in standard-of-care treatments (addition of docetaxel)

### **19.1.17 Amendments Made To Protocol Version 15.0 (Mar-2016)**

Throughout protocol – re-structure of the treatment-related information for ease of use

Throughout protocol – clarification on the definition of standard-of-care

Throughout protocol – typos have been corrected

Addition of TMG members

Table of contents updated to reflect any changes to the protocol

New section for summary of trial added in table format

Section 1. Revised format for the summary of treatment groups, with the new transdermal oestradiol arm also added

Section 2. Clarification regarding research treatments that have previously reported or completed recruitment, section updated to include the “transdermal oestradiol comparison”

Section 3. New sections added for the “transdermal oestradiol comparison” and future planned biomarker-selected comparisons

Section 4.1.4. Change in definition of adequate renal function

Section 4.3. New section added for the biomarker-screening pilot, selection criteria removed for “research RT comparison”

Section 4.4.1. Change in definition of adequate renal function

Section 4.4.2. New section added for the patient selection criteria specific to the “transdermal oestradiol comparison”

Section 4.5. Screening procedure tables and figure added for clarification.

Section 4.5.1. New section added for biomarker-screening pilot investigations prior to randomisation.

Section 5.1.1. New section added for the biomarker-screening pilot registration.

Section 6. New sections added for the “transdermal oestradiol comparison”

Section 7. Amendment throughout sections and sub-sections to include assessment and procedures specific to “transdermal oestradiol comparison”

Section 7.1.4.B. Section added on cardiovascular outcomes for the “transdermal oestradiol comparison”

Table 18. Table added to clarify follow-up assessments

Section 8. Section updated for “transdermal oestradiol comparison”

Section 9. Section updated for “transdermal oestradiol comparison”

Section 12.1.1.D. Section added on participant feedback from investigations and additional analyses

Section 15. Section updated for “transdermal oestradiol comparison” and biomarker-screening pilot

### **19.1.18 Amendments Made To Protocol Version 16.0 (Oct-2017)**

Summary of trial- Table 1: Schedule of Assessments has been added

Abbreviations & Glossary- new terms have been added

Section 1- Table 4: Abiraterone information updated as results of primary analysis published

Section 4.3 - Biomarker timelines redefined, the length of prior hormone therapy has increased to reflect change in turnaround time for testing

Section 4.6 - Biomarker screening information updated

Section 6.2 – Clarification on safety monitoring required for patients receiving trial abiraterone added . Abiraterone overdose information altered for clarity.

Section 6.3.4 - Drug interactions updated to specify that tamoxifen is contraindicated in combination with abiraterone, enzalutamide and transdermal oestradiol.

Section 6.5 – Detail on requirements at site to demonstrate compliance with per-protocol required safety monitoring added

Section 7 – Schedule for assessments updated, removal of table 19

Section 7.1 - Clarification on additional safety monitoring required for patients receiving trial abiraterone added

Section 7.4 - Table 20 QoL information removed and added to Table 1: Schedule for Assessments

Section 10.1.1- Central monitoring of consent information added

Section 11 – Re-structured and re-worded for clarity on reporting requirements for safety data captured on the SAE CRF. Explanation provided for exempted events and definitions added. Table 28 and Box 1 updated and Figure 1 added.

Section 11.2- Updated SAE exceptions, SAE flow chart added for clarity

Section 11.3 - Update of investigator assessments and notification checklist for expedited safety reporting

Section 11.4 - Update of wording of CTU responsibilities

Section 17.4 - Sub-study information added to include Disease Volumetric sub-study

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